

**TEAM ADAPTATION AND MINDFUL BOUNDARY MANAGEMENT:
THE DYNAMICS OF INTERNAL AND EXTERNAL BALANCING**

by

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Dedication

To the love of my life, Dan, for your encouragement from the first moment I considered this pursuit to the point of completion; for the capacity you created for me to have balance between work, life, and school; and for the example you offered in your own work ethic and for pursuing something big – I love you and thank you for the life we have together. To Chuck and Anna for tolerating my endless weekends of studying and writing. Thank you for accepting me into your lives. In spite of my time spent on this degree, I have thoroughly enjoyed watching you grow into the wonderful young adults you are now and am excited for the grand adventures ahead of you. You were often a source of inspiration to persevere in setting and finishing a goal. To my parents for instilling a belief in me that if I always gave something my best effort I would be successful. Thank you for encouraging my love for learning throughout my life and for allowing the focus with which you ‘gifted’ me to flourish. To my in-laws (to be), thank you for your strong encouragement for me to make Case Western my first and only choice.

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List of Abbreviations

CCBS: Command and Control Boundary Spanning
CFA: Confirmatory Factor Analysis
CMB: Common Method Bias
CSQCA: Crisp-Set Qualitative Comparative Analysis
EFA: Exploratory Factor Analysis
IBS: Inclusive Boundary Spanning
MOT (or Mot): Motivation
NPD: New Product Development
QCA: Qualitative Comparative Analysis
QI: Quality Improvement
QM: Quality Management
QUAL: Qualitative
QUAN: Quantitative
S: Systems
SA: Strategic Alignment
SEM: Structural Equation Model
TQM: Total Quality Management
U.S.: United States

Team Adaptation through Mindful Boundary Management:
The Dynamics of Internal and External Balancing

Abstract

by

HEATHER GROOMS

Teams must constantly balance internally focused interactions through Adaptation and externally focused processes of spanning boundaries to facilitate change. This dynamic of team Adaptation has not been studied. In empirical studies on team Boundary Management or Adaptation, attention is given to laboratory based experiments and field studies of new product development teams in high-tech industries. This research focuses on continuous quality improvement (QI) teams which are comprised of a dynamic and shifting set of members otherwise attuned to their professional daily activities not directly related to the project. The teams of interest operate in highly institutionalized and regulated service industries of healthcare. The overarching research question in this study asks what factors influence team adaptation and how do teams effectively achieve internal and external balance in their QI projects, and to what extent does this contribute to project success. To answer this question, a developmental sequential mixed methods study is conducted that utilizes qualitative analytics through grounded theory based theme development and thematic analysis as well as quantitative analytics of Qualitative

Comparative Analysis (QCA) and Structural Equation Modeling. The first study is a mixed methods design utilizing a grounded theory approach for theme development followed by QCA to articulate complex causal interactions among identified factors influencing Adaptation. The sample includes 23 physicians / physician leaders / and hospital administrators who provide 39 team project examples for evaluation. The second study is a Quantitative design utilizing factor analysis to discern factors identified in study one in order to assess internal and external factor effects on Adaptation. A survey resulted in 215 responses for analysis from an expanded sample of team members in institutionally structured organizations. The third study is a Qualitative design utilizing thematic analysis to expand an understanding of the dynamic nature of Boundary Management mechanisms and team Adaptation patterns. The sample includes 13 team members providing 15 team project examples from 15 hospitals some of which are high performing teams and others are not. This research reveals that effective quality improvement project teams demonstrate Adaptation versatility by using combinations of several strategies to connect to the surrounding organization. By doing so this research contributes to the team literature as it expands descriptions of Boundary Management and Adaptation; it combines factors that have not been previously studied together and finds them to contribute to Adaptation. It also focuses on teams *in situ* in a context with previously limited attention. For management practice, the study provides practical explanations for team Adaptation as a process that can be mindfully guided through mechanisms of Boundary Management.

Key words: Team Adaptation; Cognitive and Affective Adaptation Patterns;
Team Boundary Spanning; Mindful Boundary Management Mechanisms; Strategic
Alignment; Systems Thinking; Quality Improvement; Healthcare

CHAPTER 1: INTRODUCTION

“The Heart Center’s Medical Director kicked off the improvement team he asked me to gather by first reminding us about the importance of the work we were about to begin. Patients who experience a heart attack type known as ST-Elevated Myocardial Infarction (STEMI) are at high risk for death or long-term poor outcomes due to heart muscle death if the blocked artery is not opened with balloon angioplasty as quickly as possible. Stroke and trauma refer to the time from which a patient arrives at a hospital to the time of intervention as the ‘golden hour’ – the optimal time to create the best outcome for the patient. The national goal for “Door-to-Balloon Time” (hospital arrival to balloon angioplasty) was moving from 120 minutes to 90 minutes. Our leader stood before us and said our goal was 60 minutes and we were going to do it by creating smooth, standardized processes and collaboration across multiple departments inside the hospital as well as through a new technology on the market that we would be first in the city to implement with various community based Emergency Medical Systems. It was up to our group to figure out how to make it all work and achieve 100% of all Door-to-Balloon Times of 60 minutes or less. The problem was our own departments didn’t work together that well, we knew of no easy methods for changing physician practices and behaviors, and we had no experts on the team in the area of project managing the implementation of technology. Go team.”

The example above is from a quality improvement (QI) project team I facilitated in 2005. While this project had a larger scope than many QI projects, the team challenges were consistent with those commonly experienced – something new is being developed through a group of people working together to design changes then implement those changes across varying levels of the organization. Whether an organization is developing products, seeking innovation, or spreading quality in response to complexity and change, teams are one organizational form that have been called upon as a primary source for meeting organizational priorities (Balkundi & Harrison, 2006; Kirkman & Rosen, 1999; Kozlowski & Ilgen, 2006). The challenge teams face is that they function in “environments that are adaptive, dynamic, structurally complex, with multiple variables from more than one level present at any time” (Ilgen, 1999: 137).

Teams¹, therefore, serve an important purpose in organizations in adapting to change and they do so through internal interactions and dynamics within the team as well as interactions across varying levels of the organization in which they are embedded through resources and energy flowing across boundaries (Arrow, McGrath, & Berdahl, 2000f). Such demands require internal awareness and sensitivity to potential needs and impact on people outside the team (Schippers, Den Hartog, & Koopman, 2007) as well as openness (LePine, 2003) and responsiveness to boundary feedback internally (Ancona & Caldwell, 1992a; Schippers et al., 2007) in order to adapt team interactions and goals. Team adaptation is a complex topic as it calls for balancing internal and external dynamics (Coghlan, 1994: 16) which is why this research focuses on team adaptation through boundary management.

Brief Team Context Background

The context in which teams interact with the boundary (i.e. individuals and groups outside the team) and its effect on adaptation is important (Kozlowski & Bell, 2008). The types of teams evaluated in the context of adaptation and boundary management has generally covered team settings such as knowledge teams (Woolley, 2009), action teams (LePine, 2003, 2005; Marks, Zaccaro, & Mathieu, 2000), new product development teams (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Carlile, 2004; Faraj & Yan, 2009; Hoegl & Weinkauff, 2005), and broadly generalized teams (Schippers et al., 2007; Wiedow & Konradt, 2011). When industries are represented (i.e. those studies not

¹ The words team and group are often used interchangeably in the literature and will also be used as such here Ancona, D. G. 1990. Outward Bound: Strategies for Team Survival in an Organization. ***The Academy of Management Journal***, 33(2): 334-365. Teams and groups refer to organizational members who see themselves and are seen by others as a group who interact dynamically and interdependently to carry out an organizational task and/or work toward a valued goal Ancona, D. G. 1990. Outward Bound: Strategies for Team Survival in an Organization. ***The Academy of Management Journal***, 33(2): 334-365..

conducted in a lab-setting), manufacturing is heavily emphasized, but service industries and public administration have been occasionally included, too (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003; Schippers et al., 2007; Wiedow & Konradt, 2011).

My context of interest is project teams, with a primary focus on QI teams in U.S. healthcare organizations, which have been under-represented in the literature but are an important and frequently depended upon source of improvement in healthcare (Shortell, 2004). Healthcare is also an important context as a highly scrutinized and regulated industry, due in part to its visibility through public reporting (CMS.gov, 2013; DartmouthAtlas.org, 2014; Werner & Bradlow, 2010); its impact on the US economy (17.2% of the Gross Domestic Product (GDP)) (Martin, Hartman, Whittle, Catlin, & Team, 2014); and strict regulations from accrediting agencies like The Joint Commission and Center for Medicare and Medicaid Services, as well as laws such as the Patient Protection and Affordable Care Act (Government, 2012). QI teams, in general, differ in some essential ways from NPD teams. NPD teams work together over extended periods of time (Brown & Eisenhardt, 1995; Edmondson & Nembhard, 2009) and benefit when they are protected from other day-to-day routines, where increasing levels of separation are needed the more innovative the product (Barczak & Wilemon, 1989; Ernst, 2002). QI teams, in contrast, are typically short-lived whose members have other full-time responsibilities – such as physician, nurse, or administrator. They also meet infrequently (Higgins & Routhieaux, 1999), and serve as the core unit tasked with implementing QI innovations at multiple levels of the organization (Alexander, Weiner, Shortell, Baker, &

Becker, 2006; Horbar et al., 2001; Leape et al., 2006; Mills & Weeks, 2004; Ohlinger, Brown, Laudert, Swanson, & Fofah, 2003).

Research Gap

Despite the interest in team adaptation and boundary management, empirical studies are lacking. In particular, team adaptation research lacks robust empirical study of teams *in situ*. Furthermore, the extent to which interactions with the external environment are linked to team adaptation is poorly understood. Little is known of adaptive behaviors or patterns among teams that have a mandate to design and implement organizational change initiatives (Bate, Mendel, & Robert, 2008). QI teams in hospitals are frequently tasked with an improvement initiative that could include process changes or clinical outcomes improvement affecting one or more departments and/or medical specialty areas indicating social change is at play as well (Batalden & Davidoff, 2007; Berwick, 2008). It is not unusual for those constituents to be unaware of the need for change or who identified the priority, in addition to having concerns about how the team's actions will affect their daily work.

Current QI team studies, however, provide little insight into the effects of the adaptive processes teams deploy to implement changes (Douglas & Judge Jr, 2001) or engage people across levels of the organization (Arrow et al., 2000f; Hackman & Wageman, 1995; Higgins & Routhieaux, 1999; Øvretveit et al., 2002). Team adaptation studies, in general, are conceptual and theoretical offering a variety of models of adaption (Burke, Priest, Wooten, Diaz Granados, & Salas, 2009; Burke, Stagl, Salas, Pierce, & Kendall, 2006; Kozlowski, Gully, Nason, & Smith, 1999; Kozlowski & Bell, 2008; Marks, Mathieu, & Zaccaro, 2001) but provide a lack of clarity about the core factors and

processes that promote adaptation (Kozlowski & Bell, 2008). I expected that both interactions with the external environment and internal team adaptations due to the heterogeneity of the teams in QI settings would be pronounced and complicated and worthy of empirical study.

Past research suggests that teams which perform well manage their boundaries by coordinating tasks (Ancona, 1990; Ancona & Caldwell, 1990; Marrone, Tesluk, & Carson, 2007; Sundstrom, de Meuse, & Futrell, 1990), obtaining resources (Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Joshi, Pandey, & Han, 2009), and exchanging information (Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Joshi et al., 2009; Marrone, 2010). Although boundary management studies indicate that teams are rated well by others as high performers when they span boundaries (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Marrone, 2010), these studies do not clarify how boundary management influences team adaptation leading to stronger performance.

Furthermore, the context in which empirical studies have taken place relevant to adaptation or boundary spanning has left a gap in service-oriented fields which operate with teams whose members contribute only a small amount of their work-time to the project. Team adaptation literature has mostly depended on laboratory-based action style teams whose members have specialized roles and cooperate in short-term events requiring improvisation and uncertain outcomes (Entin & Serfaty, 1999; LePine, 2003, 2005; Marks et al., 2000; Sundstrom et al., 1990; Woolley, 2009). The boundary spanning literature has been focused on manufacturing or high-tech new product development teams whose responsibilities, time, and functions are primarily dedicated to developing a product (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Ancona &

Caldwell, 1992a; Faraj & Yan, 2009). Both activities and theoretical domains provide an opportunity to understand how teams made up of professionals with potentially competing priorities (Davidoff & Batalden, 2005) interact with their environment, adapt, and produce a positive project result.

Motivation for Research

I have spent my entire career in healthcare in roles such as a Physical Therapist treating patients; as a Quality Improvement professional spanning various parts of hospitals and clinical specialties; and as a Manager and Director for ambulatory and hospital based units, respectively. In those various positions I have participated in and led multiple teams whose responsibility it was to improve a process or clinical outcome. The changes developed by those teams affected multiple individuals, hospital or clinic based sites, and departments. I have also been on the receiving end of similar projects where I had nothing to do with the teams. In spite of being trained and/or certified in multiple improvement methodologies for improvement, I continued to be challenged in achieving successful results – much less lasting results. Anecdotally, I found the most successful projects were those that engaged the most people beyond the team by seeking their expertise, opinions, and collaboration. None of the methodological approaches I learned and used, however, gave insight to how to do that best, nor did they clarify what to do with any input from outside the team that didn't seem to match our espoused theory for improvement. I wanted to be able to explain what the optimal mechanisms were for connecting to the larger organization and what a team should do with those connections, particularly in the healthcare setting.

I started this research journey by exploring the domain of systems-thinking as I had developed an interest in that area through earlier educational experiences. The business of healthcare is complex and research confirms the fragmented nature of healthcare in general (Charmaz & Olesen, 2003). Systems thinking promotes understanding the whole rather than breaking problems into pieces (Senge, 2006) and is interested in the interactions among system components influencing emergent behavior over time (Senge, 2006), which seemed relevant to teams working with larger components of an organization. From my quality improvement training I knew Deming (Deming, 1994) considered the knowledge of interrelationships between people and structures of a system as critical to effective management. Most organizations place their focus on events, yet it is the recognition of behavioral patterns and more importantly, the systemic, structural explanations that cause behavior that are most helpful (Senge, 1990). These concepts still left questions for me about how interrelationships with the organization are actively generated and what behavioral patterns or structural explanations promoted a team's ability to adjust or adapt its efforts. Such questions led me to dive into the team literature through which I began to recognize that behavioral patterns have been revealed in how teams adapt (Burke et al., 2006) in order to produce successful outcomes and the interrelationships with the organization were necessary through boundary management (Ancona, 1990; Ancona & Caldwell, 1992a). This three study journey prompted me to examine what we already knew about influences on and patterns of adaptation as well as how teams purposefully manage their boundaries to promote success. Because of the gaps noted earlier, I recognized an opportunity to contribute new understanding about what influences team adaptation and how teams

actively connect to the larger organization through boundary management. In my case, the focus of such activity was QI teams in healthcare.

Research that investigates teams *in situ* can benefit in particular the team adaptation literature, because so few studies are empirical and field based. Each of the three studies in this research called upon team members from teams in the field who could give insight to their lived experiences. By revealing contributory factors of adaptation and revealing patterns of adaptation over the life of a project, this research is expected to provide teams with new information that allows them to examine their sensitivity to the larger environment and how they can internally adjust their approaches to team work and also externally engage other people in their effort. Because there is an external component to adaptation, this research explores active boundary management as to provide teams with mechanisms that promote adaptation as well as seeks to confirm the importance and benefit of managing external boundaries. This is also the first study, of which I am aware, that considers boundary management as a key input to and source of pattern development in team adaptation. Finally, U.S. healthcare has a large impact on the economy, is highly scrutinized and regulated, and is responsible for people in their most vulnerable states. Teams seeking to improve clinical outcomes and the systemic processes of care have thus a strong imperative to be successful in their efforts.

CHAPTER 2: BACKGROUND AND LITERATURE REVIEW

Groups are adaptive systems engaging their environments in “two-way interchanges” requiring groups and their environments to adapt to each other (Arrow, McGrath, & Berdahl, 2000d: 28). Because of this need for adaptation and the interactions occurring outside the team, two main theoretical areas – team adaptation and boundary management – impacted this research. Furthermore, the idea of teams as systems and teams interacting with systems led to the inclusion of systems thinking for additional insights on groups and environments influencing each other. Each of these three theoretical areas is outlined here, highlighting the concepts that provided a foundation of understanding for these core theoretical areas. The individual studies – in Chapters 3, 4, and 5 – provide an in-depth literature review relevant to the research conducted.

Team Adaptation

Because adaptation is a concept of ongoing change, it is challenging to study and put in a box. The literature review revealed a variety of definitions and explanations of adaptation. All team related descriptions were dynamic in nature, such as a change in response to a cue or stream of cues leading to an outcome for the team (Burke et al., 2006); a patterning of change between groups and the embedding context (Arrow, McGrath, & Berdahl, 2000a); “cognitive, affective, motivational and behavioral modifications made in response to the demands of a new or changing environment or situational demands” (Baard, Rensh, & Kozlowski, 2014: 50); and a dynamic change occurring over time (LePine, 2005). In addition to definitions of adaptation, consideration is also given to the wide range of inputs promoting adaptation such as task demands and characteristics (Cannon-Bowers, Tannenbaum, Salas, & Volpe, 1995; Kozlowski &

Ilgen, 2006); team learning (Burke, Salas, & Diaz, 2008; Edmondson & Roloff, 2009; Kozlowski & Bell, 2008); teams as a network within networks providing for learning and knowledge sharing (Hansen, Mors, & Løvås, 2005; Kozlowski et al., 1999); cognitive and regulatory processes working in concert with team behaviors (Kozlowski & Ilgen, 2006; Priest, Burke, Munim, & Salas, 2002); and team member characteristics, in particular, knowledge, skills, ability, and attitude (Burke et al., 2006; Cannon-Bowers et al., 1995; Kozlowski et al., 1999). This research ultimately adopted a definition of adaptation as the extent to which a team generates a functional outcome by purposeful patterning of change through cognitive and behavioral changes between the team and its embedding context. To better understand the dynamic nature of adaptation and the balancing acts teams must adopt in pursuing goals while responding to their environments, this research also examined internal and external factors as well as the mechanisms of adaptation.

Internal and External Factors of Adaptation

Teams are triggered to adapt by a variety of demands or changes which can be expected or unexpected (Burke et al., 2006; Kozlowski & Bell, 2008) as well as internally or externally driven (Entin & Serfaty, 1999; Kozlowski et al., 1999). Adaptive teams flexibly modify internal processes (Klein & Pierce, 2001) by changing cognitive states or goal-inspired actions while segmenting tasks to manage demands (Burke et al., 2006; Woolley, 2009). Additional internal factors influencing adaptation include interpersonal processes such as assigning meaning to member actions (Salas, Rosen, Burke, & Goodwin, 2009); conflict and affect management as well as confidence building within teams (Marks et al., 2001); development of shared mental models and

cognitive states (Burke et al., 2006; Marks et al., 2000); and team reflection (Schipper et al., 2007; Wiedow & Konradt, 2011). Teams, however, are not isolated silos (Ilgen, 1999) meaning external factors influencing adaptation were also examined. Some examples of external influences include shifts in the external task or communication environment (Kozlowski et al., 1999; LePine, 2005; Marks et al., 2000); a new outcomes focus (Woolley, 2009); increasingly uncertain, ambiguous, or time pressured situations (Entin & Serfaty, 1999; Klein & Pierce, 2001); and changes in organizational systems such as structure (Kozlowski & Bell, 2008). But what do teams do with these internally and externally driven influences to adapt? This question inspired an examination of the mechanisms of adaptation.

Mechanisms of Adaptation

In reviewing the literature, I found general statements or descriptions of adaptation occurring through cognitive, affective, and/or behavioral mechanisms (Burke et al., 2009; Kozlowski et al., 1999; Marks et al., 2001); however, it had not been clearly categorized or defined along those lines. This research, in its final stages, attempted to do so revealing the following. Cognition includes recognition and recursive awareness in addition to the knowledge, skills, and abilities to recognize the need for change or improvements (Burke et al., 2006; Klein & Pierce, 2001; Kozlowski et al., 1999; Kozlowski & Bell, 2008; LePine, 2003, 2005; Salas et al., 2009). Affect as a component of adaptation is particularly weakly described, but given that teams must be sensitive to their environments, it was important to try to elicit an understanding from the literature. To that end, I found descriptions of feelings about team progress and environmental feedback manifested in openness to experience, desire for collective cohesion, and beliefs

about effectiveness (Entin & Serfaty, 1999; Klein & Pierce, 2001; LePine, 2003; Marks et al., 2001; Salas et al., 2009; Schippers et al., 2007). Finally, adaptation behavior descriptions are wide-ranging and therefore difficult to categorize in small groups, but ultimately resulted in this synthesized definition: Response to the environment manifested through feedback seeking behaviors to assess gaps between actual and desired states; enacting and executing processes relevant to tasks and communication; and interpersonal processes to share knowledge, demonstrate dependability, and modify network configurations (Burke et al., 2006; Ilgen, Hollenbeck, Johnson, & Jundt, 2005; Klein & Pierce, 2001; Kozlowski et al., 1999; Kozlowski, Watola, Jensen, Kim, & Botero, 2009; Kozlowski & Bell, 2008; LePine, 2003, 2005; Marks et al., 2001; Schippers et al., 2007; Wiedow & Konradt, 2011; Woolley, 2009).

The preponderance of the team adaptation literature is theoretically rather than empirically based. Models of adaptation covering team characteristics and cycles of assessment, planning, execution, and learning as well as emergent states of shared mental models and situation awareness have been proposed (Burke et al., 2009; Burke et al., 2006; Klein & Pierce, 2001; Kozlowski et al., 1999; Kozlowski & Bell, 2008; Marks et al., 2001). The empirical studies that have been conducted have been primarily lab-based experiments (Entin & Serfaty, 1999; LePine, 2003, 2005; Marks et al., 2000; Woolley, 2009) rather than focusing on complexity of team behaviors *in situ*. While these studies offer a variety of useful mechanisms of adaptation for consideration, the lack of empirical studies of teams in the field opened a wide opportunity to examine and identify the influences on team adaptation, particularly in light of the definition-based concept of adaptation as a functional change occurring through cognitive and behavioral changes

between the team and its embedding context. I wanted to understand the nature of how changes emerge due to the interaction between the team and its embedding context, which led me to the boundary spanning literature.

Boundary Management

Successful teams have a purpose that extends beyond the setting of the team itself and to conduct their work they must interact and coordinate with functions, groups, and individuals outside the team (Ancona, 1990; Ancona & Caldwell, 1990; Marrone et al., 2007). Complexity in their environments extends the need to actively manage external relationships (Marrone, 2010) so it is not surprising a team's success can often be best predicted by the team member's ability to coordinate external efforts and related resources (Joshi et al., 2009). Terms for this activity are used interchangeably, including boundary activity, boundary spanning, and boundary management (Ancona & Caldwell, 1990; Choi, 2002; Faraj & Yan, 2009; Joshi et al., 2009). At its core, team boundary spanning is the interaction activity that creates relationships with people beyond the team enabling the team to meet its goals (Joshi et al., 2009). Such interactions and associated behaviors include representing the group to others, coordinating tasks, and gathering information (Ancona, 1990; Ancona & Caldwell, 1990; Marrone, 2010) – all of which have been identified as key drivers in team performance and even organizational innovation (Ancona, 1990; Ancona & Caldwell, 1988; Marrone, 2010). In addition to knowing what boundary spanning is, how boundary management is conducted and what influences it also needed to be understood for this research and is summarized next.

Boundary Management Influences and Mechanisms

Multiple levels including individual, team, and organizational levels influence boundary management. At the individual level, individual characteristics such as member skills and personalities (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988) in addition to member roles and self-efficacy (Marrone et al., 2007) in conjunction with “affective, cognitive, and behavioral responses to the environment” (Joshi et al., 2009: 734) provide one level of input to boundary management. The team level includes a wide variety of influential characteristics including the nature of tasks (Ancona & Caldwell, 1988); team level attitudes, behaviors, and perceptions (Joshi et al., 2009); motivation and competency (Marrone, 2010); agreement about strategic and purposeful management of the environment (Marrone et al., 2007); and composition and development (Choi, 2002; Marrone et al., 2007). Finally, instrumental organizational characteristics promoting boundary management include organizational structure and climate (Joshi et al., 2009) as well as organizational tendencies for individuals to simultaneously serve on multiple teams (Ancona & Caldwell, 1988). These varying levels of external influences informed why teams engage in boundary management, but the mechanisms of how they do so also had to be considered. By and large there were four categories offered in the literature including: 1) persuading others to support the team through representation of the team’s work (Ancona, 1990; Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003; Faraj & Yan, 2009; Marrone et al., 2007); 2) coordination of task activities across the organization (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Marrone, 2010); 3) obtaining information from various sources across the organization (Ancona & Caldwell, 1990; Ancona & Caldwell,

1988; Ancona & Caldwell, 1992a; Marrone, 2010; Marrone et al., 2007); and 4) protecting the team from being overloaded (Faraj & Yan, 2009; Marrone et al., 2007).

Outcomes of Team Boundary Management

Team performance is influenced by iteratively interpreting environmental meanings, strategizing actions, and executing plans (Burke et al., 2006; Marks et al., 2000) while continuously scanning and engaging various levels of the organization (Druskat & Wheeler, 2003; Joshi et al., 2009; Marrone, 2010) as to accomplish team goals – all components of boundary management, which is an established predictor of positive team performance (Ancona, 1990; Ancona & Caldwell, 1988; Marrone, 2010). Teams benefit from boundary management activities by demonstrating increased team effectiveness, reducing overload on individual members (Marrone et al., 2007) and improving team psychological safety (Faraj & Yan, 2009). Because boundary management means teams cross levels in the organization, others benefit as well, such as increasing knowledge transfer within organizations and improving organizational learning (Marrone et al., 2007) and ingraining shared values across organizational units (Choi, 2002). The result of teams demonstrating increased boundary management activities is enhanced viability and higher ratings of performance from outsiders when they experience teams depending on them and emphasizing external interactions to diagnose outsiders' needs and experiment with solutions (Ancona, 1990; Marrone et al., 2007).

The extant literature on boundary spanning noted above is valuable for understanding general conceptions of why teams might reach out to others as well as how it is categorized such as in activities teams undertake to interact with the larger

organization while in pursuit of team goals. These reflect opportunities for teams to adapt such as in moments or characteristics of team motivation (Marrone, 2010), purposeful management of the environment (Marrone et al., 2007) information obtained (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Marrone, 2010; Marrone et al., 2007), getting the team's work noticed (Ancona, 1990; Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003; Faraj & Yan, 2009; Marrone et al., 2007), or coordination tasks and resources across the organization (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Marrone, 2010).

What I was unable to glean from the literature, though, is what combination of factors, styles of boundary management, or changes in boundary management occur based on the response from the boundary. Because adaptation occurs from an interaction with a group's context, I anticipated boundary management to significantly influence and explain mechanisms of how teams adapt. The concepts of boundary management and adaptation were not found together in my review of the extant literature providing an opportunity to add new insight to influences on adaptation and outcomes of boundary management.

Systems Thinking

Because my research interest and questions were oriented toward understanding how teams in complex, institutional settings adapt through interactions with the larger organization, systems-thinking was also explored. As businesses have increased in complexity, a systems perspective has become more critical (Senge & Massarik, 1987). This complexity within systems has also contributed to the expanded use of diverse,

cross-functional teams (Edmondson & Nembhard, 2009), which is reflective of the teams of interest in this dissertation. Systems thinking is conceptualized as a network of interdependent components that work together to achieve an aim of the organization (Deming, 1994). This concept was important to consider while collecting and analyzing data about teams working interdependently with others, in addition to the components of organizational structures that may influence a teams' efforts. Systems thinking also reflects a shared commitment to goals, ideas, issues, and management processes (Senge, 2006), reflective of what adaptive teams need to develop. Systems thinking further informs this work as it gives primacy to the whole, to relationships, and to seeing interrelationships rather than to individuals or things (Maani, 2002; Senge, 2006), similar to boundary management. "Businesses and human endeavors are systems" (Senge, 2006: 6) and are interconnected through people whether at the individual, group, or organizational level.

Teams in this research could be considered as systems and most certainly were interconnected across levels. Additionally, I anticipated teams focused on improvement efforts would intervene in a variety of processes and systems and wanted to understand if or how teams took this into consideration and whether they adapted their plans or interactions because of any systems lens they utilized. Since the systems-thinking literature can be broad and based on 'wholes, not parts' (Maani, 2002; Senge, 2006; Senge & Massarik, 1987), I was unsure of whether this stream of literature could be broken down to the level of a single factor influencing adaptation. Knowing, however, that groups are considered complex systems with open boundaries (Arrow et al., 2000f; McGrath, Arrow, & Berdahl, 2000), that projects are temporary systems (Grabher, 2002;

Manning, 2008), and teams are bound “dynamically and interdependently with [their] context and its subsystems” (Decuyper, Dochy, & Van den Bossche, 2010: 127), I elected to incorporate an understanding of systems thinking to support maintaining an openness to the many experiences team members might describe as impactful on their efforts.

Research Questions

To advance the understanding of the importance and interaction of team boundary management and team adaptation in improvement project teams, I explore the following research questions:

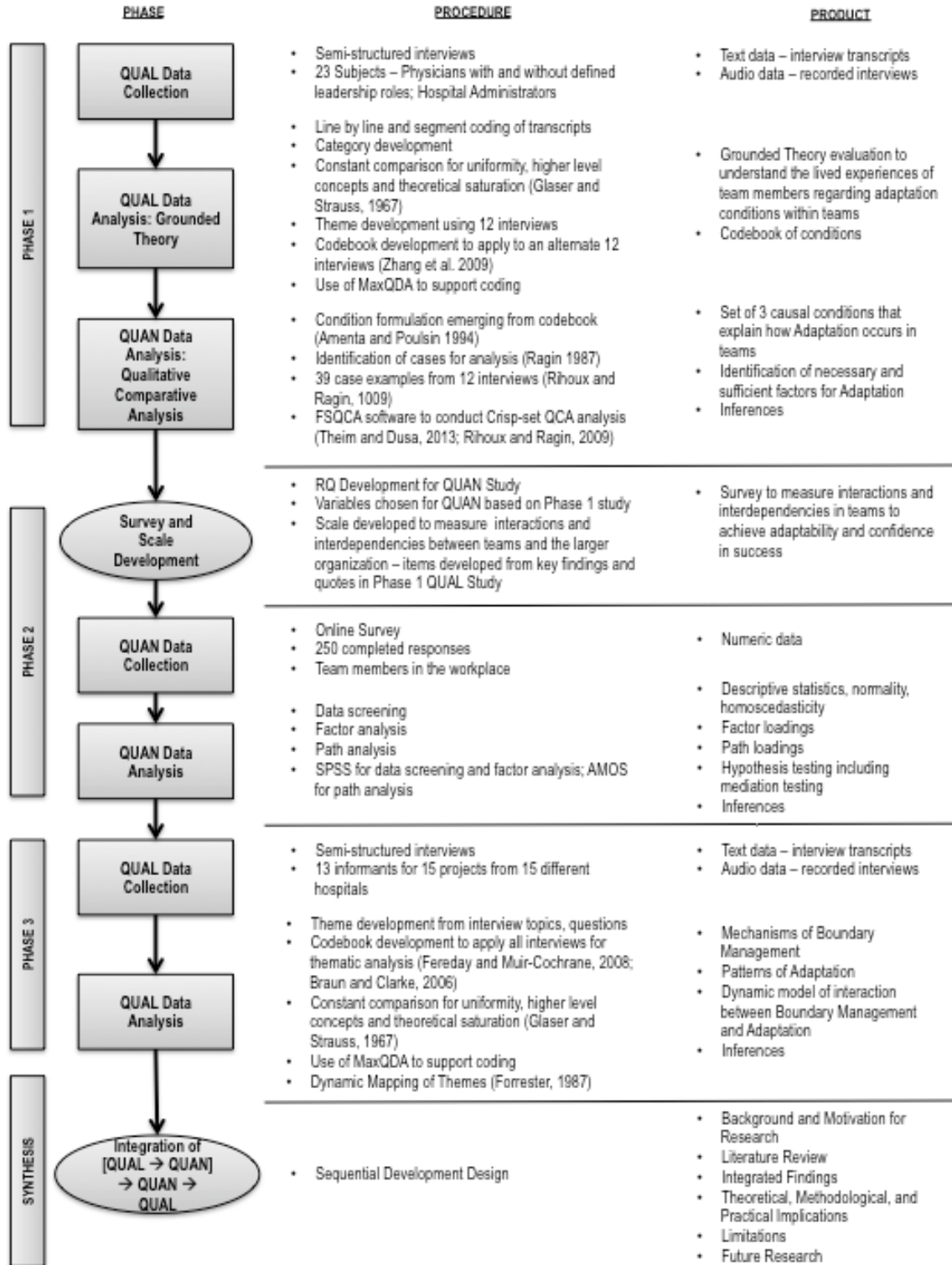
1. What are the factors of team adaptation and how do teams effectively achieve internal and external balance in their QI projects and to what extent does this contribute to project success? (Overarching dissertation question)
2. What combinations of factors influence effective QI team adaptation? What types of interactions between the team and its surrounding organization influence successful engagement with adaptation? (Study 1)
3. What factors of group interactions and interdependencies influence team adaptability and potency? (Study 2)
4. How does team adaptation emerge through boundary management? What are the mechanisms of boundary management that facilitate gaining support from outside the team in settings that are highly institutionalized and regulated? (Study 3)

Research Method and Design

This study adopts a mixed methods approach in addressing the above questions. The goal of a mixed methods approach is that the combination of qualitative and

quantitative research approaches “provides a better understanding of research problems and phenomena than either approach alone” (Molina-Azorín, 2009: 49). My research design was developmental as it started with an initial qualitative study using a grounded theory approach (Glaser & Strauss, 1967) that sought to identify how meanings are formed in and through experiences in team settings (Corbin & Strauss, 2008) relevant to team adaptation. I adopted in this study the idea of teams interacting with the larger organization and knew that related research discoveries would raise additional questions and inspire additional studies. The design strategy, therefore, was continual development that used the results from one study to inform and develop the next method for the next study in order to increase the validity of inquiry and related constructs (Greene, Caracelli, & Graham, 1989). Overall, my approach follows this design: ([QUAL→QUAN] → QUAN→ QUAL) which is a typical sequential development research design. Figure 1 represents the phases, procedures, and products of the overall dissertation research design.

Figure 1: Research Design Model



The first study ([QUAL → QUAN]) includes a mixed methods design that asks what combination of factors influence effective QI team adaptation. Physicians, physician leaders, and hospital administrators participating in QI in a team context comprise the sample. These roles, selected for study participation, provide insight to relevant clinical and administrative perspectives on how teams interact with the organization as to influence engagement with QI. Within their teams, there is a need to produce a change in the organization but at the same time control the diversity of perspectives in the presence of a broad participation, which would likely require team adaptation. Due to lack of empirical studies and related theory we utilize grounded theory (Corbin & Strauss, 2008; Glaser & Strauss, 1967) as to identify first salient themes that emerge from observing ‘teams in the wild’ (Burke et al., 2008; Salas, Cooke, & Rosen, 2008) when they pursue adaptations in their quality improvement efforts. Qualitative Comparative Analysis (QCA) (Ragin, 1987) method was then used to articulate complex causal interactions among identified factors that result in successful and not successful adaptations.

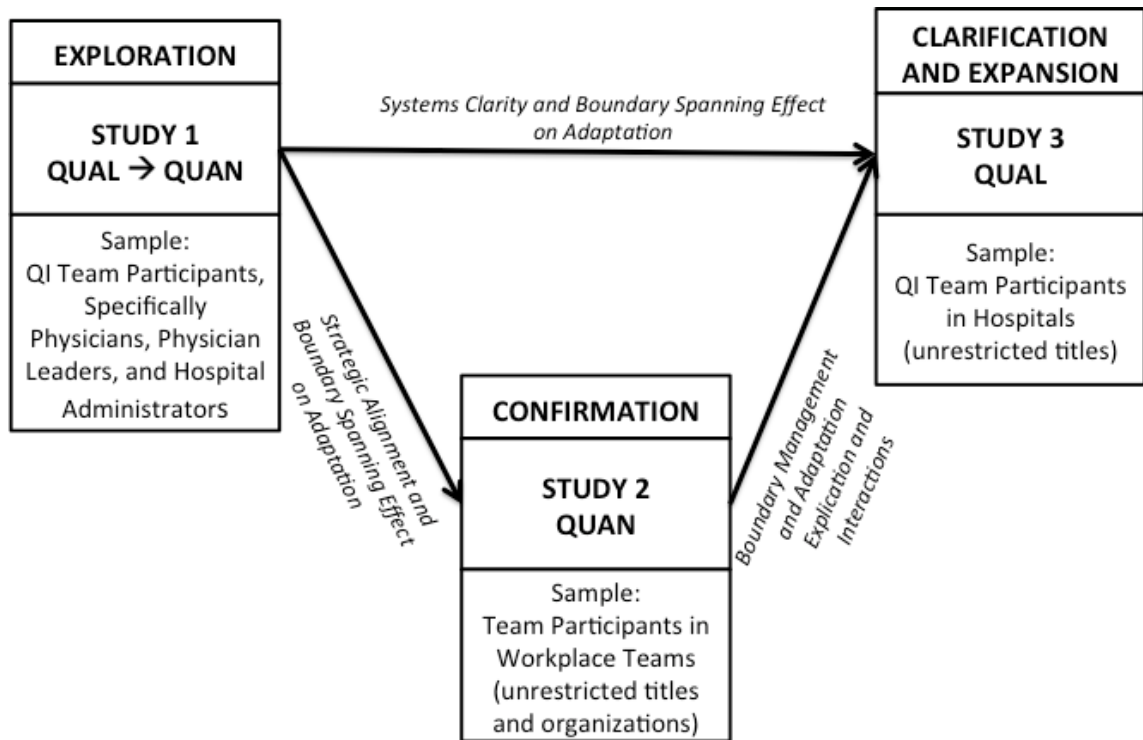
We next advance the sequential development design by asking what factors of group interactions and interdependencies influence team adaptability and potency. To do so, a survey-based statistical study is conducted as the second study ([QUAL → QUAN] → QUAN which seeks to confirm the influence of factors shaping adaptation. Such factors were identified and generalized based on qualitative findings (Creswell & Plano Clark, 2011) in the first study. The sample is expanded here to include members participating in teams who need to connect with the larger organization, but it is not strictly confined to hospitals or QI contexts. The reason for this expansion was to ensure that a large enough sample was collected to guarantee statistical conclusion validity and

demonstrate the generalizability of the factors and their effects for teams. The sample is a convenience sample garnered through personal contacts and social media sites relevant to my professional interests, followed by snowball sampling (Singleton & Straits, 2010). The context remains largely institutionally structured as represented by respondents from hospitals and educational settings.

The research journey is completed by qualitative thematic analysis ([QUAL → QUAN] → QUAN → QUAL), which seeks to expand my understanding of specific themes of how boundary management mechanisms and team adaptation patterns influence each other by specifically asking how team adaptation emerges through boundary management. This is accomplished by analyzing and systematically observing thematic interactions (Boyatzis, 1998) between key constructs and themes that were identified in the previous studies. Interviewing QI team members in hospital settings, again, creates the sample. The team members interviewed are those with a leadership or facilitator role as to provide deeper insight to the internal and external activities and experiences of a QI team. Sampling criterion also ensures an equivalent number of high and low performing teams for comparative analysis.

No one study alone could give a full insight as to how teams balance internal and external demands through boundary management and adaptation. Each phase of the research, however, led to new deeper understanding and insights while honoring the overarching dissertation question and intent of contributing new knowledge to team adaptation research and providing teams with actionable evidence. Figure 2 provides a general overview of how each study informed the next with italicized text highlighting factors of interest that were carried from one study to the next.

Figure 2: Overall Study Flow



The first mixed methods study, found in Chapter 3, describes themes and their combinations, which can promote team adaptation in healthcare QI teams. Chapter 4 contains the second study, which measures the influence of internal and external focuses of teams on their adaptation. The third study, found in Chapter 5, offers detailed explanations of how team adaptation emerges through boundary management in QI teams. Each study can be examined individually as well as considered together as a robust contribution to the team research. Chapter 6 integrates the findings and discusses conclusions and relevance. Chapter 7 offers contributions, limitations, and future research.

Research Findings

This research reveals:

- 1) Effective QI project teams are those who demonstrate adaptation versatility by using various combinations of several strategies to connect to the larger organization;
- 2) Teams who are internally focused increase confidence in their ability to succeed, but they may not effectively connect with the larger system as to adapt, which can reduce their performance;
- 3) Teams that are externally focused incorporate the larger organization's priorities and make connections to people outside the team, which influences their ability to adapt as well as supports confidence in success. These teams are adaptable, but the downside is that they are less confident in their success than internally focused teams;
- 4) In high performing teams, adaptation and boundary management form a feedback loop where a team interacts with its boundary by being sensitive to the boundary's needs. The team 'mindfully' adjusts interactions to achieve external engagement, which ultimately supports the team in achieving its goals. These results indicate the importance of balancing internal and external concerns within teams. Teams must understand and interact with components of the larger organization as to achieve their goals. At the same time teams must internally adjust their understanding of the boundary's needs and responses to mindfully generate strategies that will engage the boundary with the project plans to be developed and implemented so that goals are successfully achieved.

These findings are significant, because the data originate from people participating in complex teams in the field adding external validity to adaptation studies. Most team adaptation studies have been theoretical (Burke et al., 2006; Ilgen et al., 2005; Klein & Pierce, 2001; Kozlowski et al., 1999; Kozlowski & Bell, 2008; Marks et al., 2001; Salas et al., 2009) and lack direct studies of teams. Additionally, the list of antecedents to adaptation has grown diverse. In general, research is inconclusive about the core factors and processes that predict and promote adaptation. Relevant identified factors include: task demands and characteristics (Cannon-Bowers et al., 1995; Kozlowski & Ilgen, 2006); team learning (Burke et al., 2008; Edmondson & Roloff, 2009; Kozlowski & Bell, 2008); teams as a network within networks providing for learning and knowledge sharing (Hansen et al., 2005; Kozlowski et al., 1999); cognitive and regulatory processes working in concert with team behaviors (Kozlowski & Ilgen, 2006; Priest et al., 2002); and team member characteristics, in particular, knowledge, skills, ability, and attitude (Burke et al., 2006; Cannon-Bowers et al., 1995; Kozlowski et al., 1999). This dissertation's research, however, empirically identifies factors of strategic alignment, systems clarity, and boundary management as both antecedents to and feedback systems with team adaptation, the combination of which has not been previously offered in the extant literature.

Furthermore, the teams represented were primarily charged with improving or changing something that would affect a variety of people outside the team but within the organization. Prior studies have not treated interaction as an activity that is progressively developed to engage otherwise disinterested parties because most other studies utilize product development teams who have clear paths of coordination with other groups

(Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Faraj & Yan, 2009; Joshi et al., 2009; Marrone et al., 2007) working on individual aspects of ultimately a shared goal of producing a good or service. This study focuses on healthcare quality improvement teams who are tasked with changing processes and practices that other people carry out and who, prior to the project, often have no knowledge that this change is sought and considered a desirable pursuit, which offers a new context for the extant literature.

Teams that perform well have been identified as those who effectively manage their boundaries (Ancona, 1990; Ancona & Caldwell, 1988; Marrone, 2010) through functions such as coordinating complex tasks for product or service development (Ancona, 1990; Ancona & Caldwell, 1990; Marrone et al., 2007; Sundstrom et al., 1990); obtaining resources (Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Joshi et al., 2009); and exchanging information (Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Joshi et al., 2009; Marrone, 2010). This study found that high performing teams balance internal and external focuses by employing team adaptation as a dynamic process expressed in mindful boundary management's structural, cognitive, and behavioral mechanisms and through progressive development of engagement, which provides a new explanation for how high performing teams achieve success.

The remainder of this dissertation is organized as follows. Chapters 3, 4, and 5 each include the progressive studies conducted. Chapter 6 integrates the findings of the three studies and makes conclusions. Chapter 7 provides implications, contributions, and limitations of the research and suggests future research possibilities.

CHAPTER 3: ADAPTATION CONDITIONS WITHIN TEAMS: THE ROLE OF SYSTEMS, STRATEGIC ALIGNMENT, AND BOUNDARY SPANNING

Introduction

Teams operate in multilevel, dynamic, and complex environments, which requires balancing of internal and external demands and interactions as to accomplish team goals (Arrow et al., 2000d; Ilgen, 1999; Kozlowski et al., 1999; Kozlowski & Ilgen, 2006).

Groups are simply adaptive systems engaging their environment in “two-way interchanges” requiring that the group and the embedding environment adapt to each other (Arrow et al., 2000d: 28). Adaptation – the patterning of change at the nexus between groups and their embedding contexts (Arrow et al., 2000a) – then, forms the heart of this balance and also serves a desired outcome for team processes.

Team adaptation has received increased attention in the extant literature (Burke et al., 2006; Klein & Pierce, 2001; Priest et al., 2002; Wiedow & Konradt, 2011); however, it is a complex matter with a large variety of suggested inputs such as knowledge, skills, abilities, goal difficulty, leadership, and member behaviors, just to name a few (Kozlowski et al., 2009; LePine, 2005; Priest et al., 2002). Adding context to the mix – including situational, organizational, team characteristics and structures – increases the complexity of successful adaptation (Cannon-Bowers et al., 1995; Kozlowski & Bell, 2008). Factors found to contribute to successful adaptation include communication, information sharing, challenging norms through an open environment, and strong network ties (Kontoghiorghes, Awbre, & Feurig, 2005; Kraatz, 1998). We also know that organizations with mechanistic and/or hierarchical structures often suppress behaviors that promote adaptation (Chakravarthy, 1982; Weick, 2009). To date, however, much of the work on team adaptation has been theoretical in nature (Burke et al., 2006; Coghlan,

1994; Klein & Pierce, 2001; Kozlowski et al., 2009; Salas et al., 2009) likely due to the many inputs that could be empirically evaluated and the challenge of studying adaptation processes over time. Complexities of teamwork and adaptation suggest that many interacting factors in combination are likely to promote effective team adaptation (Burke et al., 2006; Goodwin, Burke, Wildman, & Salas, 2009).

Teams serve many functions in organizations. They are, among others, called to implement organizational initiatives and to enact related adaptive processes (Cooney & Sohal, 2004; Douglas & Judge Jr, 2001; Eisenhardt & Tabrizi, 1995; Hackman & Wageman, 1995), or to engage in introducing new innovative ideas, behaviors or product in the organization. Current team literature on adaptation has mostly studied manufacturing and technology firms where teams are involved in innovative behaviors associated with new product development (NPD) (Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003; Hoegl & Weinkauff, 2005; Sydow & Staber, 2002; Van Der Vegt, Emans, & Van De Vliert, 1999; Zeller, 2002). Less is known of adaptive behaviors among teams that have a mandate to design and implement organizational change initiatives. One example of such teams is healthcare quality improvement projects. They differ in some essential ways from NPD teams. NPD teams conduct the majority of their work together and demonstrate increasing performance over the course of time – up to 3 to 5 years (Brown & Eisenhardt, 1995; Edmondson & Nembhard, 2009). They benefit when they are protected from other day-to-day routines, where increasing levels of separation are needed the more innovative the product (Barczak & Wilemon, 1989; Ernst, 2002). Quality improvement (QI) teams, in contrast, are typically short-lived whose members have other full-time responsibilities – such as physician, nurse, or administrator. They

also meet infrequently (Higgins & Routhieaux, 1999), and serve as the core unit tasked with implementing QI innovations at multiple levels of the organization (Alexander et al., 2006; Horbar et al., 2001; Leape et al., 2006; Mills & Weeks, 2004; Ohlinger et al., 2003). Current QI team studies, however, provide little insight into the effects of the adaptive processes they deploy to implement changes (Douglas & Judge Jr, 2001) or engage people across levels of the organization (Arrow et al., 2000f; Hackman & Wageman, 1995; Higgins & Routhieaux, 1999; Øvretveit et al., 2002).

Our specific organizational context is healthcare – American hospitals, to be specific. Healthcare quality is highly scrutinized and regulated due, in part, to its visibility through public reporting (CMS.gov, 2013; DartmouthAtlas.org, 2014; Werner & Bradlow, 2010); (CMS.gov, 2013); its impact on the US economy (17.2% of the Gross Domestic Product (GDP)) (Martin et al., 2014); and strict regulations from accrediting agencies like The Joint Commission and Center for Medicare and Medicaid Services, as well as laws such as the Patient Protection and Affordable Care Act (Government, 2012). How adaptations are accomplished across varying levels within healthcare organizations in such context has been analyzed through descriptive accounts (Godfrey, Melin, Muething, Batalden, & Nelson, 2008; McKinley et al., 2008), but has received little empirical or theoretical attention (Bate et al., 2008). Healthcare QI initiatives in general involve social (Berwick, 2008), operational and process change (Batalden & Davidoff, 2007).

While multidisciplinary teams are viewed as a key factor in quality management (QM) (Cooney & Sohal, 2004; Daily & Bishop, 2003; Hackman & Wageman, 1995), there is little analysis how these teams connect with employees beyond the team or how

they interact with other units to bring about the improvement. Added to this gap is a lack of knowledge of the performance of QM teams where the professionals have competing priorities (Davidoff & Batalden, 2005) and face the challenge of engaging others to implement the change (Batalden & Davidoff, 2007). Douglas and Judge (2001) studied QM implementation success and its effects on organizational performance in some hospitals, but did not explore the effects of teams on facilitating quality improvement. As noted most quality improvement team studies have focused on manufacturing and only two studies have covered service firms with a small representation from healthcare (Daily & Bishop, 2003; Kaynak, 2003). These studies did not, however, examine whether there were differences between manufacturing and service industries in regard to effective QM by teams.

Our research question is: what combinations of factors influence effective QI team adaptation in a health care context. Furthermore, we ask: what interactions between teams and the surrounding organization influence successful engagement with adaptation? To answer these questions, we review the literature on adaptation and its salient inputs. We utilize a mixed methods approach to address these questions. Due to lack of empirical studies and related theory we utilize grounded theory (Corbin & Strauss, 2008; Glaser & Strauss, 1967) to identify salient themes that emerge from observing ‘teams in the wild’ (Burke et al., 2008; Salas et al., 2008) as they pursue adaptations in their quality improvement efforts. Qualitative Comparative Analysis (QCA) (Ragin, 1987) method is next used to analyze the complex interaction among identified themes that result in adaptations. We will finish with a report of our findings

and conclude with a discussion of the critical results as well as identify scholarly and practical contributions of the study.

Theoretical Framework

Adaptation

Adaptation (Baard et al., 2014) has been varyingly defined at multiple levels as: 1) Organizational adaptation – the extent to which an organization can adapt to changes quickly (Kontoghiorghes et al., 2005); 2) Complex adaptive systems inclusive of adaptation between teams and the larger system – a patterning of change at the interchange between groups and embedding contexts (Arrow et al., 2000a); 3) Team adaptation – “a change in team performance in response to a salient cue or cue stream, that leads to a functional outcome for the entire team” (Burke et al., 2006: 1190) and 4) Individual and team performance adaptation both of which include “cognitive, affective, motivational and behavioral modifications made in response to the demands of a new or changing environment or situational demands” (Baard et al., 2014: 50). The combined definition we adopt treats adaptation as the extent to which a team generates a functional outcome by purposeful patterning of change through cognitive and behavioral changes between the team and its embedding context.

Teams are one critical unit of adaptation. They do not exist in isolation and they span across individual, group, and organizational levels. Therefore they need to constantly balance between internal and external demands in “environments that are adaptive, dynamic, structurally complex, with multiple variables from more than one level present at any time” (Ilgen, 1999: 137). Therefore addressing change and

implementing organizational initiatives requires teams to be capable of adaptation (Burke et al., 2006) internally and through facilitation.

The Many Facets of Adaptation

Adaptation, how it arises, and its effects give insights to capabilities teams need to possess to be adaptive. A key outcome of adaptation is changing something within the team such as tasks, goals, cognitive structures, or composition in response to varying inputs or stimuli such as learning, innovation, or problems (Burke et al., 2006). Adaptive team performance emerges through team level processes generated by member interactions or between the team and its environment. These may span multiple levels such as member-team, team-unit, or team-team interactions (Burke et al., 2006; Kozlowski & Ilgen, 2006). Adaptation is also informed by members' awareness resulting in actions signaling a need for change (Burke et al., 2006) and it may be leveraged through strong leadership, clarity of goals, and constant feedback (Kozlowski & Ilgen, 2006).

The bulk of the team adaptation literature emanates from conceptual and theoretical analyses where adaptation is primarily treated as an outcome or as part of a full adaptive cycle (Burke et al., 2006; Ilgen et al., 2005; Klein & Pierce, 2001; Kozlowski et al., 2009; Kozlowski & Bell, 2008; Salas et al., 2009). In such analyses, the list of antecedents has grown diverse. In general research is inconclusive about the core factors and processes that predict and promote adaptation. Relevant identified factors include: task demands and characteristics (Cannon-Bowers et al., 1995; Kozlowski & Ilgen, 2006); team learning (Burke et al., 2008; Edmondson & Roloff, 2009; Kozlowski & Bell, 2008); teams as a network within networks providing for learning and knowledge

sharing (Hansen et al., 2005; Kozlowski et al., 1999); cognitive and regulatory processes working in concert with team behaviors (Kozlowski & Ilgen, 2006; Priest et al., 2002); and team member characteristics, in particular, knowledge, skills, ability, and attitude (Burke et al., 2006; Cannon-Bowers et al., 1995; Kozlowski et al., 1999). Table 1 synthesizes how these factors have been theoretically treated and how they have contributed to team adaptation research.

Table 1: Team Adaptation Antecedents and Concept Combinations

Antecedent	Outcome Studied	Effect on Outcome	Study
Regulatory Processes	Team Adaptation	Goal difficulty requires commitment and behaviors oriented toward goal striving to support the iterative process of self-regulation. Furthermore, individuals must allocate limited cognitive and behavioral resources to accomplish multiple goals, which is discretionary but critical to team performance making regulatory processes to influence adaptation multilevel in teams	(Kozlowski & Ilgen, 2006)
Task Demands, Characteristics, and Competencies	Adaptation as a dynamic process	Task demands determine workflow structure and coordination. Task demands are cyclical contributing to team regulatory skills, which promote adaptation. Use of information from the task environment to adjust strategies through behaviors and reallocation of intra-team resources.	(Kozlowski & Ilgen, 2006) (Cannon-Bowers et al., 1995)
Learning	Team Adaptation and Adaptive Cycle	Learning is facilitated through team training, which includes feedback information and motivational frames such as goal or performance orientation. Training enables teams to combine expertise, information, and resources to accomplish tasks. Team tasks are linked to the embedding environment necessitating appropriate team processes for resolution leading to outputs that cycle back in an adaptive loop. Learning can occur both before adaptation and as a consequence of adaptation. Learning translates, differentiates, and integrates knowledge allowing teams to “execute cognitive and behavioral processes” adjusting actions to the environment driving adaptation.	(Kozlowski & Bell, 2008) (Burke et al., 2006: 1190)

A relatively common theme in literature is the prevalence of obtaining information from the environment and applying team's cognitive and behavioral resources to adapt and accomplish some type of outcome – whether it is task completion, regulation, or learning (Burke et al., 2006; Cannon-Bowers et al., 1995; Kozlowski & Bell, 2008; Kozlowski & Ilgen, 2006). These studies are less clear how such information is obtained and how it is processed and by whom. Therefore, we are curious what teams actually derive from their environments, and how they utilize this information in order to change and adapt. Additionally, what behaviors are needed to engage with the environment in order to accomplish the intended result? In this regard the literature lacks actionable clarity to help answer such questions.

There are few empirical studies specific to team adaptation and related outcomes. LePine's (2005) study found that successful adaptation was influenced by a team's composition regarding its cognitive ability (producing positive adaptation) as well as interaction between goal difficulty and high performance orientation (producing negative adaptation) or between goal difficulty and high learning orientation (producing positive adaptation). This study, however, was specifically focused on role structures and their role in adaptation. It is also lab-based conducted with college juniors and seniors, which does not fully reflect real teams in the field. The other two empirical studies available utilize the same scale for adaptation developed in one of the two studies. Schippers et al. (2007) developed scales for team reflection, feedback seeking behavior, proactive personality, reflector learning style, and adaptation. The result of the study simply confirmed construct validity but did not predict any influence on adaptation by some of the included factors. In addition, face and content validity are in doubt because the scale

was developed by interviewing two managers about their perceptions of what is important in the context of adaptation (Schippers et al., 2007). Wiedow and Konradt (2011) proceeded to use a portion of this scale as an antecedent to predict team performance. They did find that team adaptation positively influences team performance. Overall, the lack of rigorous empirical studies and the paucity of field research to define and understand team adaptation, in addition to lack of a clear theoretical model of adaptation's inputs and properties, prompted us to pursue this topic through a field study where we examined clear group adaptations in their embedding contexts. In this inquiry we note the importance of teams interacting with their environmental contexts, specifically using boundary-spanning activities to promote adaptation (Cannon-Bowers et al., 1995; Schippers et al., 2007).

Team Boundary Spanning

Boundary spanning forms an important aspect in adaptation as it deals with how information flows across the team boundaries. Complexity related to the environment often creates for teams the need to actively manage their external relationships as to coordinate their efforts and bring together disconnected parties (Marrone, 2010). No wonder, in complex settings, a team's success can be often best predicted by the team members' ability to coordinate such external efforts and related resources (Joshi et al., 2009). Boundary spanning behaviors engage members in managing interactions within the environment and can include individuals or other teams within, or external units to an organization (Joshi et al., 2009; Marrone, 2010). Boundary spanning, itself, is thus a meso-level activity crossing the micro (individual) and macro (organization) levels (Joshi et al., 2009). This further emphasizes the complex nature of interactions teams need to

often manage. At the micro level team members' behaviors and responses to the environment can shape future interactions and boundary spanning (Joshi et al., 2009). At the macro level, "organizational structure, culture, and processes" can influence boundary spanning activities (Joshi et al., 2009: 734). This is similar to our earlier observation that organizational context influences adaptation (Chakravarthy, 1982; Kontoghiorghes et al., 2005; Kraatz, 1998).

Boundary spanners typically interact with the environment in various roles as to obtain information, coordinate tasks, as well as to obtain direction and assistance (Marrone, 2010; Singh, 1998). Boundary spanning behaviors are concerned with representing the team, carrying out its task coordination, and seeking information outside the team (Marrone, 2010). Some research has categorized boundary spanning behaviors accordingly as ambassadorial, task coordination, scouting, or guarding roles (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a) and identified related boundary spanning behaviors of informing, parading, and probing (Ancona, 1990). Here the extent to which team members enact each role indicates the extent to which teams are actively engaged with or separated from people around them.

These studies focus on behaviors and activities that embed the team in the environment and enable engaging with people outside the team – such as ambassadorial and probing behaviors. These have been also linked to better team performance (Ancona, 1990; Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a). However, the focus of these studies has not been on the interrelated factors of eliciting change and adapting, rather they have been primarily focused on understanding just teams' boundary spanning

activities (Ancona, 1990; Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003; Faraj & Yan, 2009; Marrone, 2010).

Boundary spanning literature has also mainly examined permanent teams in manufacturing and/or high-tech new product development (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Carlile, 2002, 2004; Faraj & Yan, 2009). Quality improvement teams as noted are somewhat different in their environmental connections, so the behaviors and knowledge these teams use to interact vertically and horizontally to facilitate change in complex systems is likely to be different. Such differences are likely to inform new concepts for adaptation or introduce alternate combinations of factors that promote adaptation facilitated by teams. Borrowing from Pawson and Tilley's (1997: 57) social system change model of "*outcome = mechanism + context*" we wonder what combinations of mechanisms, such as boundary spanning in the healthcare QI team context contribute to successful adaptation outcomes.

Methods

Design

Our research questions ask: What combinations of factors influence effective Quality Improvement team adaptation? Furthermore, what interactions between teams and the surrounding organization influence engagement with adaptation activities? Our research design was driven by our questions; the lack of empirical studies relevant to team adaptation; and the lack of team studies dedicated to our context of quality improvement teams. Therefore, to answer these questions we utilized a sequential mixed methods design (Qual→Quant) in which we first adopt a grounded theory approach (Corbin & Strauss, 2008; Glaser & Strauss, 1967) to identify and define salient themes in

team adaptation, and then applied Qualitative Comparative Analysis (QCA) (Ragin, 1987) to identify different combinations of causal conditions that are either necessary or sufficient (Berg-Schlosser & De Meur, 2009; Ragin, 1987) to produce a positive change – adaptation – related to an identified quality improvement initiative.

A qualitative research approach was deemed appropriate, as we sought to understand how meanings are formed through and in experience and to identify rather than test variables (Corbin & Strauss, 2008: 12). Team adaptation is still a nascent field with few empirical studies and because “little rich, detailed and evocative data” exist to “shed light on the phenomenon” (Edmondson & Mcmanus, 2007: 1162). Because of this, we initially conducted data collection and analysis using a grounded theory approach (Corbin & Strauss, 2008; Glaser & Strauss, 1967) which uses data collection context to ground concepts and minimize the risk of misrepresenting meaning (Corbin & Strauss, 2008). Furthermore, grounded theory reveals organizational action in a context, allowing the complexity of situations and outcomes to be identified and examined (Ng & Hase, 2008). Related microanalysis of interviews allowed us to break open the data, focus on pieces that seemed relevant and assisted in remaining open to concepts outside the initial research frame (Corbin & Strauss, 2008). This was followed by general analysis which took a broader perspective on the data and supported developing higher level concepts that were elevated into key factors (Corbin & Strauss, 2008).

We then extended beyond the grounded theory model to evaluate cases (i.e. specific project examples of group facilitation of adaptation) to compare the conditions or causes that fit together, or not, in various settings (Ragin, 1987). This analysis was based on the higher-level concepts that emerged from the grounded theory approach. We used

QCA method to detect a set of causal conditions through systematic comparison (Ragin, 1987) and compared conditions that promoted adaptation across all cases through a set of identified variables as to “establish common causal relationships” (Rihoux & Ragin, 2009: 2). Comparisons occurred at the case level where the combinations of characteristics of cases were evaluated against each other allowing for a holistic interpretation (Ragin, 1987). In this regard, QCA provides a rigorous computational method for such between-case assessment.

Overall, QCA is appropriate in small (10-15) and intermediate N (up to 100) cases (Rihoux & Ragin, 2009). At the same time the holistic nature of QCA (all cases are analyzed simultaneously) allows for the retention of complexity of causality while allowing for a higher level of generalizability (Rihoux, 2006). The fundamental assumption of QCA is that varying combinations of conditions can produce the same outcome depending on the context (Ragin, 1999). The strength of QCA is in identifying the minimum set of those combinations that produce an outcome. Given the richness of the stories and examples shared by interview participants, we decided to apply QCA to our intermediate n case examples of improvement projects (n = 50 case or project examples included for evaluation) to provide more parsimonious insights to the complex nature of causal conditions influencing adaptation by quality improvement teams.

Sample

Our sample consists of interview data of 23 U.S. physicians and hospital administrators who led or participated in QI team initiatives. We included these informants as the point of focus because of their frequent participation and leadership of teams, their expected interactions with multiple parts of the organization, and their

significant positions of influence. There are three subsets in our sample: physicians without clear leadership roles or administrative titles (n = 5); physicians with administrator or leadership roles as all or part of their work (n = 10); and non-physician administrators, from Director through C-Suite (n = 8). The participants were drawn from hospitals primarily in urban settings in the Midwest. In total, there were 18 males comprising 78% of the sample and 5 females for 22% of the sample. The average age was 47 years old. Each of the individuals included had significant experiences relating to quality improvement initiatives in a hospital in their roles in the prior two years. The physicians were either employed by or had other professional arrangements (such as admitting privileges) with a hospital. Physician leaders had similar ties to the hospital. Non-physician administrators were directly employed by the hospital.

Participants were initially sourced through personal contacts from the interviewer's current and prior working relationships in healthcare. This accounted for 11 of the participants. Additional participants were identified through "snowball sampling" (Singleton & Straits, 2010: 178). Snowball sampling assumes "members of a target population know each other" and has been used to "create sampling frames" (Singleton & Straits, 2010: 178). To add to our sample, participants were asked for names and contact information of colleagues who have also participated in QI initiatives. Five additional interviews were completed as a result of the eleven personal contacts providing introductions to their colleagues. The remaining seven participants were identified via introductions provided by non-participating healthcare friends and colleagues. By utilizing recommendations from our interview participants we built a sample that provided a depth and breadth of experiences.

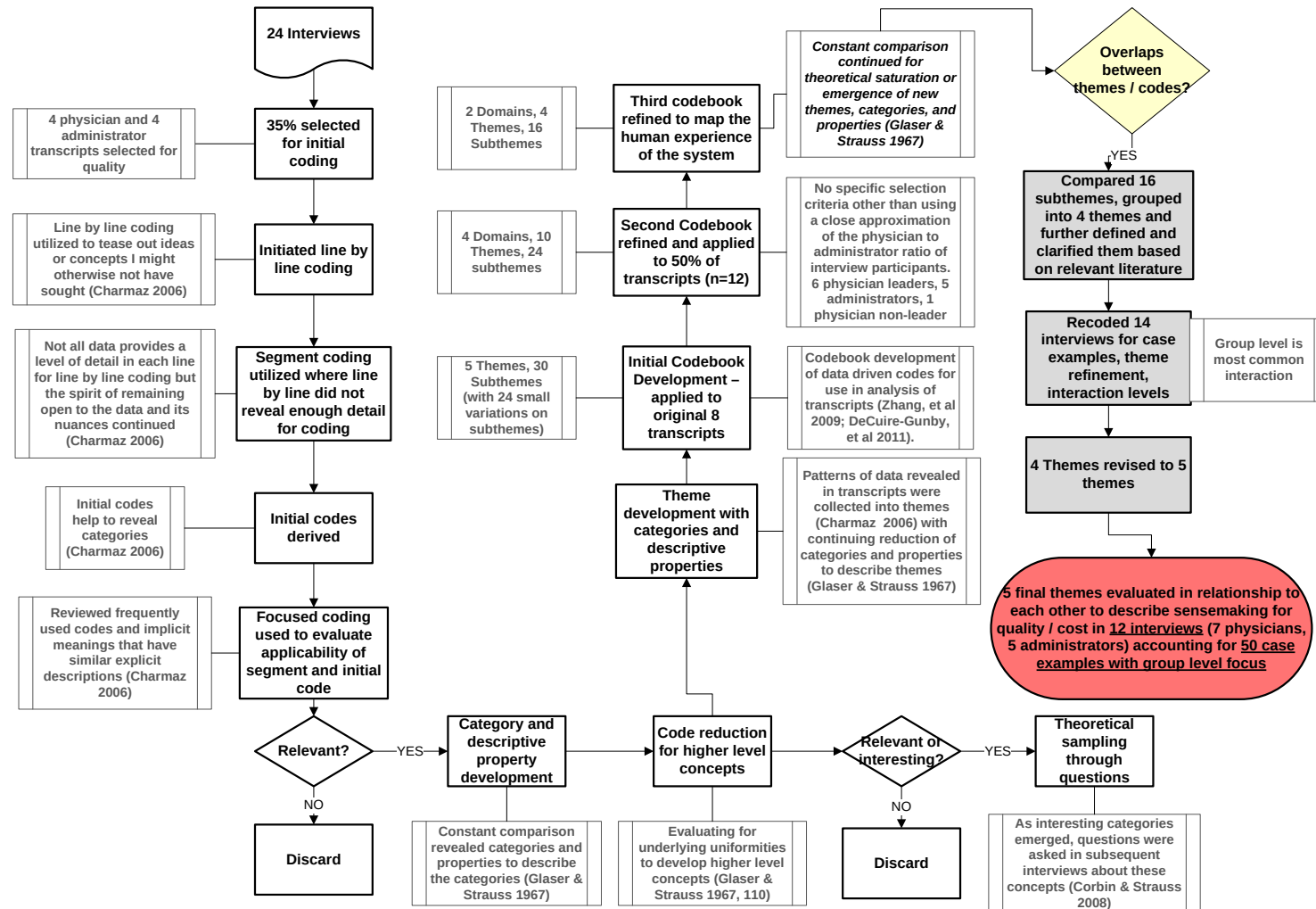
Data Collection

Data collection occurred between April 2011 and September 2011. Semi-structured interviews were conducted to obtain a rich history of lived experiences. Each interview lasted approximately 60 minutes and was guided by a protocol (See Appendix A for the protocol). The majority of interviews occurred face-to-face, except in instances of time or geographical barriers, in which case phone interviews were employed. The interviews were conducted in the environment of choice for the participant. All interviews were recorded with a digital recording device. The participants were advised of the recording and they were also assured of the confidentiality of the interview. They were offered the right to stop the interview at any time. The recordings were provided to a single reputable, professional transcription service for verbatim transcription resulting in 480 pages of transcripts. Transcripts were validated for accuracy and reliability in two steps. The first was through the transcriptionist highlighting, in the transcript, unfamiliar terms and/or phrases or those that were not clearly understood. The second validation step was through the researcher, who also conducted the interviews, listening to the recordings to fill in gaps highlighted by the transcriptionist as well as comparing the transcript text for accuracy upon receipt of the transcription.

Data Analysis

The overall flow of data analysis is summarized in Figure 3 below. We will next outline the logic of each step identified in Figure 3.

Figure 3: Coding, Theme Development, and Refinement Process



Data collection and analysis occurred concomitantly. First, 35% or 8 early transcripts were selected for initial coding due to the quality of the interview and anticipated richness of coded data as a result. MaxQDA software was utilized to support coding. Line by line coding served as the initial effort to tease out the ideas or concepts that might otherwise have been overlooked (Charmaz, 2006) by applying a personal bias to concepts of interest to the researcher. Given that not all line by line data provided a level of detail appropriate for such coding, segments were coded continuing in the spirit of remaining open to the data and its nuances (Charmaz, 2006). In that initial coding effort, approximately 1150 highly individualized codable moments were derived. We did not search for specific codes, instead, we allowed the data to reveal the concepts that were important to our interview participants regarding philosophies about quality improvement initiatives, in what ways their QI groups approached management of these issues, and who the groups interacted with to accomplish a quality improvement effort. Figure 4 summarizes the coding and theme refinements that continued prior to refining for context and causal conditions for QCA.

Figure 4: Grounded Theory Codes for Theme Development and Refinement

Code Reduction and Theme Development																																																																																																			
Emerging Themes		Theme Refinement 1 (5 Themes, 30 Subthemes + 24 variations within subthemes)	Theme Refinement 2 (4 Domains, 10 Themes, 24 Subthemes)	Theme Refinement 3 (2 Domains, 4 Themes, 16 Subthemes)																																																																																															
Physician and Administrator - Overlap - Accountability - Alignment - Change - Communication - Complexity - Cost - Culture - Decision-Making - Engagement - Ethics - Execution - Experience - Frustration - Futility - Future Orientation - Identity - Leadership - Multi-Disciplinary - Network - Perception - Quality - Relationship - System - Time		Behavioral Interactionism - Change - Communication - Conversation / Dialogue - Messaging - Decision Making - Incentive - Leadership Engaged Interactionism - Accountability - Alignment - Engagement World View - Ego - Futility - Perspective - Physician Leadership - Quality - Relational Dysfunction - Alignment - Cost - Culture Systems Integration - Complexity - Cost - Internal Systems - Network - Quality, Cost Balance - Quality – Clarity, Influences, and Engagement - System Development Time - Meaningfulness - Mindlessness - Urgency - Values	Domain - Emotion Theme: Relational <table><tr><td>Positive</td><td></td><td>Negative</td><td></td></tr></table> Theme: Engagement <table><tr><td></td><td>Motivation</td><td></td></tr><tr><td></td><td>Discouragement</td><td></td></tr></table> Theme: Perception <table><tr><td>Values</td><td></td><td>Others</td><td></td></tr></table> Domain - Behavior Theme: Communication <table><tr><td>Verbal</td><td></td><td>Non-Verbal</td><td></td></tr><tr><td></td><td>Messaging</td><td></td><td></td></tr></table> Theme: Interactions <table><tr><td></td><td>Resistors</td><td></td></tr><tr><td></td><td>Attractors</td><td></td></tr><tr><td></td><td>Alignment</td><td></td></tr><tr><td></td><td>Accountability</td><td></td></tr></table> Domain - Cognition Theme: System Elements <table><tr><td></td><td>People Groups</td><td></td></tr><tr><td></td><td>Internal Organization</td><td></td></tr><tr><td></td><td>External Systems</td><td></td></tr><tr><td></td><td>Principles</td><td></td></tr></table> Theme: Processes <table><tr><td>Business</td><td></td><td>Clinical</td><td></td></tr></table> Theme: Interaction or Links <i>(subthemes of linking domains and elements)</i> Domain - Time Theme: Influence <table><tr><td>Positive</td><td></td><td>Negative</td><td></td></tr></table> Theme: Surrogate <table><tr><td></td><td>Cognition</td><td></td></tr><tr><td></td><td>Emotion</td><td></td></tr><tr><td></td><td>Behavior</td><td></td></tr></table>	Positive		Negative			Motivation			Discouragement		Values		Others		Verbal		Non-Verbal			Messaging				Resistors			Attractors			Alignment			Accountability			People Groups			Internal Organization			External Systems			Principles		Business		Clinical		Positive		Negative			Cognition			Emotion			Behavior		Domain - People Theme: Intensity Interactions <table><tr><td>Ethics – Moral Judgment</td><td></td></tr><tr><td>Negative Emotional Responses</td><td></td></tr><tr><td>Positive Emotional Responses</td><td></td></tr><tr><td>Perception</td><td></td></tr></table> Theme: Reactionary Interactions <table><tr><td>Communication</td><td></td></tr><tr><td>Decision-Making</td><td></td></tr><tr><td>Ethics – Moral Character</td><td></td></tr><tr><td>Ethics – Moral Motivation</td><td></td></tr><tr><td>Negative Behavioral Responses</td><td></td></tr><tr><td>Positive Behavioral Responses</td><td></td></tr></table> Domain - Structure Theme: Contextual Inputs <table><tr><td>Ethics Awareness</td><td></td></tr><tr><td>Organizational Inputs</td><td></td></tr><tr><td>Time</td><td></td></tr></table> Theme: Knowledge of Forces <table><tr><td>Legal</td><td></td></tr><tr><td>Processes</td><td></td></tr><tr><td>Regulatory</td><td></td></tr></table>	Ethics – Moral Judgment		Negative Emotional Responses		Positive Emotional Responses		Perception		Communication		Decision-Making		Ethics – Moral Character		Ethics – Moral Motivation		Negative Behavioral Responses		Positive Behavioral Responses		Ethics Awareness		Organizational Inputs		Time		Legal		Processes		Regulatory	
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The initial codes were then reviewed to reveal categories (Charmaz, 2006) according to philosophies, QI management activities, and interactions. The first categories included 41 physician categories and 32 administrator categories, with 24 categories overlapping groups. The learning generated from this analysis was that system and interaction themes were among the primary experiences of these groups and should be evaluated further in the next coding effort and refinement.

Our next coding effort utilized focused coding to evaluate the applicability of each segment and its initial code (Charmaz, 2006) in order to further refine and understand these system and interaction experiences. Frequently used codes or concepts were reviewed for “implicit” meanings that had similar “explicit” descriptions (Charmaz, 2006: 57-58). Those codes that were not relevant were discarded, such as “Holy Grail Fatigue” and “Autonomy”, but those that were relevant were used to develop higher level categories and descriptive properties via constant comparison of similar codes (Glaser & Strauss, 1967). The previously separate categories by profession of informant were refined to five themes and 30 subthemes, with 24 characterizations within subthemes (as found in Theme Refinement 1 in Figure 4) and allowed us to develop an initial codebook (Zhang, Dolan, Lingham, & Altman, 2009) that was applied to both groups as noted in the coding process map in Figure 3. These revealed a need to better understand the people groups and systems comprising quality management as well as interactions occurring between people and systems.

In the next step we evaluated for “underlying uniformities” among codes, categories, and properties to develop higher-level concepts that were continually compared and assessed as relevant or interesting (Glaser & Straus, 1967: 110). This

refinement of codes included a selection of 12 transcripts or 52% and is found in Theme Refinement 2 in Figure 4. These 12 were not selected for a particular reason other than to ensure a close approximation to the ratio of physician and administrator participants. There were six physician leaders, five administrators, and one physician non-leader in this assessment. As salient categories emerged, questions were asked in subsequent interviews for additional theoretical sampling of these concepts (Corbin & Strauss, 2008). As a result of the code reduction and conceptualizing effort, patterns of data revealed in transcripts were collected into higher themes (Charmaz, 2006). The large number of domains, themes, and subthemes suggested the need to continue to reduce categories and properties that were used to further describe the themes (Glaser & Strauss, 1967) related to interactions of people and structures.

This effort resulted in a codebook of data driven codes that were divided into two domains, four themes, and sixteen subthemes (in Theme Refinement 3 in Figure 4) for use in analysis of an alternate twelve transcripts (DeCuir-Gunby, Marshall, & McCulloch, 2011; Zhang et al., 2009). As the remainder of transcripts were analyzed and coded, constant comparison continued for theoretical saturation or emergence of new themes, categories, and properties (Glaser & Strauss, 1967). While no new themes or categories emerged, we recognized overlap among categories and properties. We also focused on the heavy emphasis of interactions among people – particularly group interactions, which was considered in a new distillation effort that provided concepts for inclusion in QCA. This new refinement of the codes is represented in the right-most process map column in Figure 4.

The comparison continued in an effort to reduce the number of themes and subthemes to a set of conditions relevant to the context of cases provided by the informants, in order to identify meaningful conditions (Amenta & Poulsen, 1994) that could be applied in the final analytical effort utilizing QCA. The emphasis on systems and interactions among groups, as a central context for level of analysis, led us to combine subthemes from Theme Refinement 3 (in Figure 4) and conceptualize four overarching conditions that were then refined through triangulation both with data and literature. These four themes included: interdependence, embeddedness, systems thinking, and motivation, which were defined as follows:

- Interdependence is an action-oriented concept as found in interrelationships of activities and adjustment of tasks over time (Lenox, Rockart, & Lewin, 2006). Furthermore, it reflects a goal of interdependence guiding efforts among members (Gully, Incalcaterra, Joshi, & Beaubien, 2002) and dynamically integrating tasks into a whole (Huberman & Wilkinson, 2010).
- Embeddedness reflects linkages among social actors (Dacin, Ventresca, & Beal, 1999) and the potential for social capital to enhance or stagnate collective capabilities (Sydow & Staber, 2002). We thought this was a particularly valuable concept as it revealed whether those who were responsible for improving quality successfully interacted with relevant others in the organization (Ilgen, 1999) and thus were able to transform tasks and team efforts and relate them to heterogeneous systemic environments and needs (Manning, 2008).

- Systems Thinking is conceptualized as a network of interdependent components that work together to achieve an aim of the organization (Deming, 1994). Additionally, it reflects a shared commitment to goals, ideas, issues, and management processes (Senge, 2006).
- Motivation is described as encouragement to perform better and also to generate collective confidence and cohesion in pursuing an aim (Marks et al., 2001).

The above definitions were utilized in a final refinement during which these four themes were evaluated against 14 interviews to further specify their definitions. This evaluation gave rise to the seven final conditions and one outcome specifying QI team initiatives where adaptations were implemented in varying levels of the organization. These conditions and outcome were utilized to conduct an analysis of 12 interviews and their project (case) examples of group work to assess the combination of conditions, in QCA parlance, (Ragin, 1987; Rihoux & Ragin, 2009) that resulted in positive quality improvement adaptation. Case examples were projects – more than one project example was frequently offered – described by the interview participants – seven physicians and five administrators. Two of these interviews overlapped with the 14 interviews used for condition refinement in order to ensure their usability and meaningfulness. Constant comparison of cases continued through which we further articulated the characteristics of the conditions that could be utilized in the formal analytics of QCA. Development of these conditions and evidence from the interviews to serve as examples of the definition is outlined in Table 2.

Table 2: Condition Formulation for QCA

Four Overarching Conditions Resulting From Triangulation of Group Context, Data, and Literature	Condition and their Characterization for Use in QCA	Evidence from the Data
<u>Interdependence</u> - Action oriented concept as found in interrelationships of activities and adjustment of tasks over time (Lenox et al., 2006). - Reflects a goal of interdependence guiding efforts among members (Gully et al., 2002) and dynamically integrating tasks into a whole (Huberman & Wilkinson, 2010). - Informed by a cross section of themes (Figure 4, Theme Refinement 3), particularly subthemes of Moral Judgment and Motivation, Decision-Making, and Processes related to task development	<u>Adaptation</u> - Represents the outcome sought by groups, which was some type of change related to a quality or cost concern. - Adaptation focuses on changes in practices or processes but also represents changes in behavior and attitudes.	- A team's efforts to eliminate surgical site infections described as: "...we've gotten our SSIs down quite significantly, and it's tremendous work for the pre- and intra-op, but it's not at zero..." That group did not declare their efforts finished, but valued the changes that had occurred and the learning that could be applied to other aspects surrounding surgeries to continue to reduce the risk of infection. - Another example described the "biggest uptick in compliance with published quality metrics" when comparisons were made resulting in behavior and attitude changes as reported by "most of them decided, yeah, I'm gonna make this a priority."
<u>Systems Thinking</u> - A network of interdependent components that work together to achieve an aim of the organization (Deming, 1994). - Reflects a shared commitment to goals, ideas, issues, and management processes (Senge, 2006). - Informed by Contextual Inputs and Knowledge of Forces (Figure 4, Theme Refinement 3), particularly Organizational Inputs and Processes for Systems and Legal and Regulatory for Strategic Alignment.	<u>Systems</u> Processes and management structures, measurement systems, and technologies that informed where dysfunction occurred and whether progress toward change could be made evident.	In one example, "significant quality differences in utilization of the appropriate medications and technologies" were found through program reviews and data analyses. To drive improvement, this participant's group provided recommendations for change using infrastructure and processes that included "designated nurse practitioners who make sure that guideline-adherent, quality-assured care is administered in a timely fashion."
	<u>Strategic Alignment</u> Represents a group's awareness of larger system priorities and/or the establishment and agreement of priorities the group intends to pursue.	- A negative alignment example: A team focused on reducing costs – an organizational priority – associated with what was described as a "huge staffing problem" in a specialty area. The team developed a prediction model to better schedule staff according to patterns of utilization, but found no appetite in the organization for enforcing adoption of a new staffing pattern that would save large amounts of money. - A positive example: A project to reduce "inductions less than 38 complete weeks or 39 complete weeks of gestation" to minimize this as a "trigger, obvious trigger, to prematurity or preterm births." In this example, this was not only a priority for the team and organization it also became a priority for the community.
<u>Embeddedness</u> - Linkages among social actors (Dacin et al., 1999) and the potential for social capital to enhance or stagnate collective capabilities (Sydow & Staber, 2002). - Reveals whether those who were responsible for improving quality or cost successfully interacted with relevant others in the organization (Ilgen, 1999) and thus were able to transform tasks and team efforts and relate them to heterogeneous systemic environments and needs (Manning, 2008).	<u>Boundary Spanning – Who</u> Participants frequently specified groups of professionals, departments, teams, and sometimes individuals with whom they interacted to pursue a particular change priority.	Development of a dashboard to guide improvement and make performance transparent: We went "first to our physicians and our leaders and our board, but then to the general employees".
	<u>Boundary Spanning – Why</u> Participants noted why these groups or individuals were important interaction sources, albeit less frequently than other boundary spanning concepts.	"...so they could see how we're [performing]"
	<u>Boundary Spanning – How: Inclusive</u> Participants provided multiple examples, in story form, of how they included or engaged people outside the	Example of efficiency in the Operating Room: Physician said "if the nurses have an idea, we discuss it – say it's a great idea – let's do it tomorrow".

• Informed by Reactionary Interactions (Figure 3, Theme Refinement 3), particularly Communication, Positive and Negative Behavioral Responses	team to support the team's change initiative. Examples that encouraged such conditions as inclusion, collaboration, mutual interest, idea seeking and incorporation, deference to expertise were considered an Inclusive style of Boundary Spanning. <u>Boundary Spanning – How: Command and Control</u> In some examples, presented through stories, regarding how people outside the team were managed or addressed in order to achieve the team's outcome of interest, a directive style was utilized. We termed this Command and Control. This did not always stand alone, but at times followed an effort to be inclusive.	• Another physician who worked with a team to develop a faster lab test result provided an example in which they worked with the lab to find a mutual investment interest and then considered additional specialties that might be interested and stated "so before we do that, let's make sure that everybody's onboard." • Administrator noted "but you reach a decision point where you have to be mean" at which point they created a rules based system to which people had to adhere in order to practice at the hospital. • Another example was provided by a physician leader who was eventually able to achieve his team's goal related to a practice management change "because I absolutely insisted on it".
<u>Motivation</u> • Encouragement to perform better and also to generate collective confidence and cohesion in pursuing an aim (Ilgen et al., 2005). • Informed by Intensity Interactions (Figure 4, Theme Refinement 3), particularly Perception, but also informed by concepts of Ethics Awareness	<u>Motivation</u> Defined and specifically evaluated in the context of people external to the core group, represented by the interviewee, as being willing participants in the movement toward Adaptation.	• A positive motivation example was provided in regard to a balanced scorecard development that supported multiple groups' and the organizations' ability to assess their progress toward goals. In this, one of the clinical service lines engaged in the change showed how they wanted to participate in its development by stating "I know that this isn't a core measure, but could we put this on here? This would really help us." • Negative reactions were apparent too, though, as represented by an administrator's story of the leadership team working with clinical divisional leaders to reduce costs and the initial reactions presented by the clinical leaders in saying "You all need to change, but I'm doing OK. So he's got too many FTEs...and they spend too much money, but I'm OK".

Within each of the twelve interviews utilized, more than one team-based case example of a tangible adaptation effort was offered. Again, by case example, we mean group-based projects (both formally and informally planned work) with an intention to improve a quality issue. Overall, fifty cases were identified for use in QCA. In the final analysis, 39 case examples were retained and included in the QCA based on clarity of whether or not the outcome of interest was present or not present. That is, if it was unclear there was any movement toward or away from adaptation the originally identified case example was excluded. For example, one administrator described a team pursuing the implementation of "best practices" in a specialty area, but never specified if the best practices were adopted, only that professionals were invited to, attended, and enjoyed the monthly meetings.

Because it is preferable to use a reasonably small set of conditions in QCA when one has small (10-15 cases) or intermediate N (greater than 15 cases, less than 100 cases) (Rihoux & Ragin, 2009) so as to reduce the risk of creating more combinations than cases, we sought to reduce the number of conditions to the most feasible and logical ones (Berg-Schlosser & De Meur, 2009). Since Boolean algebra (two states exist – the condition is present or absent) is used to identify the logically possible combinations for analysis, the number of conditions is treated as 2^x (Ragin, 1987) meaning four conditions would produce up to 16 combinations, five conditions would produce 32 combinations, and up to seven conditions would produce 128 combinations (Berg-Schlosser & De Meur, 2009). We had 39 case examples to use, therefore, suggesting we should have no more than five conditions excluding the outcome, which is not included in the combination of causal conditions. Adaptation was identified as the outcome. This left two conditions to be excluded from the formal comparative analysis. Since there were four Boundary Spanning Conditions, the sensible choice was to first examine those for possible exclusion in order to retain conditions that were distinctly different from Boundary Spanning. It was apparent in our review for QCA inclusion that “Why” boundary spanning was conducted was infrequently described and “Who” was considered for boundary spanning was often more a descriptor to assist in story clarification, but “How” boundary spanning occurred was an aspect found in each Adaptation example. We, therefore, retained How-Inclusive and How-Command and Control Boundary Spanning in addition to Systems, Strategic Alignment, and Motivation for condition inclusion.

After identifying the main conditions of interest, we conducted crisp set QCA (csQCA) utilizing FSQCA software (Rihoux & Ragin, 2009; Thiem & Duşa, 2013). The basic operations of this software to analyze a set of causal conditions is outlined in Rihoux and De Meur (2009) and specified in detail in the FSQCA manual (Ragin, Drass, & Davey, 2008). Crisp set QCA utilizes Boolean algebra and uses Boolean minimization algorithm for “identifying patterns of multiple conjunctural causation” (Rihoux & De Meur, 2009: 33). The process includes creating first a raw data table, which includes the conditions of interest, including the outcome, and each individual case to be analyzed. In each case, the identified conditions are dichotomized as present (1) or absent (0) (Rihoux & De Meur, 2009). In our situation, 39 cases were added to the data table.

Once the raw table is finalized, it needs to be reduced to a truth table in FSQCA. This is a table representing configurations (combinations of conditions) that are associated with each outcome (Rihoux & De Meur, 2009) – adaptation in our case. We had no contradictory configurations², which allowed us to proceed to Boolean minimization. During this process, the software numerates all possible combinations. Only those case examples with a positive outcome are retained. Next, a measure of consistency identifies the configurations associated with each outcome – i.e. whether a condition is necessary for the outcome to be present (Rihoux & De Meur, 2009). Per Ragin et al. (2008), a consistency value of ≥ 0.75 was utilized to select the final set of

² Contradictory Configurations occur when a combination of conditions has a “1” outcome (the outcome is present) in some observed cases as well as a “0” outcome (the outcome is absent) in other observed cases. These need to be resolved prior to continuing with the Crisp Set QCA. Rihoux, B., & De Meur, G. 2009. Crisp-Set Qualitative Comparative Analysis (csQCA). In B. Rihoux, & C. C. Ragin (Eds.), *Configurational Comparative Methods: Qualitative Comparative Analysis (QCA) and Related Techniques*: 33-68. Los Angeles: SAGE Publications, Inc..

retained configurations associated with the outcome. At this point, we opted to utilize the software's capability to provide parsimonious, intermediate, and complex solutions for configurations to discuss our findings. These solution sets represent a continuum of possible minimization outcomes (Thiem & Duşa, 2013). Solution sets are those in which 1) logical remainders (or non-observed cases (Rihoux & De Meur, 2009)) are retained to become a part of the canonical union to produce a parsimonious solution, 2) logical remainders are eliminated to produce a complex solution, and 3) at least one logical remainder is retained to produce an intermediate solution (Thiem & Duşa, 2013). For more detailed descriptions of logical remainders and how these three solutions are derived, the interested reader can refer to Ragin (2009), Thiem and Duşa (2013), and/or Rihoux and De Meur (2009).

Findings

In QCA, the factors that are present (=true) as contributory to the outcome are represented by capital letters. Those that are absent (=false) are represented by lowercase letters. As noted three solutions are provided in FSQCA – parsimonious, intermediate, and complex (Ragin, 2009; Thiem & Duşa, 2013). Typically, the intermediate solution is considered superior for ensuring the inclusion of all necessary conditions that “‘make sense’ given the researcher’s substantive and theoretical knowledge are incorporated into the solution” (Ragin, 2009: 111). The intermediate solution is presented, below, providing initial insight for theorizing about what produces changes through healthcare quality improvement efforts in the group context. The asterisk (*) represents the logical operator ‘AND’ in Boolean multiplication which is applied in QCA to assess causally relevant combinations of conditions (Ragin, 1987). The solution sets below indicate the

combination of conditions – their presence or absence – that are associated with the presence of our outcome, Adaptation.

1. SYSTEMS * STRATEGIC ALIGNMENT * INCLUSIVE BOUNDARY SPANNING
2. SYSTEMS * STRATEGIC ALIGNMENT * command and control boundary spanning * motivation
3. SYSTEMS * MOTIVATION * STRATEGIC ALIGNMENT * COMMAND AND CONTROL BOUNDARY SPANNING

These combinations represent “multiple conjunctural causation” meaning more than one path leads to the same outcome, allowing for the complexity (Berg-Schlosser, De Meur, Rihoux, & Ragin, 2009: 8) that exists in group efforts to effect changes in an organization. In the remainder of this section, we will describe these conditions and their combinations in more common language; identify necessary and sufficient conditions; and examine how these combinations explain how adaptation occurs through groups in healthcare organizations.

The first combination of conditions, SYSTEMS * STRATEGIC ALIGNMENT * INCLUSIVE BOUNDARY SPANNING (S*SA*IBS), yields three conditions presenting together leading to adaptation. Groups that effectively identify processes, infrastructure, or measurement systems (SYSTEMS) that influence their capability to achieve change as well as obtain agreement about priorities to pursue (STRATEGIC ALIGNMENT) and utilize collaborative / cooperative / engaging behaviors to influence and embed change beyond the team (INCLUSIVE BOUNDARY SPANNING) effect change in processes and/or behaviors (ADAPTATION).

The second combination, SYSTEMS * STRATEGIC ALIGNMENT * command and control boundary spanning * motivation (S*SA*ccbs*mot), indicates that for the presence of Systems and Strategic Alignment in combination to influence change, both command and control boundary spanning and motivation must be absent. In other words, identifying processes, infrastructure, or measurement systems combined with agreement on priorities at the group or organizational level can stand alone to drive change. For this to occur, though, both directing or controlling others and collective or willing participation by others outside the team are absent.

The third combination, SYSTEMS * MOTIVATION * STRATEGIC ALIGNMENT * COMMAND AND CONTROL BOUNDARY SPANNING (S*MOT*SA*CCBS), appears to hold an opposite occurrence from the second. In this combination, directives and a controlling approach combined with participation from others beyond the team materialize. These conditions must occur in the presence of agreed upon priorities (strategic alignment) and an identification of processes, infrastructure, or measurement systems (systems) meaningful to achieving the desired improvement.

Before considering examples of these combinations and theorizing about how these varying conditions can work together to produce adaptation, we first review an important facet of QCA – necessity and sufficiency. The presence of Systems, Strategic Alignment, and Boundary Spanning represent necessary and sufficient conditions for Adaptation to occur. What we mean by this is that a combination of factors may be sufficient for a path to lead to a desired outcome, but may not be necessary as alternative paths may also achieve the desired outcome (Berg-Schlosser et al., 2009). A factor or

combination of conditions may be necessary as it or they are present in all combinations, but may not be sufficient, alone, to achieve the outcome (Berg-Schlosser et al., 2009). Because Systems and Strategic Alignment are present in all combinations, they are *necessary* for Adaptation, but *not sufficient* meaning they must be combined with the presence or absence of other conditions in order to lead to change. Boundary Spanning is sufficient, but not necessary, because there is a need for it to be clearly present or absent in each combination, but it does not stand alone nor is the same type of boundary spanning always present to lead to Adaptation. Motivation is neither necessary nor sufficient as it is only clearly present or absent in two of the three combinations.

In each case example, the informant describes a strategic priority that serves as the foundation for the improvement that is to be pursued. Within that priority are processes and management systems that support their capability of achieving a desired endpoint. What differs is when and how boundary spanning is used to assist in getting to that desired endpoint and whether or not motivation is described as further input to the outcome.

There are two examples of our first combination - SYSTEMS * STRATEGIC ALIGNMENT * INCLUSIVE BOUNDARY SPANNING. In these examples, different systems and different priorities are at work, but the priorities are clear and there are systems that play a role in achieving their desired outcomes. Both demonstrate an effort to include people and to create an environment of collaboration to achieve the outcome. In Table 3 we summarize the condition elements of the case.

Table 3: Case Examples for S*SA*IBS Combination

	S*SA*IBS – Example 1	S*SA*IBS – Example 2
Systems	Surgical case management – distribution between surgical center and hospital	“....went through process of accreditation for Chest Pain Society” (statement suggests this was used to guide their improvement efforts)
Strategic Alignment	Efficiency of the OR and surgeons’ time	Door to Balloon Time reduction sought
Inclusive Boundary Spanning	“If the nurses have an idea, we discuss it – say it’s a great idea – let’s do it tomorrow”	“....able to break down walls between various departments”
Adaptation	“We put it [ideas] to action immediately”	“....vastly improved door to balloon time”
Comments	The physician leadership group wants efficiency. They choose to do the bulk of their cases in the environment where they can have more control, where they can welcome ideas and discuss ideas and ultimately act in the way they want the organization to run – with efficiency.	The group used a known process for improving their Door to Balloon Time (which is a regulatory metric and good clinical care). The leader indicates it was the breaking down of walls or the building of collaboration between departments involved in this clinical process that led to improvement.

This combination suggests that one equifinal³ path to Adaptation includes clarity about what the priority is, not only in the group but across other interested parties as well. The group driving the improvement or changed outcome must also be aware of the process or management structure meaningful to the outcome. The additional condition needed, though, is to include people outside the team in a positive manner that encourages ideas, discussion, and cooperation. There were other case examples that included these conditions, but also intermittently included the additional condition of Motivation, but not Command and Control Boundary Spanning. Because Motivation is

³ Equifinality “means that different paths can lead to the same outcome” Berg-Schlosser, D., De Meur, G., Rihoux, B., & Ragin, C. C. 2009. Qualitative Comparative Analysis (QCA) as an Approach. In B. Rihoux, & C. C. Ragin (Eds.), *Configurational Comparative Methods: Qualitative Comparative Analysis (QCA) and Related Techniques*: 1-18. Los Angeles: SAGE Publications, Inc.

intermittent, it is not included in this equifinal path, since a purpose for conducting QCA is to reach minimal formulas that cover the observed cases (Rihoux & De Meur, 2009).

Another equifinal path exists, however, that does not indicate the need for boundary spanning activities in order to move toward a desired outcome. That path is SYSTEMS * STRATEGIC ALIGNMENT * command and control boundary spanning * motivation (S*SA*ccbs*mot). Table 4 summarizes the case example.

Table 4: Case Example for S*SA*ccbs*mot Combination

	S*SA*ccbs*mot
Systems	"And we are still in kind of a paper-based mode at this hospital." "And if you have to hand collect it, it really is labor intensive."
Strategic Alignment	"We use [scorecards] as part of credentialing. We report it to med exec on an annual basis." It is an agreed upon understanding that scorecards are used for credentialing purposes.
Command and Control Boundary Spanning	Distinctly absent
Motivation	Distinctly absent
Adaptation	Positive performance is expected as the leadership group will be giving the physicians what they want – data - "they're being hypocritical if they don't act on it" and because they must be credentialed in order to practice at the hospital.

In this second equifinal path, Systems and Strategic Alignment are again present, but are also able to work together without other conditions to achieve the outcome. In order for these conditions to do so, however, the command and control boundary spanning style and motivation are absent. The strategic alignment to use a system for a purpose that affects the team and associated groups provide the necessary components to accomplish the desired performance. As in the S*SA*IBS combination, there were other examples where this particular combination occurred with the presence of an additional condition (Inclusive Boundary Spanning), but since that condition was not consistently covered with this combination, it was excluded in the minimization result. This means that Inclusive Boundary Spanning may occur in the presence of this combination, but it is

not necessary for this combination to effectively lead to Adaptation. The necessary condition holds that Command and Control Boundary Spanning and Motivation both be absent for Systems and Strategic Alignment, alone, to lead to Adaptation

Finally, the third equifinal path – SYSTEMS * MOTIVATION * STRATEGIC ALIGNMENT * COMMAND AND CONTROL BOUNDARY SPANNING

(S*MOT*SA*CCBS) – could be considered the opposite of the first two. In this combination, all conditions must be present. The reason we do not see that Inclusive Boundary Spanning does not have to be absent is similar to the previous combinations, it is at times present, but not consistently so it is not included in a minimization result.

Table 5 summarizes the case example.

Table 5: Case example for S*SA*CCBS*MOT Combination

	S*SA*CCBS*MOT
Systems	Process of documentation to describe clinical choices made (supports metrics of performance) – “....you have to document contraindications to not being given fibrinolytics, right?” “That really is kind of a physician documentation issue and so that does go back to the physicians’ daily workflow.”
Strategic Alignment	“We’re trying...to become a Stroke Center of Excellence”
Command and Control Boundary Spanning	“As administrator, it’s a constant struggle for the rank and file emergency physician that I administrate over to impress upon them that this really does impact you”. The physician leader also uses comparison of physician performance to motivate behavior change.
Motivation	Physicians “internalized it well”. “....they understand it. Everyone else is doing better, I’m gonna do better”.
Adaptation	Physicians decided to make this a priority.

This case example provides insight to the interrelationship of factors representing agreement about a strategic priority, the processes that support the priority, and the style of boundary spanning necessary to obtain participation in the effort as the combination

represents. In order to pursue the larger priority of obtaining the status of Stroke Center of Excellence, along with meeting other regulatory metrics, documentation had to be completed in a particular manner. This incorporates both a strategic priority and the processes that support meeting the priority. Physicians, however, needed further impetus beyond regulatory metrics and an organizational designation to encourage a behavior and practice change. This group needed to appeal to a sense of competition as well as to direct the physicians to change behavior indicating the use of a Command and Control Boundary Spanning style as well as noting they had been motivated to make the changes.

Organizations are complex as are healthcare quality concerns. Creating change through groups, therefore, requires a complex interplay of conditions to move not only the group tasked with the change effort, but those around them as well, toward the desired direction. QCA allows for recognizing such complexity. Situations have special context, people across the organization are responsive to different tactics, and groups can manipulate specific conditions to achieve a desired outcome through differing paths. Identifying and understanding meaningful processes, management structures, and measurement systems along with agreed upon priorities are necessary to lead toward a desired change. Boundary Spanning, in two different styles, is sufficient to support Adaptation, but must occur in combination with Systems and Strategic Alignment. This creates a challenge for groups to be aware of the context in which their improvement effort is positioned as well as sensitive to the types of interactions that will engender the necessary engagement of people outside the team to achieve the desired outcome.

Discussion

We began our study asking what combinations of factors influence effective quality improvement team adaptation. Additionally, we were interested in what interactions between teams and the surrounding organization influence engagement with adaptation activities. We carried out a sequential mixed methods (Qual → Quan) study utilizing semi-structured interview data to develop themes through a grounded theory approach and by using qualitative comparative analysis (QCA) to assess the combinations of factors that led to a result. Our study showed that QI teams facilitate adaptation through changing constantly a portion of the organization's processes or practices. They accomplish this by combining strategic alignment (i.e. alignment with organizational priorities) and systems clarity (i.e. processes, infrastructure, and measurement systems) with behaviors that engage people outside the team (boundary spanning) in their efforts. Additionally, these teams focus on varying levels of the organization – that is individual or groups of departments – rather than the improvement being centered on the team.

Many of our findings associated with individual factors influencing adaptation align with team and quality management literature. The concept of systems influencing adaptation is reflected in the quality management literature where process management and data quality / reporting serve as core characteristics of quality management and have been shown to have positive effects on performance measures (e.g. quality, firm, and employee performance) (Kaynak, 2003; Nair, 2006; Sadikoglu & Zehir, 2010; Sousa & Voss, 2002). The quality management literature, however, does not clearly combine

systems with team facilitation of strategic alignment or boundary spanning to effect change in the larger organization.

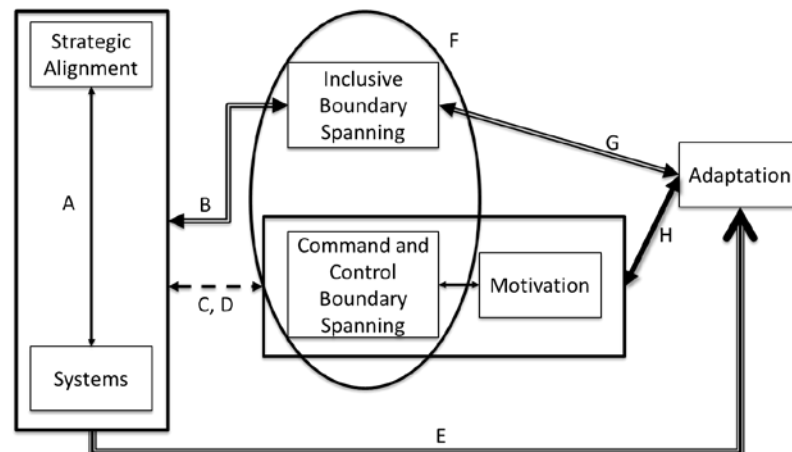
The team literature points to the effect of goals on team performance and while it does not specify alignment with organizational priorities, goal oriented studies are informative and relevant. Previous studies demonstrate that the orientation (individual vs. collective) and difficulty of goals promotes team effectiveness and adaptation within teams (Knight et al., 2001; LePine, 2005; Porter, 2008), which is similar to our finding of strategic alignment influencing adaptation in the organization. Ours may be among the first to specify alignment with organizational priorities as an input for teams to effectively promote adaptation at a different level of the organization.

Where adaptation can be construed as performance, the literature on boundary spanning affirms our finding of a relationship between these factors. Team boundary spanning that actively engages others beyond the team – vertically and/or horizontally – results in better team performance (Ancona, 1990; Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a). Similar to our findings, one review specifically indicated that boundary management is a positive contributor to the success of organizational change initiatives (Marrone, 2010). Boundary spanning studies do not, however, combine systems perspectives and strategic alignment, like our study, as factors that work in conjunction with boundary management to influence change beyond the team.

Our study offers a new lens on these factors and their relationships by demonstrating the combinations that result in adaptation. We offer Figure 5 as a representation of the varying combinations and paths that breed adaptation. For instance, we found the combination of strategic alignment and systems are *necessary* for

adaptation to occur (A, E) but are *not sufficient*. Instead, these factors need to be supported in combination by various behaviors of boundary spanning, with boundary spanning serving as sufficient factors (circle F) while individual characterizations of boundary spanning were not deemed as necessary (A, B, G and A, C, D, H) – that is boundary spanning needs to work in conjunction with strategic alignment and systems perspectives. Motivation plays surprisingly a small role, but is neither necessary nor sufficient indicating there may be additional opportunities to better understand boundary spanning’s influences on behavior and attitude. We will discuss additional insights of particular importance in the following results.

Figure 5: Factor Combinations Influencing Adaptation



Teams Facilitate Adaptation at Micro-, Meso- and Macro-system Levels

Our findings paint the complex nature of teams facilitating and embedding change within organizations. Multiple combinations of factors can result in the desired outcome as demonstrated in the QCA. By employing a grounded theory approach to identify core concepts we recognized that the outcome of interest was about changing something at varying levels of the organization in order to facilitate an improvement. The stories

shared revealed a core purpose for the involved groups reflected team and organizational level adaptation through pursuit of a production of change (Wiedow & Konradt, 2011) to find resolutions for diverse conditions (Klein & Pierce, 2001) in order to respond to environmental cues and changes (Burke et al., 2006; Kraatz, 1998). Examples of those adaptations, their purposes and their varying levels included such instances as creating an organizational culture of improvement through the development and use of scorecards; developing a new nursing unit to manage a particular population of cardiology patients; and improving processes and practices across supporting departments to obtain a ‘center of excellence’ designation to respond to a competitive environment.

While similarity exists in adaptation concepts to the team literature, alas, it has been more focused on adaptation within the team (Burke et al., 2006; Klein & Pierce, 2001; Kozlowski et al., 2009) rather than assessing how the team supported or facilitated other levels of the organization to adapt in such a way as to implement improvements, adopt new processes, or put a strategic priority into action. Adaptation within teams simply does not fully explain how teams create change – adaptation – among people outside the team and at varying levels of the organization. In contrast, our research found that *teams are central to facilitating adaptation between the micro- and meso-system levels* – that is at specific locations where care is provided (e.g. microsystem = hospital departments), and collections of interrelated departments (meso-systems such as found in Centers of Excellence) within the organization (Nelson et al., 2008). Examples from our study included micro-system level adaptations driven by teams in clinical specialty departments like obstetrics and otolaryngology or clinical units like the neonatology intensive care unit. A meso-system example was represented in a team’s pursuit of a

center of excellence designation that required coordination across cardiology, angiography, emergency medicine, and nursing units.

We found adaptation, as represented in our data, shared traits with the definition found in organizational adaptation literature which describes adaptation as change in strategy or structure (e.g. change in throughput processes and documentation) in response to or to fit environmental changes (e.g. pursuit of Chest Pain Center of Excellence to address regulatory requirements and local competition) (Chakravarthy, 1982; Kraatz, 1998). Because this literature does not, however, indicate who facilitates these changes, we have the opportunity to contribute to a layer of adaptation knowledge that is bridged by teams embedding a change, as a functional outcome, at micro- and meso- levels. Larger system levels share similarities with our findings in that there is a “coordinated and cooperative response to change” (Gulati, Lawrence, & Puranam, 2005), there are capabilities demonstrated of moving with and shaping cycles of change in response to the environment (Benner & Tushman, 2003), and organizations are viewed as “malleable” and can be altered “with evolving conditions” (Eisenhardt & Tabrizi, 1995). Teams in our study interacted with the environment identifying areas of opportunity for change – each adapting to the other. By using cycles of change, cooperative responses, and evolving conditions teams facilitated an adaptation at the micro- and meso- levels. The primary combination of factors that resulted in adaptations included strategic alignment, systems perspectives, and boundary spanning, our core themes, as demonstrated in Figure 5 (A,B,G and A,C,D,H).

Teams Utilize Interrelated Factors to Facilitate Adaptation

Initially, we set out to evaluate each of these constructs individually, and how each is represented in the literature as it relates to adaptation or aiding a desired change. What we found, however, is that generally a combination of factors work together interdependently to result in an outcome. For instance, strategic alignment influencing adaptation at the microsystem level in healthcare reflected Kosnik and Espinosa's (2003) micro-level changes acting as the agent of change and critical for carrying out a macro-system's strategic priorities. Stages and themes – such as understanding and developing interlinked processes, as well as developing reciprocal relationships like our study's systems and boundary spanning – further connect micro-level with a macro-level priorities (Kosnik & Espinosa, 2003). We additionally see strategic alignment described in the management literature at the firm level as an alignment of multiple factors such as strategy, structure, culture, and environment to influence business performance or as aligning the organization with its environment and arranging for supportive resources (Bergeron, Raymond, & Rivard, 2004). This concept of aligning strategy and systems at multiple levels to affect performance is in line with Kaplan and Norton's (2001) report of how to develop a strategy driven organization. Furthermore, process management specifically supports organizational efficiency and at least incremental innovation and change, but to bring about innovative adaptation, it must occur in combination with other innovation factors (Benner & Tushman, 2003). Such organizational examples support one of our key findings that *Systems, Strategic Alignment, and Boundary Spanning, in combination, are explanatory factors influencing Adaptation* as found in our Figure 5 model combinations of A,B,G and A,C,D,H. We differ from the extant literature,

however, in that central to our findings is the use of teams to facilitate adaptation rather than primarily pointing to strategy implementation or other types of changes and adaptations undergirding organizational performance without clarity of who is involved and how adaptations are embedded.

Our contribution to the literature is centered on *teams as the functional units* that identify and develop process and infrastructure systems, who recognize and relate their work to organizational priorities, and who connect people within and across departments and layers of the organization to *facilitate an adaptation at a micro- or meso- level*. The quality management literature calls on teams to implement improvements, but it lacks clarity in how they do so (Cooney & Sohal, 2004; Daily & Bishop, 2003; Hackman & Wageman, 1995; Sousa & Voss, 2002). Literature combining systems, strategic alignment, and boundary spanning are also unclear about how teams facilitate an outcome simultaneously at varying levels of the organization. Piercy (2009) demonstrated the importance of boundary spanning both within and across organizations to enhance strategic relationships among supplier organizations. In order to be successful, internal relationships need to be strong and are enhanced through “process management, internal partnering strategies, and internal marketing activities” (Piercy, 2009: 857). While teams are not specifically identified, Piercy’s concepts for success (2009) reflect our combination of systems, strategic alignment, and boundary spanning, which led to successful adaptation through teams. The inter-organizational literature further indicates the need for articulating a vision, coordinating activities, and motivating change as Volkoff, Chan, and Peter Newson (1999) found in the evolution of technological inter-organizational systems. Here, teams did a play role in this evolution through professional

and social networks; however, the point of reference was across organizations (Volkoff et al., 1999). These relevant findings lend credence to our study's strategic alignment, systems perspectives, boundary spanning, and our minimally represented factor of motivation (Figure 5 A,B,G and A,C,D,H) influencing adaptation. Furthermore, such findings allow us to contribute to the conversation by specifying teams as critical units that facilitate desired outcomes at micro and meso levels within organizations through the interaction and interdependence of systems identification and development, strategic alignment with organizational priorities, and boundary spanning.

Boundary Spanning Isn't Easy

Boundary spanning has a unique and consistent role in supporting systems and strategic alignment to generate adaptation in our results. Our analysis indicated the presence of different types of boundary spanning behaviors – such as inclusive or command / control as found in Circle F in Figure 5 – and all can play a role in combination with the other factors for facilitating improvements. A bulk of the team and organizational level literature highlights boundary spanning as critical to performance and, accordingly, uses descriptors that are collegial in nature. Collegial behaviors are conveyed in relationship development or giving and seeking information (Ancona, 1990; Druskat & Wheeler, 2003; Joshi et al., 2009); developing knowledge based trust and demonstrating distributive justice to facilitate cooperation and goodwill (Ireland & Webb, 2007); as well as ambassadorial boundary spanning in which information is shared with and support is sought from senior leadership (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a). These all are represented in our inclusive style of boundary spanning in which our informants described boundary spanning engagement behaviors in terms of

cooperation, collaboration, inclusion, mutual interest, and deference to expertise. We additionally found a boundary spanning style we called command and control type that was highly directive. We found that command and control boundary spanning only works with systems and strategic alignment and must include motivation (obtaining agreement to participate in a change initiative) in order for adaptation to occur as found in the A,C,D,H combination in Figure 5. The team literature is silent about similar boundary spanning behaviors. In fact, we were only able to locate one relevant organizational study in which individual boundary spanners acted as change agents to remove or reconfigure intra-organizational boundaries (Balogun, Gleadle, Hailey, & Willmott, 2005). What we do not know from our study is what triggers indicate a need to shift from an inclusive style of boundary spanning to a directive one, nor do we know whether ‘agreement to participate’ (motivation) is voluntarily offered or begrudgingly accepted.

Limitations

Our informants were organizational leaders who have the power to influence change and who also led and participated in groups with an intention of improving or changing something at varying levels of the organization. Using such informants may limit our ability to generalize our findings to teams led by or comprised of mid-level management and front-line staff because of power or influence differences. When temporary project teams, with a similar charge, are led by or are comprised of non-senior leaders, it may influence:

- The capabilities within boundary spanning behaviors since non-senior leaders may not be in a position to utilize the directive, command and control styles that emerged in this study with senior leaders;

- The importance of systems and strategic alignment since these may be identified for or imposed upon a mid-level management or front-line staff team rather than initiated by a senior leader who leads or is part of the project team as present in our study.

Additionally, hospitals are hierarchical, in nature. These combinations may not be directly reflected in flatter project based organizations. While we believe logic would suggest that a combination of 1) clearly understanding systems of importance to a project, 2) agreement on improvement / project / or larger organizational priorities, and 3) active boundary spanning to engage others would be applicable in new product development teams, such teams were not represented in our sample so we cannot claim these findings to directly correlate.

Implications for Future Research and for Management

Research Implications

This study revealed a set of interdependent factors influence adaptation indicating both research and practice need to allow for the complexity of interrelated factors in order to understand how to generate adaptation. The organizational level literature reflects similar combinations of factors as our study in regard to producing a change (Benner & Tushman, 2003; Kaplan & Norton, 2001; Volkoff et al., 1999); however the team literature provides little guidance for such combinations of factors. This may be in part because the team literature tends to center on team performance or adaptation within the team (Burke et al., 2006; Edmondson & Nembhard, 2009; Wiedow & Konradt, 2011) rather than changes the team produces in others leading to new performance in larger system levels – that is, micro- and meso-systems.

In further examining crossing levels, we do see teams spanning and managing their boundaries (Ancona & Caldwell, 1992a; Joshi et al., 2009; Marrone, 2010) and that they are embedded in the organization (Ilgen, 1999; Ilgen et al., 2005) as influential for team performance. There are two opportunities for future research in this realm that began to emerge in this study, however. One is relevant to embedding adaptation at a larger level outside the team. It would be valuable to know if the same boundary spanning practices that support team performance also enhance micro- or mesosystem level performance. Another related opportunity is to study when, why, and how boundary spanning becomes more directive rather than inclusive as was found in our study. Our finding indicates a directive style has its place, but it must occur in combination with additional factors, including obtaining a level of agreement from others to participate in the improvement initiative. Furthermore, inclusive boundary spanning does not stand alone in producing a desired outcome. It too must co-exist with factors of understanding systems as well as alignment with organizational priorities. In summary, future research should closely examine team boundary management in terms of who teams connect with, why those people are selected, and the ways and means by which they engage people in micro- and meso-system levels to embed changes designed by the team. It would additionally be valuable to explore measurement capabilities for the same or similar combinations found in this study and the extent to which they influence adaptation in the system.

Management Implications

There are four insights from this study which we propose offer value to management practitioners guiding teams facilitating adaptation in the organization. First,

it is important to both clarify alignment with a strategic priority and apply a systems lens to implement a change or adaptation. By that we mean, tie the change to an organizational priority or goal and understand the processes and infrastructures that are being affected and /or developed. Second, be clear about the level where adaptation is intended to take place. While the team needs to work together well, we also recommend teams distinguish the embedding context where the adaptation is going to occur in order to determine who needs to be engaged in the change activities, leading to our third point, which is to actively seek input and cooperation from people beyond the team who will be affected by or responsible for sustaining adaptations. Finally, while a directive style of boundary spanning may have its place, we are not suggesting this as an initial or primary approach. Our interview participants were senior leaders in the organization who were in a position to take a directive approach when needed. It may contribute to advancing adaptations, but its judicious use is strongly recommended as we primarily saw groundwork laid in a more inclusive style with a potential triggering event that prompted a decision to adopt a directive posture. This is an area of opportunity for further research prior to offering additional management recommendations.

Conclusions

Reflective of complex adaptive systems, we found combinations of systems-thinking, strategic alignment, boundary spanning, and motivation render explanatory influence on the emergence of adaptation. It is the interaction of conscious attention to the system and behaviors between groups and individuals that influence change. By this we mean understanding organizational priorities and system infrastructure / processes combined with behaviors of engaging people outside the team result in micro and meso-

system level changes. We propose that identifying and empirically evaluating such combinations contributes to quality management and boundary spanning studies, at the team level, as neither have incorporated these combinations in order to assess influences on improvement implementation at micro- and meso-system levels in organizations. Additionally, we learned boundary spanning behaviors may be inclusive or directive and that there appears to be intention behind the style of engagement. Both styles engender improvement implementation, that is adaptation, but they only work in combination with attention to systems and strategic alignment. Furthermore, in order for command and control boundary spanning to positively support systems and strategic alignment to promote change, the groups have to identify and instill motivation in others – they have to obtain agreement to participate – willingly or begrudgingly. Adaptation reflected the teams as complex adaptive systems definition of a patterning of change (Arrow et al., 2000a) but it was outwardly focused on how the group could generate change that included and/or affected others at larger levels of the organization. Team performance is reflected in the ability to embed changes and elevate performance at other levels; therefore, teams must be well-versed in the processes and infrastructure that will be affected by their planned adaptations, aligned with organizational priorities, and competent in connecting with others who will be affected by the team's efforts.

CHAPTER 4: EMBEDDING IN ORGANIZATIONS: INDIVIDUALS' PERSPECTIVE OF THE EFFECTS OF INTERNAL VERSUS EXTERNAL FOCUS ON TEAM ADAPTABILITY AND POTENCY

Introduction

Teams regularly face the challenge of balancing internally focused team interactions and externally focused embeddedness within the larger organization. They do not function in a vacuum (Joshi, 2006). Team participants intersect with the organization through interdependent tasks, interconnected relationships, and shared or reciprocal effects on the organization or vice versa (Manning, 2008). Teams are “open and complex systems” (McGrath et al., 2000: 98) existing in “adaptive, dynamic and structurally complex” environments within multiple levels of organizations – a complexity which “cannot be ignored” (Ilgen, 1999: 137). These levels include individual members, other teams or departments or divisions, outside resources, and the organization at large (McGrath et al., 2000). Ancona (1986) proposes that group processes include the way in which members interact with each other as well as with those external to the boundaries of the team. Organizations, then, are the social systems within which teams⁴ and their members (Ilgen, 1999) embed themselves through a continuous process of engagement with each other and with external stakeholders for various task and team feature reproduction (Manning, 2008).

The internal function and behavior of teams is well studied from a variety of perspectives such as (but not limited to) team learning and conversation (Druskat &

⁴ Throughout this study, we will use the words team and group interchangeably as both “refer to a bounded system composed of a set of interdependent individuals organized to perform specific tasks that affect others” Choi, J. N. 2002. External Activities and Team Effectiveness Review and Theoretical Development. *Small Group Research*, 33(2): 181-208, Choi, J. N. 2002. External Activities and Team Effectiveness Review and Theoretical Development. *Small Group Research*, 33(2): 181-208.

Kayes, 2000; Edmondson, 1999; Edmondson, 2003; Edmondson & Nembhard, 2009; Lingham, 2004), psychological safety (Edmondson, 1999) / healthy teams (Druskat & Wolff, 2001), cultural diversity (Lingham, Richley, & Serlivos, 2009); team interaction (Lingham et al., 2009), and input-process-output (Hackman, 1987; Ilgen et al., 2005; Oetzel, 2001) within an organizational system. Using Oetzel's (2001: 21) definition of a system as a "set of component parts that have interdependent relationships," group membership is based on individual perceptions in regard to how the individual perceives him or herself and others in the group at hand (Lichtenstein, Alexander, Jinnett, & Ullman, 1997). Teams in which individuals participate are systems with inputs-processes-outputs with a reciprocal influential relationship among these components (Ilgen et al., 2005; Oetzel, 2001). Employees belong to a variety of organizational and identity groups embedded in the larger social structure, representing these in intergroup exchanges to varying degrees based on the focal context (Alderfer & Smith, 1982; Lichtenstein et al., 1997). Additionally, employees identify their networks with the organization and "being embedded in a network of reciprocated, multiplex exchange relationships at work" can influence perception of the organization (Hayton, Carnabuci, & Eisenberger, 2011: 237). Embeddedness includes intergroup effects across interpersonal, group, and social system levels and concerns "how system and subsystem dynamics are affected by supra-system events and vice versa" (Alderfer & Smith, 1982: 61). Closely aligned with our study of interest is the concept of teams as embedded in multiple levels of systems.

Empirical studies of the complex nature of team interactions in combination of internal and external forces is challenging and has little presence in the literature. Most studies have been either qualitative or network analysis based (Joshi et al., 2009).

Measuring a non-linear, complex system is difficult and may only provide a snapshot of interactions and interdependence, which may have contributed to the lack of empirical studies. Additionally, measuring adaptability is still lacking even in the face of acknowledging its importance (Burke et al., 2006). “The gap between the need to think and act interdependently and our ability to do so sits at the heart of all the most difficult problems we face today” (Senge, Smith, Kruschwitz, Laur, & Schley, 2008: 17).

The internal and external focuses of individuals and teams in organizations has been conceptualized in a number of articles (DeFillippi & Arthur, 1998; Manning, 2008; Salas et al., 2008; Salas, Sims, & Burke, 2005), however, empirical studies are lacking. Individuals interacting with groups interacting with organizations need to think and act interdependently. Having a means of evaluating such interactions and interdependencies may give directional guidance to individuals and teams on how to improve or support interactions within systems.

The major aim of this study is to attempt to measure the factors affecting teams embedded in organizations – informed by individuals participating in teams. Specifically, we seek to evaluate the effects of the internal and external focus of teams – that is, ‘within-team’ interaction as well as connection with organizational priorities and with others in the organization – on adaptability, the ability to adjust tasks and goals, and potency, which is confidence in the team’s ability to achieve its goals. To that end, our research question asks what factors of group interactions, interdependencies, and interconnections influence adaptability and potency?

This paper is organized in the following manner. First, we will review theories and supporting literature for systems thinking and team boundary spanning, which played

a key role in informing this study. Then, we will outline the research model and associated hypotheses associated with the constructs in our survey. This will be followed by the analyses conducted through factor analysis and structural equation modeling to report the findings and support for our hypotheses. Finally, we will provide our insights gleaned from the findings as well as discuss any limitations and future research to further this work.

Theory / Literature Review

In thinking about individuals embedded in teams that are embedded in organizations we looked to two theoretical areas that have most closely informed ideas of embedding groups in organizations - systems thinking (Arrow et al., 2000f; Senge, 2006; Senge et al., 2008) and team boundary spanning (Ancona, 1990; Ancona & Caldwell, 1992a; Joshi et al., 2009). Systems thinking and team boundary spanning have an interest in the internal and external workings of groups and serve as a bridge for understanding these complex adaptive systems. Systems thinking assists in illuminating interactions, interconnections and interdependencies that are important for managers and team members alike to mindfully approach those factors, which support embedding groups into the larger organization. Systems thinking is about assimilating and organizing detail complexity, which results in focusing attention on the variables of importance among the vast information available in organizations (Senge, 2006). Team boundary spanning considers the potential inputs and value to a team's efforts to manage interconnections that exist outside the team either within or external to the organization (Marrone, 2010).

Systems Thinking

The world, in general, is shaped by networks of interactions and interconnections (Senge et al., 2008), so it is reasonable to consider the same across the varying levels of individuals, groups, and organizations. One of Senge et al.'s (2008) core competencies of systems thinking – the crucial nature of crossing boundaries – parallels a component of our study and suggests one approach to supporting the multiplex environment in which individual team members participate and observe their teams functioning within organizations as complex adaptive systems (Arrow et al., 2000f). Plsek and Greenhalgh (2001) define complex systems as a “collection of individual agents with the freedom to act in ways that are not always totally predictable, and whose actions are interconnected so that one agent’s actions changes the context for other agents” (p. 625). Rather than leading to chaos, though, consideration of cooperation reminds us that the potential for people to meet again – and it is likely they will in an organization - encourages interactions that will likely benefit the actors in the moment as well as in future connections (Axelrod, 2006).

As businesses have increased in complexity, a systems perspective has become more critical (Senge & Massarik, 1987). This complexity within systems has also contributed to the expanded use of diverse, cross-functional teams (Edmondson & Nembhard, 2009). Stress increases as complexity increases (Edmondson & Nembhard, 2009), which is an important consideration for how adaptable a group can be and the degree of confidence they can experience. Edmondson and Nembhard (2009) note the importance Weick, in 1993, placed on frequent communication and interaction to reduce the tendency to unravel in stressful situations. This has the potential to be applicable both

within and outside of teams. Identifying the factors that support adaptability and strength of confidence – whether it is a focus internal or external to the team – may shed light on interconnections and interdependencies of importance.

Systems thinking further informs our work as it gives primacy to the whole, to relationships, and to seeing interrelationships rather than to individuals or things (Maani, 2002; Senge, 2006). “Businesses and human endeavors are systems” (Senge, 2006: 6) and are interconnected through people whether at the individual, group, or organizational level. Impacts on systems require an understanding of the forces at play in a system (Senge, 2006). It is important for teams to have a supportive interconnected embedding context in order for members to accomplish their work through tasks and tools indicating that both initial conditions and interrelations are important factors (Arrow et al., 2000f). The focus of our study is to understand the interaction and interdependent forces at play, including the initial condition of team interaction, and the relational contexts of alignment with the organization as well as the connections members have with others as these relate to adaptability and potency. System thinking provides a context of understanding for both consideration of system forces or needs (such as strategic alignment) and the internal and external interactions necessary to adapt and promote confidence in success.

Team Boundary Spanning

A group is expected to have “an existence and purpose apart from serving as a setting and apart from the individuals who compose it” (Ancona, 1990: 336). In dynamic and complex organizations, a team’s success can be predicted by the team members’ ability to coordinate efforts and resources externally (Joshi et al., 2009). Team boundary

spanning surrounds activities that engage members in managing interactions within the embedding environment, which can include people or other teams within or external to an organization, in order to complete their work (Marrone, 2010). Defining boundaries and boundary activities serve as a mechanism for more closely integrating a group with the organization and its environment (Choi, 2002). Boundary spanning can include such behaviors as representing the group to others, coordinating tasks, and information gathering (Ancona, 1990; Ancona & Caldwell, 1992a; Marrone, 2010) – all of which have been identified as key drivers in performance and organizational innovation (Marrone, 2010). Team members also play roles in coordinating efforts and knowledge across organizational functions, simultaneously facing challenges in maintaining effective internal processes which points to the importance of understanding the mechanisms of capably fulfilling internal dynamics and external demands (Marrone, 2010).

“Team boundary spanning is defined as interactions that are aimed at establishing relationships and interactions with external actors that enable the team to meet its overall goals” (Joshi et al., 2009: 732). Social interaction among organizational members includes structural, relational and cognitive dimensions (Chua, 2002). Boundary spanners interact with various role senders to obtain information, coordinate task performance, as well as obtain direction and assistance (Marrone, 2010; Singh, 1998). Boundary spanning, itself, is a meso-level activity crossing both the micro and macro levels of an organization (Joshi et al., 2009). At the micro level, a bottom-up antecedent factor, members’ behaviors and responses to the environment can shape interactions and boundary spanning efforts (Joshi et al., 2009). At the macro level, a top-down contextual

factor, “organizational structure, culture, and processes” can be influential in boundary spanning activities (Joshi et al., 2009: 734).

While considering boundary spanning, we are not ignoring internal team activities; rather we are in agreement with Ancona’s (1990) assertion that teams that manage their external interconnections and interdependencies “should perform better than those that are only able to manage their internal dynamics” (Ancona, 1990: 336). This is further supported by Choi (2002) who noted that teams do not function in a vacuum nor can they accomplish their work without maintaining a boundary. While there is a commonality between internal and external activities in the purpose of meeting collective performance goals, external processes are distinct in that they reach beyond the boundaries of the team into the embedding environment (Marrone, 2010). In boundary spanning, group members interact internally and externally and over time develop cycles of actions and processes for both that enable long term performance success (Ancona & Caldwell, 1992a). Such boundary activities have been found to influence how group members view the embedding environment and how external stakeholders perceive the group (Ancona & Caldwell, 1992a).

It is also useful to be aware of the inputs and influences on team boundary spanning. Social network research indicates individual attributes, demographics, and personality can influence relationships and networks that are relevant to boundary spanning (Joshi et al., 2009). There can also be a temporal influence on interdependence such that environmental characteristics may fluctuate as may project cycles leading to the need to increase external activities (Choi, 2002). Task complexity or uncertainty as well as dependence on other teams related to these complexities and uncertainties can also

influence the need to engage in external activities (Choi, 2002; Joshi et al., 2009). Furthermore, of interest to this study are influences such as leader behavior and member attributes. Team leaders' "attributes, perceptions and behaviors" can critically influence the extent to which teams interact with the embedding environment (Joshi et al., 2009: 740). Member attributes such as functional background, tenure, and age when aggregated at team level can also influence boundary spanning. While not directly included in this study, it is important to note that studies have also found organizational structures such as hierarchies and lateral relationships as well as informal processes and environmental climate have an effect on boundary spanning (Joshi et al., 2009).

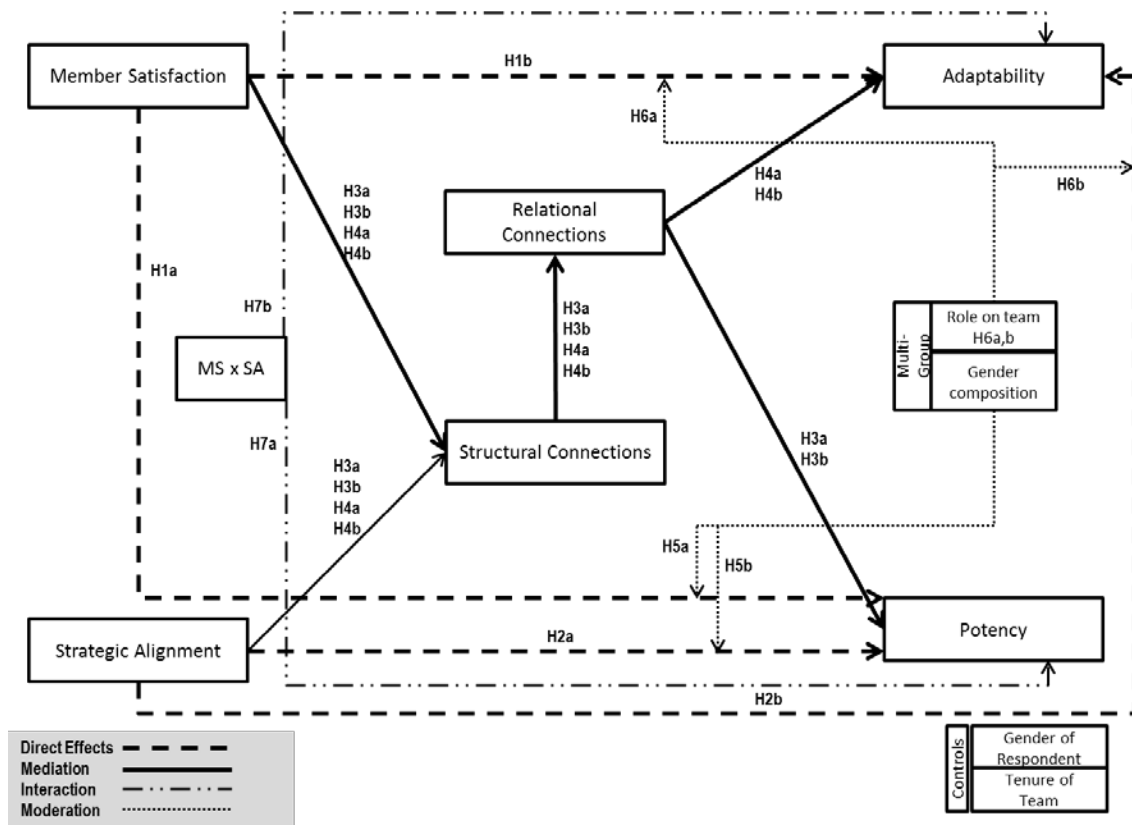
It is worth additionally emphasizing the value that boundary spanning offers. As knowledge work becomes more complex, tasks are no longer easily performed linearly and instead require interdependent and coordinated action (Marrone, 2010). Teams can serve as bridges across disconnected parties acting as conduits for information transfer and innovation (Marrone, 2010). Returning to the importance of maintaining a balance between internal and external activities for synergy and effectiveness, internal activities may support boundary spanning activities and external activities may infuse pride and collective efficacy (Choi, 2002). Outsiders, too, tend to rate teams as high performers when they depend on outside resources and promote their team's activities to people external to the team (Ancona, 1990). Finally, study results support improved knowledge creation and success as a result of social interaction outside the team (Ancona, 1990; Chua, 2002). For reasons such as social interaction outside the team, maintaining a balance between internal and external activities, and response to uncertainties –

adaptability – we looked to the boundary spanning literature for support for our hypotheses.

Research Model and Hypothesis Development

Given the lack of empirical studies measuring the combined internal and external forces of teams (Joshi et al., 2009) as well as measures of adaptability as a consequence (Burke et al., 2006), we sought to contribute to the teams as complex adaptive systems literature through a causal analysis. Both Systems Thinking and Team Boundary Spanning contain elements of understanding and connecting to the context in which a group is situated as well as knowing who and relating to others in the organization that can serve as partners to achieving an aim. We propose that these concepts are important inputs to being able to adapt tasks, goals, and management of priorities to successfully perform in groups. Our research model includes Member Satisfaction as a measure of internal focus of teams, while Strategic Alignment provides insight to an external focus of teams. Structural and Relational Connections are concerned with to whom and how team members connect to others outside the team as part of achieving team aims. Adaptability is treated both as an outcome of internal and external interactions. Potency provides a measure of performance as it has been recognized as a predictor variable for performance (Collins & Parker, 2010; Gully et al., 2002). We established our model and hypotheses assuming full mediation as well as the potential for interaction between the independent variables.

Figure 6: Conceptual Model



Direct Effects

The ever present reality of change and the complex interdependence (Burke et al., 2006) of individuals, teams, and organizations interacting together to enact tasks and accomplish goals highlights the importance of the experience of adaptability among members of teams. Complexity science suggests trying multiple approaches and allowing time and attention to reveal what works (Plsek & Greenhalgh, 2001). Teams that are adaptable are often capable of reducing barriers (Burke et al., 2006), indicating an attentiveness to what works best, both of which can enhance performance.

Well integrated teams are those whose members feel comfortable with the team, participate in team interactions, and feel positive about “the team’s overall goals and

functioning” (Lichtenstein et al., 1997: 415). Additionally, members interact with multiple levels; of interest here are levels of teams and organizations (Ilgen, 1999). This multi-level interaction requires a level of awareness of shared commitments across levels to ideas, issues, goals, and management processes (Senge, 2006) . Hence, one could argue that an end product or output of a team is based on its capacity to increase adaptability and potency. A team strong in the belief of its capability to succeed tends to be more effective – team potency – which is predictive of performance (Collins & Parker, 2010; Gully et al., 2002). We further contend that Adaptability can and should be evaluated as a Dependent Variable. Burke et al. (2006: 1192) conceptualized adaptability as an important dependent variable owing to its nature as a multilevel phenomenon in which team members or teams “use their resources to functionally change current cognitive or behavioral goal-directed actions or structures to meet expected or unexpected demands.”

Accordingly, we first offer the following hypotheses that effective team interactions, measured through Member Satisfaction and Strategic Alignment with the organization, each directly and positively influence Potency and Adaptability.

Hypothesis 1a. Member Satisfaction has a direct, positive effect on Potency

Hypothesis 1b. Member Satisfaction has a direct, positive effect on Adaptability

Hypothesis 2a. Strategic Alignment has a direct, positive effect on Potency

Hypothesis 2b. Strategic Alignment has a direct, positive effect on Adaptability

Mediation

Embedding in an organization is conceptualized as a continuous process in which members of teams experience their teams interacting with various components of the

organization, engage in tasks, and relate to multiple systemic environments (Manning, 2008) including people outside the team. Social systems are developed and social contexts for the team are brought about by social practices (Manning, 2008) both within teams and external to teams. Employees and members of teams have social exchanges and are embedded in networks (Hayton et al., 2011). These exchanges are multiplex and involve both instrumental and socio-emotional resources (Hayton et al., 2011).

“‘Who you know’ affects ‘what you know’” (Chua, 2002: 376). People can become siloed in their attention to within-team interactions, but people who bridge networks have more diverse information, more opportunities, and access to broader vision (Burt, 2004). Acquaintances across the organization may increase exposure to multiple perspectives. Those to whom a member is close, are more likely to be motivated to provide assistance and are more easily available (Granovetter, 1983). Effective coordination, driven by who knows what type of information along with the development of relationship-specific learning and understanding, enhances how well people interact with each other (Reagans, Argote, & Brooks, 2005). Relational connections are also characterized by the level of care or trust generated between organizational relationships and by norms of cooperation (Chua, 2002). These are important for giving rise to mutual trust, access to help, exchange of information, openness, teamwork, and even a tolerance to failure (Chua, 2002). Hayton et al.’s (2011) notion that “favorable instrumental and expressive exchanges tend to promote each other” (p. 236) supports our conception of the interconnected influences that promote Adaptability and Potency. In particular, we suggest that Member Satisfaction (an internal focus) and Strategic Alignment (an external focus) will inform who to know and influence how they relate to those outside the team

resulting in improved adaptability and increased potency. We propose, though, that Structural Connections influencing Relational Connections will carry the effect of Member Satisfaction and Strategic Alignment, each, on Adaptability and Potency. To that point, we offer the following hypotheses.

In regard to the effects of Member Satisfaction and Strategic Alignment on Potency, we offer the following hypotheses:

Hypothesis 3a. Member Satisfaction → Structural Connections → Relational Connections → Potency. Member Satisfaction positively influences the level of Structural Connections and Relational Connections, which positively influence Potency. We contend that Member Satisfaction will not directly influence Potency in the presence of Structural and Relational Connections.

Hypothesis 3b. Strategic Alignment → Structural Connections → Relational Connections → Potency. Strategic Alignment positively influences the level of Structural Connections and Relational Connections, which positively influence Potency. We contend that Strategic Alignment will not directly influence Potency in the presence of Structural and Relational Connections.

Hypothesis 4a. Member Satisfaction → Structural Connections → Relational Connections → Adaptability. Member Satisfaction positively influences the level of Structural Connections and Relational Connections, which positively influence Adaptability. We contend that Member Satisfaction will not directly influence Adaptability in the presence of Structural and Relational Connections.

Hypothesis 4b. Strategic Alignment → Structural Connections → Relational Connections → Adaptability. Strategic Alignment positively influences the level of Structural Connections and Relational Connections, which positively influence Adaptability. We contend that Strategic Alignment will not directly influence Adaptability in the presence of Structural and Relational Connections.

Moderation

Successfully embedding in organizations is influenced by interactions among team members and with others outside the team. Diversity within a team may affect performance. Studies have suggested that exposure to diversity on a team can capitalize on skills, expertise, and stimulate divergent and flexible thinking (Pearsall, Ellis, &

Evans, 2008). Other studies have shown increased demographic diversity to increase friction, decrease creativity, and negatively influence the exchange of creative ideas (Pearsall et al., 2008). “Gender composition at the team level and dissimilarity from the majority of the team with respect to gender at the individual level both influence important individual and organizational outcomes” (Baugh & Graen, 1997: 368). Types of roles women fill also influences outcomes; however, it is less clear how gender diversity affects men’s responses or processes within teams (Baugh & Graen, 1997).

Studies in regard to gender diversity and/or gender influence offer various views and influences on teams related to gender. Mixed results have been shown in the level of contribution between male teams and female teams (Ivanova-Stenzel & Kübler, 2011). In a prisoner’s dilemma experiment, women cooperated more in the beginning, but in regard to compensation, competition led to higher performance among men than women (Ivanova-Stenzel & Kübler, 2011). Dufwenberg and Muren (2006) found groups are more generous and treated people more equally when women are in the majority, but the most generous teams were those with two men and one woman. “Women, on average, are more likely to focus on group processes and interpersonal relationships and men, on average, are more likely to focus on achieving specific outcomes” (Lichtenstein et al., 1997: 420). In another study, male majority teams rated their team effectiveness lower than homogenous teams, but external evaluators did not share that result (Baugh & Graen, 1997).

We propose that gender majority may influence adaptability and/or potency. Most of the studies reviewed had an interest in the existence of gender diversity, but not necessarily a gender majority. Also, most of the studies were conducted in an

experimental setting rather than asking members of existing work teams about their experiences. In general, there was little evidence to drive a specific choice in which gender majority would likely moderate a particular relationship but Lichtenstein et al (1997) at least provided reasonable guidance. With the concept of women being more focused on interpersonal relations, our closest parallel for a variable relationship of interest is in Member Satisfaction's relationship with Potency. Likewise, as men are more focused on achieving outcomes, we posit a male majority team is likely to moderate the relationship between Strategic Alignment and Potency. Hence, we offer the following hypotheses:

Hypothesis 5a. The positive relationship between Member Satisfaction and Potency will be stronger for female majority team composition than for male.

Hypothesis 5b. The positive relationship between Strategic Alignment and Potency will be stronger for male majority team composition than for female.

Team leadership and team membership can provide different perspectives and offer different contributions to teams. The behavior of group representatives, whether leader or member, "is both cause and effect of the total pattern of intergroup relations" (Alderfer & Smith, 1982: 39). We proposed the perspectives of team leaders vs. team members may have a moderating influence on the relationships between Strategic Alignment and Adaptability as well as Member Satisfaction and Adaptability.

Hoegl and Gemuenden (2001: 436) in studying teamwork quality and the success of innovative projects started with the foundational concept that the "success of work conducted in teams depends (beyond the quantity and correctness of the task activities) on how well team members collaborate or interact". Their findings included considerable agreement between team members' and team leaders' general perspectives on team

performance (Hoegl & Gemuenden, 2001). There was a significant difference, though, between team members and team leaders in terms of rating teamwork quality – team members attributed good performance to good team processes; whereas the relationship between team performance and teamwork quality was considerably lower for team leaders (Hoegl & Gemuenden, 2001).

Other studies have also recounted the varying influences of team leaders and members in regard to network influences, team design, and coaching. Both leaders and members can influence the task performance of a team through their position in organizational networks (Balkundi & Harrison, 2006). The informal power of leaders, which is partly provided by networks, can be used to provide teams with valuable information and to guide them toward common goals (Balkundi & Harrison, 2006). It can also be just as important for members to have a central location in networks because if they lack task information or fail to pass along critical information, it can have a negative impact on the team (Balkundi & Harrison, 2006). Finally, leaders, specifically, can positively influence group processes through team coaching as well as both performance and processes through design factors such as providing clear direction; outlining task interdependence, goals, and strategy norms (Balkundi & Harrison, 2006).

In light of the varying influences and perspectives that have been observed among team leaders and team members, we offer the following hypotheses.

Hypothesis 6a. The positive relationship between Member Satisfaction and Adaptability will be stronger for Members than for Leaders.

Hypothesis 6b. The positive relationship between Strategic Alignment and Adaptability will be stronger for Leaders than for Members.

Interaction

As in Hoegl and Gemuenden's (2001) study, we assert that how well team members interact is an important antecedent for team outcomes – in our study, the outcomes are Potency and Adaptability. Members must be mindful of the emotions of members of the internal group as well as those of individuals outside the group who influence their work (Druskat & Wolff, 2001). Teams that are well integrated have members who feel comfortable sharing points of view with other members and feel positive about the team's overall goals and how it functions (Lichtenstein et al., 1997). Member Satisfaction, itself, encompasses such a multidimensional nature of interaction (Oetzel, 2001) and is proposed as a significant factor directly influencing a team member's perception of the ability to adapt as well as confidence in a capability to succeed.

Strategic Alignment suggests that goal interdependence guides efforts among members (Gully et al., 2002). In our study, we are interested in the team's awareness of how their goals align with the organization's goals and strategic priorities. Sydow et al. (2004) noted the potential for tension between a desire for autonomous project management among team members and their embeddedness in the organization which may call for integration of their work into organizational routines and coordination efforts. The ability of a team to adapt to the environment within which it operates, which for us includes the organization's goals and priorities, is an important factor in teamwork (Salas et al., 2005).

We contend the interaction of effective member interaction as expressed through Member Satisfaction and alignment with the organization's goals and priorities will

influence adaptability and confidence in achievement capabilities. We, therefore, propose the following hypotheses:

Hypothesis 7a. Strategic Alignment (external focus) amplifies a positive relationship between Member Satisfaction (internal focus) and Potency.

Hypothesis 7b. Strategic Alignment (external focus) inversely moderates the relationship between Member Satisfaction (internal focus) and Adaptability.

Controls

We elected to control for gender of respondent and tenure of the team. Gender was selected as a basic element of social categorization and identity (Lichtenstein et al., 1997) which may influence perceptions of interactions, adaptability, and potency. Teamwork is often temporary, but group longevity has been shown to moderate the effect of the relationship between diversity and task conflict as well as between diversity and performance, which may influence adaptability or potency (Edmondson & Nembhard, 2009). Dynamics in teams change over time (Choi, 2002) and time together as a team may help team members develop a shared understanding of task requirements and develop interpersonal familiarity (Balkundi & Harrison, 2006).

Research Design and Statistical Analysis

We used a quantitative study conducted via a survey consisting of five-point Likert-type scales to test and validate our hypotheses represented in the hypothesized model (Figure 6). A survey was designed to measure the factors influencing adaptability and potency of work teams as informed by individual participants. SPSS and AMOS (version 20.0) were used for the statistical analyses. SPSS was used for basic statistics, conducting the Exploratory Factor Analysis (EFA), and determining reliability of constructs using Cronbach's alpha. AMOS was used for covariance based Structural

Equation Modeling (SEM). SEM was used to simultaneously integrate the measurement and structural models into a unified model that provides a better estimation than linear regression of estimates for measurement and structural relationships in Covariance-Based SEM (Gefen, Straub, & Rigdon, 2011).

Construct Operationalization

With the intent to consider both the internal interactions experienced by individuals in teams and their perceptions of the teams' interactions and interconnectedness to the larger organization, we chose Likert-type scales – validated and new – to represent these facets. The scales and associated items are available in Table B1 in Appendix B. Two of the scales used were validated and reported in the literature – Member Satisfaction and Team Potency (Collins & Parker, 2010; Oetzel, 2001). Four were newly designed. The newly designed scales were based on the authors' earlier qualitative study indicating team adaptation resulted from boundary spanning, strategic alignment, and systems identification (Grooms & Lyytinen, 2014). These scales were further refined based on literature related to teams embedding in organizations, interdependencies among systems, and systems thinking.

Adapted Scales

Member Satisfaction. Oetzel's (2001) Member Satisfaction scale was selected to represent effective team interaction as it was developed to encompass the multidimensional nature of satisfaction in regard to "satisfaction with the group, the group's decision, and the group's interaction" (Oetzel, 2001: 35). Additionally, in Oetzel's (2001) study, the reliability using Cronbach's Alpha for satisfaction in three separate task studies (0.90, 0.91, 0.92) was high. The scale items were modified only to

represent the present tense to use the scale as an input or antecedent versus the past tense used as an outcome or consequence. This construct was intended to serve as an Independent Variable.

Potency. The Potency construct was selected due to its usefulness in predicting performance (Collins & Parker, 2010; Gully et al., 2002). Collins and Parker (2010), reference Guzzo et al. 1993 and Lester et al. 2002 as demonstrating this potency scale as reliable, one-dimensional, and having good inter-rater reliability. In a meta-analysis, Gully et al. (2002) demonstrated a 0.37 effect size on performance supporting previous theories that potency is predictive of performance. Because we pursued an anonymous study of individuals participating on teams at work, it would not have been feasible to obtain objective team performance data. In lieu of that, we used team potency as a surrogate for its predictive nature.

New Scales

The Strategic Alignment, Relational Connections, Structural Connections, and Adaptability constructs were conceptualized from key findings in the authors' (Grooms & Lyytinen, 2014) qualitative study regarding conditions of adaptation of Quality Improvement teams in the organization. Items were initially drafted from statements captured in the interviews conducted in the qualitative study, refined, reviewed against literature for concept parallels, and tested. The items retained for use in the survey demonstrated participant understanding in modified paper and electronic Q-sorting activities (Block, 1961) and adequate grouping in an Exploratory Factor Analysis of the tested questions. Details of item development are described with instrument development.

Items, descriptions, supporting literature for item refinement, and final Cronbach's Alpha for each newly developed scale is available in Table B1 in Appendix B.

Instrument Development

For data collection, we used a survey that included 39 Likert-type items and ten demographic questions. The survey consisted of two validated scales – one for Member Satisfaction and one for Team Potency. These two scales accounted for sixteen items. The remaining 23 items made up four new scales to assess Structural Connections (four items), Relational Connections (five items), Strategic Alignment (eight items), and Adaptability (six items). We chose to mix validated and new scales to evaluate the relationship between existing measures and newly developed measures (Hinkin, 1998).

The validated scale of member satisfaction was selected as a measure of internal team interaction. As noted as earlier, given its strength of reliability and its ability to measure affective response to the environment of group interactions (Oetzel, 2001), we chose this as our internal team interaction scale. Member Satisfaction is a five-point Likert-type scale using Strongly Disagree to Strongly Agree. The other validated scale was Team Potency and was selected to serve as a surrogate for performance due to its predictive capabilities (Collins & Parker, 2010; Gully et al., 2002). Team Potency is a five-point Likert-type scale ranging from 'To No Extent' to 'To a Great Extent'.

In choosing to quantitatively assess measurements of factors supporting teams embedding in systems resulting from our qualitative study, we referred to interview statements from key findings from the study results. These statements were grouped according to similarity and used to develop item statements. They were initially sent to a group of 53 individuals participating on known project teams for testing. 17 people

completed the sample survey, which was too small a sample to verify the results. We did, however, use the results of an Exploratory Factor Analysis (EFA) to guide our next revision of items as there were items that loaded on factors at >0.5 and were aligned with our intended constructs of systems thinking, embeddedness, and interdependence. Five-point Likert-type scales using Strongly Disagree to Strongly Agree were used. Five-point scales were selected as opposed to using more options as increasing the number of options beyond five is not shown to increase validity or reliability (Clark & Watson, 1995).

Due to a concern about sampling those who were the intended population for the survey, we chose to pursue a route of obtaining feedback for further revisions through verbal and written review as well as a modified electronic Q-Sort technique with doctoral cohorts to test item connection to constructs (Block, 1961). We additionally sought feedback about the complexity or simplicity of the language and the presence of double-barreled questions that could confuse the participant (Hinkin, 1998). These tests with doctoral cohort students resulted in two more revisions of items prior to obtaining agreement from the doctoral committee chair to use the final revisions for use in the survey. A graphical representation of scale development for these four new scales can be found in Figure B1 in Appendix B.

Sampling / Data Collection

Our sample was informed by our qualitative study (Grooms & Lyytinen, 2014), which indicated team adaptation results from identification of systems at play, strategic alignment with organizational priorities, and boundary spanning to interact with the larger organization. Accordingly, we surveyed individuals who self-identified as

participants in workplace teams. The survey, supported by an online survey software, Qualtrics (www.qualtrics.com), was distributed via 139 individual e-mails; three group e-mails for doctoral cohort classes (school generated group e-mail addresses are blinded) for an approximate number of 60 additional individuals directly contacted; plus two social media site postings. The invitation included a brief explanation of the study, IRB information, and the anonymous survey link. It also included an invitation for participants to share the anonymous survey link with their contacts who might be appropriate, accounting for snowball sampling (Singleton & Straits, 2010). The data collection period lasted seven weeks beginning at the end of September 2012 through mid-November 2012.

The survey was opened and initiated by 342 people. An initial data screening revealed 250 records appropriate for retention to conduct factor analysis tests as will be described in detail in the Findings. While we can report 73.1% of people opening the survey link completed the survey, we are not able to assess the percent of completions compared to total invitations due to the anonymous nature of the survey, the use of social media as an invitation platform, and the effect of snowball sampling (Singleton & Straits, 2010). Nonresponse bias is a common concern in surveys as it may affect the validity of inferences (Sivo, Saunders, Chang, & Jiang, 2006). It is commonly evaluated through differences in early and late responders and extrapolation of those who less readily respond as measured through responses to successive waves of questionnaires (Armstrong & Overton, 1977). Given that our direct messages, social media postings, reminder notes, and snowball sampling techniques occurred throughout the seven weeks of keeping the survey open coupled with the anonymous nature of our survey, we could

not evaluate differences in early and late responders or extrapolation through successive waves.

Of the 250 records retained to initiate analyses, the following demographics describe our survey participants. Details of demographic items can be found in Table B2 in Appendix B, but a summary of age, gender, and the highest represented groups by item follows:

Table 6: Demographics Summary

Age: Average = 46.3 yrs. Range: 22 – 78	Gender: Male = 62.6%, Female = 37.4%	Team Type: Project Team 32.5%
Organization Type: Healthcare = 57.4%	Role on Team: Team Member = 51%	Frequency of Team Meetings: Weekly = 34.5%
Tenure of Team: One to Six Months = 30.9%		Education Completed: Graduate = 51.8%

Data Analysis

The data were analyzed in two stages – factor analysis followed by structural equation modeling. First, factor analysis was conducted to determine items for retention; second, the structural model was built and analyzed. Factor Analysis was undertaken, first as an exploratory factor analysis (EFA), then confirmatory factor analysis (CFA). The EFA demonstrated acceptable standards of adequacy, validity, and reliability, which were important to achieve in the presence of two validated scales and four newly developed scales. Once the EFA was deemed acceptable, the decision was made to proceed to a CFA with all items retained from the EFA for refinement in the measurement model. Once the CFA achieved good model fit and demonstrated convergent and discriminant validity along with reliability, we tested for common method bias. This was followed by evaluating for configural and metric invariance by gender

majority composition of teams and role of respondent on the team considered during the survey. These serve as our moderators. Composites were constructed in order to complete Structural Equation Modeling in which mediation, multi-group moderation, and interaction moderation testing were conducted.

Pre-Analysis Data Screening

Data screening was conducted to identify missing data, evaluate for outliers and test normality. 39 survey items were included in the screening.

Missing Data

Among the 342 surveys were opened, 254 surveys were initially identified as complete, indicating that <10% of items were blank and there was no singular blank variable. Upon closer examination, three were problematic with half or more of a variable blank. These three respondents were removed with 251 total responses retained. All missing items were considered missing completely at random as the pattern of missing data were not dependent on any other variable or on values of the variable itself (Hair, Black, Babin, & Anderson, 2010). Where there were fewer than 10% of items missing and/or no complete variables missing, median imputation was utilized because the variables used Likert scales.

Data Screening

Univariate and Multivariate Outliers

For the full survey, there were 203 outliers across 65 records and 39 items. There were 11 records that were among the outliers for $\geq 10\%$ of their responses, but only one had a pattern of repeating values for multiple variables so it was removed for a total of 250 records retained. No other records represented apparent patterns to the responses.

These other records with outliers were not removed in the data screening phase as there was no clear theoretical indication that these responses were truly aberrant or not representative of a larger population (Hair et al., 2010). We chose to evaluate the structural model variables prior to eliminating outliers.

Upon examination of the variables utilized in the structural model, 26 univariate outliers were revealed through evaluation of outliers on boxplots. There were 24 multivariate outliers as evaluated by a $p < 0.05$ result for Mahalanobis d-squared. 15 outliers were found in both univariate and multivariate analyses. We chose to eliminate all 35 total outliers to obtain more accurate effect sizes and better model fit leaving us with 215 complete records for use in structural model testing.

Normality

Since all variables are Likert-type scales, we have no reason to exclude variables for skewness or kurtosis unless they exhibit no variance. Thus, we tested standard deviation as an alternate for skewness or kurtosis. For scales of 5 or 7 intervals, we expected to find no variables with standard deviation less than 0.5, which would have indicated over 68% of the responses fell on the mean. Our results show that all interval variables had standard deviations greater than 0.649 with the majority being greater than 0.8. This indicates there are no univariate normality issues that should affect our analysis.

For completeness, the variables utilized in the structural model were evaluated for skewness and kurtosis. Left skewness was detected for seven items with values outside the acceptable range of -1.0 and +1.0. Kurtosis was detected for 7 items (6 of which also demonstrated left skewness) with values exceeding +1.

Linearity

We tested linearity using OLS Regression and evaluating the significance for all direct effects in our model. The results demonstrate that the relationships are generally sufficiently linear at $p < 0.05$.

Homoscedasticity

We tested for homoscedasticity by evaluating the relationship between residuals (error) and the predicted values for the variable of interest through a Scatterplot. All variables used in the model were tested and resulted in homoscedasticity as represented by consistency between the error variances and the predicted variables (Hair et al., 2010).

Multicollinearity

The Variable Inflation Factor (VIF) was tested for all independent (exogenous) variables resulting in an indication that these variables have little to no correlation with each other with a VIF result < 3.0 . VIF < 10 can be considered indicative of the lack of multicollinearity (Griffith & Harvey, 2001; Stine, 1995).

Exploratory Factor Analysis (EFA)

An EFA was conducted using Maximum Likelihood as the method of extraction to determine unique variance among items and correlation to the anticipated six factors. Promax rotation was utilized and selected due to the expectation of correlation among factors. The EFA was conducted to evaluate whether the observed variables loaded together as expected, demonstrated adequate correlation, and met criteria for validity and reliability. We initially utilized Eigenvalues to determine the number of factors with the expectation there would be six, which matched the result. For completeness, we followed

the common practice of testing +1 and -1 fixed factors (testing seven and five factors). Neither test improved the pattern matrix.

Adequate explanation of six factors was obtained at 64.191% of variance explained and 5% non-redundant residuals greater than 0.05. There were seven of 26 items that had communalities <0.5 , but were retained for testing in a confirmatory factor analysis due to either approaching 0.5 or due to having only three items in a factor. KMO and Bartlett's test for sampling adequacy was also significant.

Convergent validity was met with all items but one at >0.50 . The one item <0.5 at 0.480 provides no concern because it is considered adequate for our sample size (Hair et al., 2010). Discriminant validity was also met. Items had no cross loadings and the correlation matrix shows no correlation above 0.70. There was one Heywood case, but testing with the use of Principal Axis Factoring for extraction as a cross check for this item resulted in no Heywood cases. The EFA pattern matrix is in Appendix B in Table B4.

All factors had good reliability measured with Cronbach's Alpha; each was >0.70 . All items are reflective of their constructs because they are generally interchangeable, internally consistent, and even small item factors have high reliability as each item should adequately represent the latent factor (Jarvis, MacKenzie, & Podsakoff, 2003). Because validity and reliability are adequate, confirmatory factor analysis was used to determine if any further item reductions would be necessary prior to moving to a structural model.

Measurement Model - CFA (Including Common Method Bias)

Model Fit

The measurement model included 39 items that were reflective of six first order factors. Due to high modification indices, MS7, Pot2, and Pot3 were removed. Additionally, due to a convergent validity concern, Emb4 was deleted. The final fit prior to testing Common Method Bias and group Invariance was excellent (CMIN/DF = 1.382 with $p=0.0$; CFI = 0.977; TLI = 0.973; RMSEA = 0.039; PCLOSE = 0.948; SRMR = 0.0437). With the unmeasured common latent factor present (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003), model fit continued to be excellent (CMI/DF = 1.27 with $p=0.009$; CFI = 0.986; TLI = 0.981; RMSEA = 0.033; PCLOSE = 0.99; SRMR = 0.0323). Each step taken in CFA reduction is available Table B5 in Appendix B as is an image of the reduced measurement model (Figure B2, Appendix B) with the unmeasured latent factor.

CFA Validity and Reliability

Discriminant validity is met as demonstrated by the square root of the AVE of each factor is greater than the correlation of any other items (Hair et al., 2010).

Convergent validity is met on each factor as demonstrated by an AVE >0.5 for each factor (Hair et al., 2010). Model reliability is met with each factor having CR >0.7 (Hair et al., 2010).

Table 7: Convergent and Discriminant Validity of the Constructs

	CR	AVE	Adapt	StratAlign	MemSat	StrucCon	RelCon	Potency
Adapt	0.891	0.621	0.788					
StratAlign	0.846	0.734	0.517	0.857				
MemSat	0.915	0.684	0.742	0.440	0.827			
StrucCon	0.742	0.595	0.343	0.415	0.306	0.771		
RelCon	0.799	0.571	0.563	0.542	0.610	0.415	0.756	
Potency	0.890	0.620	0.632	0.421	0.667	0.329	0.438	0.788

Common Method Bias

The data were collected using a single method – a survey, using individual respondents for all items on a single survey. Because of using this single method, a Common Method Bias (CMB) test was conducted. An unmeasured common latent factor was used to assess for CMB (Podsakoff et al., 2003). The resulting unstandardized regression weight was 0.478 and squared the result is 0.228 indicating some CMB exists. Next, the standardized regressions for each item were compared (weights with and without the common latent factor present). Four out of five Member Satisfaction items had a difference >0.2 ; one out of three Relational Connections items had a difference >0.2 ; and two out of five Adaptability items had a difference >0.2 indicating these were the items where common method bias could present a concern.

Prior to incorporating the effect of the common latent factor into composites for the Structural Equation Model, we tested validity and reliability with the common latent factor present. Significant convergent and discriminant validity issues emerged as did reliability concerns. This may be consistent with disadvantages of using an unmeasured common latent factor (Podsakoff et al., 2003) such as:

1. Inability to identify the specific cause of method bias;

2. Inability to identify if the common method variance is “inflat[ing], deflat[ing], or hav[ing] no effect on observed relationships between predictors and criterion variables” (Podsakoff et al., 2003: 893).
3. “The factor may reflect not only different types of common method variance but also variance due to relationships between the constructs other than the one hypothesized” (Podsakoff et al., 2003: 894).
4. Use of a common latent factor assumes the method factor does not interact with trait factors.

Due to these disadvantages and the effect the common latent factor has on measurement model validity and reliability, we chose to move forward with a Structural Equation Model (SEM) that does not include a common latent factor in composites of CMB-adjusted values or as a factor in the SEM. We do, however, recognize common method bias may exist and will be reported as a limitation to our results.

Invariance

Invariance testing was conducted to assess whether groups are generally similar. For multi-group moderators, we considered the gender majority content of teams and roles of survey participants on the teams reported. For Gender Majority Composition of Teams, configural invariance was achieved with good model fit. The groups are roughly equivalent according to the factor structure with invariance at the path level. Additionally metric invariance was met as demonstrated by a Chi-Square difference test p-value >0.05 indicating no difference between groups at the model level for metric invariance. Configural invariance and metric invariance for Role on the Team (leader or member) were also achieved. The Chi-Square difference test for role on team was >0.05 indicating

no difference between groups at the model level for metric invariance. Table B9 in Appendix B represents final model fit after invariance testing.

Structural Model

A structural model was built using non-CMB adjusted composite latent factors from the CFA. Composite variables were created in AMOS using regression imputation. The interaction term was created by standardizing all composite variables and multiplying Member Satisfaction and Strategic Alignment. Best fit was obtained prior to testing for mediation, total effects, moderation, or interaction. In order to achieve good fit we had to include direct paths from Member Satisfaction to Adaptability and Potency, as well as from Strategic Alignment to Adaptability and Potency. We felt these changes to be theoretically logical and likely indicate the hypothesized paths to be partial mediation rather than full. Finally, the interaction term was not significant and trimmed from the final model. The fit after removing univariate and multivariate outliers was excellent (see Model Fit below). An alternate model was tested which had very good fit, but not as good as our preferred theoretical model (see Table 8). Representative models of our model retained and the alternate model that was dismissed can be found Figures B3 and B4 in Appendix B.

Table 8: Model Fit – Structural Equation Models

Metric	Observed Value with Controls (Final Model)	Observed Value without Controls (Final Model)	Alternate (Rejected Model)	Recommended
CMIN / DF, p-value	1.25, p=0.271	2.018, p=0.109	2.514, p=0.028	Between 1 and 3; p>0.05
CFI	0.997	0.995	0.991	>0.950
TLI	0.986	0.976	0.973	>0.95
RMSEA	0.034	0.069	0.078	<0.05
PCLOSE	0.592	0.271	0.162	>0.05
SRMR	Unable to obtain due to missing values in Controls	0.0180	0.0425	<0.09

Mediation was tested using Mathieu and Taylor's (2006) recommended bootstrapping using 2000 bias corrected bootstrapping samples (at 95 BC Confidence Level) in AMOS while applying the Shrout and Bolger (2002) approach for assessing the presence of chained mediation effects. This was further informed by Fletcher's (2006) approach to assessing two mediators and also as conducted by Carlo, Lyytinen, and Rose (2012). Details behind this mediation approach are in Appendix C. Accordingly, Structural Connections and Relational Connections were tested as M1 and M2 to assess whether these factors successively mediate the independent variables and to what extent they carry the weight of mediation. Hypotheses with direct effects were tested while controlling for Gender of respondent, and Tenure of the team. Mediation tests were conducted without the presence of moderators or controls and all mediation is treated as potentially full mediation.

Multi-group moderation was examined in the full model with the influence occurring between the independent variables and the dependent variables so we could include two validated scales compared with two new scales. Multi-group moderation was

evaluated for team gender composition majority and team role of respondent (Leader vs. Member). Critical Ratios were produced for the differences in regression weights between groups. P-values were also calculated from the critical ratios to determine the strength of the difference between groups. Interaction terms were tested on the full model, but were added after testing multi-group moderation. This order of steps was utilized in order to test each hypothesis and retain clarity.

To test interaction, all variables were standardized. Since there were only two Independent Variables (Member Satisfaction and Strategic Alignment), we multiplied them to create an interaction term.

Table 9: Correlation Table of SEM Model

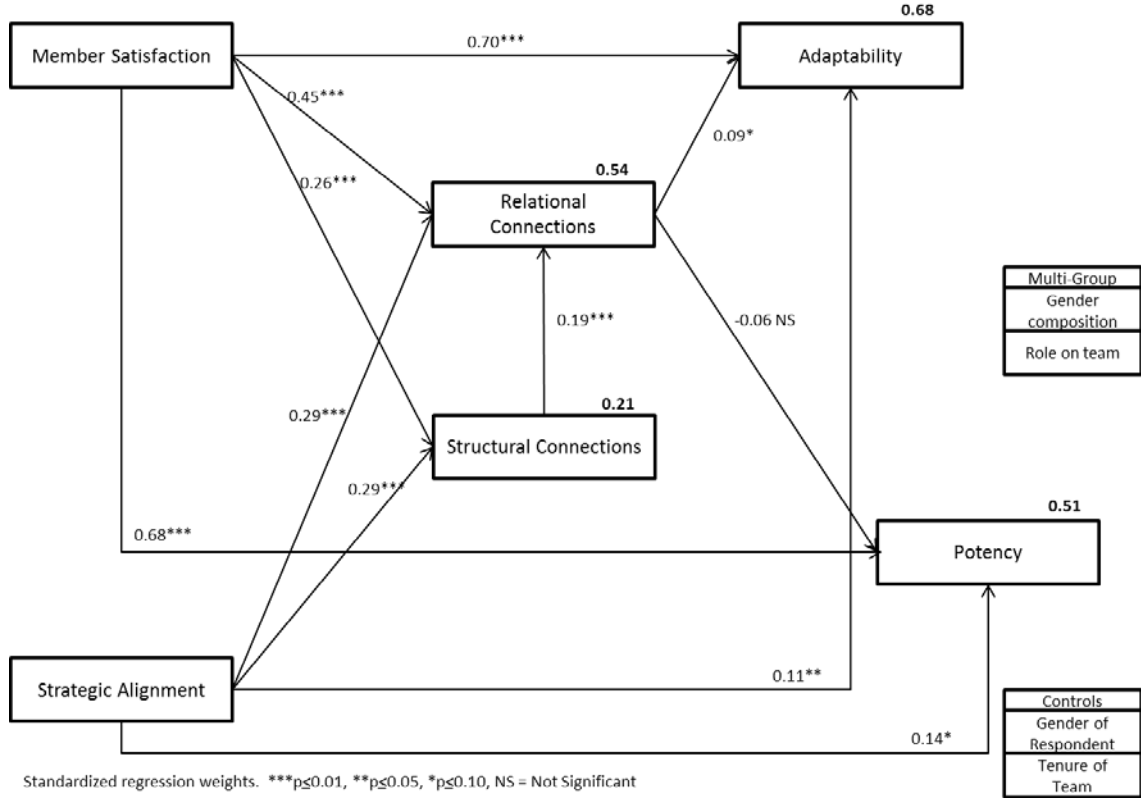
	Mean	Std Dev	StratAlign	MemSat	StrucCon	RelCon	Potency	Adapt
StratAlign	3.896	0.793	0.846*					
MemSat	3.889	0.775	0.403	0.915				
StrucCon	3.914	0.624	0.397	0.377	0.742			
RelCon	3.958	0.618	0.545	0.641	0.473	0.799		
Potency	4.189	0.646	0.386	0.703	0.351	0.456	0.890	
Adapt	3.779	0.687	0.446	0.81	0.359	0.606	0.633	0.891

*Values on the diagonal are the Construct Reliability

Results and Findings

The path standardized coefficients are represented with their associated p-values in Figure 7 below. The R-square values for endogenous factors are also included. The results for all hypotheses are described in their associated Direct Effects, Mediation, Moderation, and Interaction sections as well as in Table 10: Hypotheses Summary at the end of this section.

Figure 7: Final Model



Direct Effect (H1a, H1b, H2a, H2b)

The Direct Effect paths ($X \rightarrow Y$) were tested individually with controls included. The hypotheses were supported. Each of the hypothesized four direct paths had large positive effects, all > 0.310 and were significant at $p < 0.01$. Member Satisfaction had a larger direct effect than Strategic Alignment loading on both Potency (Std. Est. = 0.562 vs. 0.317) and Adaptability (Std. Est. = 0.665 vs. 0.374). The Controls of Gender of Respondent and Tenure of Team were non-significant in their effect on the dependent variables of Potency and Adaptability.

Mediation (H3a, H3b, H4a, H4b)

What Affects Potency?

Two paths were hypothesized to affect potency through two mediators – one initiated by an internal focus (Member Satisfaction) and one by an external focus (Strategic Alignment). Of the two paths, we found support for mediation effects in one path, that initiated by the external focus of Strategic Alignment. Member Satisfaction has only a Direct Effect on Potency as we found Indirect Effects to be non-significant (Unstd. Est. = 0.015, $p = 0.651$). We also saw a break in the hypothesized path with Relational Connections $\rightarrow$ Potency (M2 $\rightarrow$ Y) resulting in a non-significant path (Unstd. Est. = -0.037, $p = 0.652$). Strategic Alignment, however, distally impacts Potency through both Structural Connections and Relational Connections. The Indirect path ($abc + af + ec$) and the Total Effects (C) are both significant (Unstd. Est. = 0.178 and 0.560, respectively; $p = 0.001$ for both). Additionally, all direct paths evaluated were significant. Because the mediated paths are significant and mediation is partial, Strategic Alignment may at times bypass one mediator and have some effect carried by the other to affect Potency.

What Affects Adaptability?

Two paths were also hypothesized to affect adaptability through the same two mediators as above. We found support for mediation effects in both paths whether initiated by the internal focus of Member Satisfaction or by the external focus of Strategic Alignment. Both paths had significant Total Effects (C) and Indirect Effects ($abc + af + ec$): Member Satisfaction path had Total Effects of Unstd. Est. 0.666, $p = 0.001$ and Indirect Effects of Unstd. Est. 0.081, $p = 0.003$; Strategic Alignment path had Total Effects of Unstd. Est. 0.375, $p = 0.001$ and Indirect Effects of Unstd. Est. 0.247, $p =$

0.001. In both paths, the direct effect of Structural Connections to Adaptability (M1 → Y) was non-significant (Member Satisfaction: Unstd. Est. = 0.026, $p = 0.623$; Strategic Alignment: Unstd. Est. = 0.068, $p = 0.317$). Both Member Satisfaction and Strategic Alignment may at times bypass Structural Connections and be mediated by Relational Connections to affect Adaptability. Relational Connections, however, serves as an obligatory mediator for both Member Satisfaction and Strategic Alignment to affect Adaptability.

Reviewing all direct paths against the full structural model reveals unstandardized estimates and standard errors that are similar in size providing increased confidence in the results of our mediation tests. The paths' loads were generally minimally smaller in the full structural model, which is reasonable given that all paths were being analyzed at one time diluting the strength of any single path.

Multi-Group Moderation (H5a,b; H6a,b)

We found no support for our multi-group moderation hypotheses. That is there was no statistical difference in the effect of the gender majority of the team on the influence of Member Satisfaction or Strategic Alignment on Potency. There was also no statistical difference in the effect of Leaders versus Members on the influence of Member Satisfaction on Adaptability. There was a difference, however, for Leaders and Members in the influence of Strategic Alignment on Adaptability, but the results were opposite our hypothesis. The relationship between Strategic Alignment and Adaptability is stronger for Members than for Leaders. The results of moderation tests are summarized in Table 10, the hypotheses summary table at the end of this section; however, there is one additional interesting result to note. While we did not hypothesize a difference for gender

composition majority in the relationship between Strategic Alignment and Adaptability, we did see a difference where respondents reported their team having a majority of women had a positive moderation influence, but for male majority teams this was not significant. We did not trim paths as there were no paths that were insignificant across and between both testing groups (gender majority and role).

Interaction Summary (H7a,b)

We trimmed non-significant interaction paths one at a time until only significant paths remained. This resulted in the elimination of the interaction term as it resulted in no significant paths, thereby rendering our interaction hypotheses moot.

Support for Hypotheses Summary

Table 10: Hypotheses Summary

<i>Direct Effects</i>	<i>Evidence</i>	<i>Supported?</i>
H1a. <u>Member Satisfaction</u> has a direct, positive effect on <u>Potency</u>	0.703***	Supported
H1b. <u>Member Satisfaction</u> has a direct, positive effect on <u>Adaptability</u>	0.810***	Supported
H2a. <u>Strategic Alignment</u> has a direct, positive effect on <u>Potency</u>	0.386***	Supported
H2b. <u>Strategic Alignment</u> has a direct, positive effect on <u>Adaptability</u>	0.446***	Supported
<i>Mediation (See Appendix B for detailed results)</i>	<i>Evidence</i>	<i>Supported?</i>
H3a. <u>Member Satisfaction</u> → <u>Structural Connections</u> → <u>Relational Connections</u> → <u>Potency</u>	a X→M1: 0.300*** b M1 → M2: 0.251*** c M2 → Y: -0.037 NS e X→M2: 0.398*** f M1→Y: 0.109* c' X—Y (D): 0.545*** abc+af+ec X→Y (I): 0.015 NS C X→Y (T): 0.560***	Not Supported: Direct Effect only
H3b. <u>Strategic Alignment</u> → <u>Structural Connections</u> → <u>Relational Connections</u> → <u>Potency</u>	a X→M1: 0.322*** b M1 → M2: 0.283*** c M2 → Y: 0.320*** e X→M2: 0.320*** f M1→Y: 0.144** c' X—Y (D): 0.136** abc+af+ec X→Y (I): 0.178*** C X→Y (T): 0.314***	Partial: Partial Mediation
H4a. <u>Member Satisfaction</u> → <u>Structural Connections</u> → <u>Relational Connections</u> → <u>Adaptability</u>	a X→M1: 0.300*** b M1 → M2: 0.251*** c M2 → Y: 0.154**	Partial: Partial Mediation

	e X→M2: 0.398*** f M1→Y: 0.026 NS c' X→Y (D): 0.585*** abc+af+ec X→Y (I): 0.081*** C X→Y (T): 0.666***	
H4b. <u>Strategic Alignment</u> → <u>Structural Connections</u> → <u>Relational Connections</u> → <u>Adaptability</u>	a X→M1: 0.322*** b M1 → M2: 0.283*** c M2 →Y: 0.548*** e X→M2: 0.320*** f M1→Y: 0.068 NS c' X→Y (D): 0.128*** abc+af+ec X→Y (I): 0.247*** C X→Y (T): 0.375***	Partial: Partial Mediation
Moderation		
H5a. The positive relationship between <u>Member Satisfaction</u> and <u>Potency</u> will be stronger for <u>Female majority team composition</u> than for <u>Male</u> .	Female: 0.431*** Male: 0.416*** Zscore: 0.126ns	No: There is no statistical difference
H5b. The positive relationship between <u>Strategic Alignment</u> and <u>Potency</u> will be stronger for <u>Male majority team composition</u> than for <u>Female</u> .	Female: 0.038ns Male: 0.363* Zscore: 0.720ns	No: There is no statistical difference
H6a. The positive relationship between <u>Member Satisfaction</u> and <u>Adaptability</u> will be stronger for <u>Members</u> than for <u>Leaders</u> .	Leader: 0.583*** Member: 0.588*** Zscore: 0.047 ns	No: There is no statistical difference
H6b. The positive relationship between <u>Strategic Alignment</u> and <u>Adaptability</u> will be stronger for <u>Leaders</u> than for <u>Members</u> .	Leader: -0.017ns Member: 0.229*** Zscore: 2.577**	Partial: There is a difference, but it is for the opposite role
Interaction		
H7a. An increase in <u>Strategic Alignment</u> will strengthen the positive relationship between <u>Member Satisfaction</u> and <u>Potency</u> .	No interaction (path deleted due to non-significance)	No: No interaction effect
H7b. An increase in <u>Strategic Alignment</u> will strengthen the negative relationship between <u>Member Satisfaction</u> and <u>Adaptability</u> .	No interaction (path deleted due to non-significance)	No: No interaction effect

***p<0.01, **p<0.05, *p<0.1, NS = Not Significant

Discussion

An Internal Focus Disconnects Interconnections from Potency

The perception of individuals participating in teams in organizations provides insight to positioning and embedding of the microsystems of teams into macro systems (Joshi et al., 2009). Our findings indicate that although team participants may experience positive interactions within the team, it does not mean they perform well with or for the organization. Member Satisfaction served as the indicator for internal team interactions due to its multidimensional nature capturing satisfaction with the group, the group's decisions, and the group's interactions (Oetzel, 2001). Positive internal interactions of the

team, as measured by Member Satisfaction, has a strong and significant direct effect on Adaptability and Potency, however, that relationship is altered when including factors that contribute to a team embedding itself in the organization through boundary spanning of connecting with others outside the team.

In regard to Potency, interconnections (particularly Relational Connections) do not have mediational effects for Member Satisfaction influence on Potency. In this path, there is no significant relationship between Relational Connections and Potency, nor is there an Indirect Effect for Member Satisfaction in relation to Potency. In terms of Adaptability, Structural Connections do not have mediational effects for Member Satisfaction's influence on Adaptability. There remains, however a small Indirect Effect between Member Satisfaction and Adaptability indicating interconnections, at least Relational Connections, play some mediation role. This further suggests that *a team may interact well, adjust their internal tasks, and feel confident in their ability to perform, but they may or may not connect with the larger system in a way that supports performance.* A team can become isolated by being too internally focused; conversely, it can become "underbounded" if it continually scans the environment and does not settle on a course of action (Choi, 2002: 187).

Interactions and Interconnections with the Larger System Influences Adaptability and Potency

A work group is a system with inputs, processes, and outcomes (Oetzel, 2001). Our construct of strategic alignment – an input – is concerned with consideration for the larger system's priorities and aligning team activities with those priorities. Knowing who to connect to – Structural Connections – and collaborating and cooperating with those 'others' outside the team – Relational Connections – serve as processes. Our constructs of

Adaptability and Potency represent outcomes. We learned that *when team participants see the team as strategically aligned with the larger system, interconnections partially carry the influence of Strategic Alignment on both Adaptability and Potency. Knowing who to know and relating well are important factors.*

While the Total Effects were stronger (i.e. larger beta) for Member Satisfaction in relation to Adaptability and Potency, we see this as driven by the strong direct effects of these relationships. Strategic Alignment, on the other hand, results in a larger Indirect Effect on both Adaptability and Potency, which is driven by the partial mediating effects of Structural and Relational Connections. Because interdependencies and interconnections are recognized through an external focus, groups are likely better able to make changes to meet demands and they are more likely to perform well. When a team is strategically aligned with the system, added layers of complexity are not only present, but may be necessary for enhancing adaptability or potency. Embedding in the organization, with an emphasis on the system as a whole rather than disconnected parts (Senge, 2006) and as a complex pattern of interconnections across people and tasks (Arrow et al., 2000f), strongly influences adaptability and potency.

Predictors – Useful New Measures and Counterintuitive Results

It is worth noting the differences in the coefficient of determination (R^2) for Structural Connections, Relational Connections, Adaptability and Potency. This is important because it indicates the amount of variance in these variables that is explained by their antecedents (Hair et al., 2010). The higher the value, the better the antecedents predict the value of the dependent variable in a given path (Hair et al., 2010).

We found two counterintuitive findings. One was that the positive relationship between Strategic Alignment and Adaptability was stronger for Members than Leaders. We thought Leaders of teams would be more attuned to how Strategic Alignment would influence the ability to adapt to changing priorities as well as adapting tasks and goals. As it turned out, there was a negative and non-significant relationship for Leaders. Members, though, resulted in a strong (beta = 0.229) and significant relationship. A second counterintuitive finding was that Structural Connections is actually weakly predicted ($R^2 = 0.21$) by its inputs of Member Satisfaction and Strategic Alignment. We do not know if this is a matter of only being predicted by two variables or if it is due to the inputs themselves. We expected that members of teams could help make connections to those outside the team who could be helpful to the group. Additionally, it seemed intuitive that being strategically aligned would give insight to and predict connections to those outside the team that could serve as influencers to their efforts.

The remaining constructs, though, were well predicted by their antecedents. Adaptability is well predicted ($R^2 = 0.68$) by the four antecedent factors of Member Satisfaction, Strategic Alignment, Structural Connections, and Relational Connections. Potency is also well predicted ($R^2 = 0.51$), but not to the same extent as Adaptability. Relational Connections, which serves as a significant input to Potency when preceded by alignment with the organization is also well predicted ($R^2 = 0.54$) by its antecedents to an even slightly better degree than Potency.

With three new scales serving as antecedents to a well-established scale of Potency, it was important to see the externally focused constructs measured by these new scales not only positively influenced Potency, but also explained a large amount of the

variance in the measure of Potency. Furthermore, an empirical measure of Adaptability is poorly represented in the literature and has not typically been treated as a dependent variable (Burke et al., 2006). We found it can be well predicted by internal and external focuses of team participants, partially mediated boundary spanning activities of connecting to the larger organization, and it can serve as a dependent variable as well as if not better than Potency. Both boundary spanning and strategic alignment have an emphasis on connecting with others to achieve goals (Joshi et al., 2009; Senge et al., 2008). Our new scale of Relational Connections provides a measure of influence on and explanation of variance in both Adaptability and Potency – both of which contribute to achieving aims. Our results indicate internally and externally focused factors are influencers of adaptability and potency, but externally focused groups may accomplish this better.

Final Thoughts

Members of teams with an external focus – one that is strategically aligned with the organization - consistently utilize interconnections with those outside the team to influence adaptability and potency. Those who are internally oriented, however, tend to rely less on whom they know or how they collaborate outside the team to support adaptability or potency. While the internal focus of a team does directly affect the team's ability to adapt to changing priorities and tasks as well as its predictive power of performance, it does not effectively take into account the complexity of a team embedding into a system as perceived by participants. Our findings suggest that our new scales of Strategic Alignment, Structural Connections, Relational Connections, and Adaptability do help to explain the effects of the perceived internal and external focus of

teams and provide predictive capability for Potency. There is a pattern to embedding in an organization. The internally focused group clearly has the capacity to adjust tasks and goals according to changing priorities and they do demonstrate confidence in their ability to achieve goals. A disconnect to Adaptability and Potency occurs, however, when connections to others outside the team are included. The internally focused team more heavily depends on their internal interactions to manage change and develop confidence in success. Interconnections influence a group's Adaptability and Potency, though, when boundaries are spanned and the group considers its interdependence with the organization's priorities.

Limitations and Future Research

Limitations for this study include four particular elements. First, we sought to gain an understanding of teams, but were not able to obtain team level data. Instead our survey information and results are based on the perceptions of individual participants who were asked to consider a single team in which they participate at work; thus a variety of teams are represented, but no teams as wholes are represented.

Second, common method bias is present; however, not as an overwhelming concern. As noted, we anticipated the potential for common method bias to be present given that our data included singular informants for the entire survey and data set (Podsakoff et al., 2003). We were able to calculate the existence of common method bias, but chose to accept the risk of a small influence on our results rather than incorporate an unmeasured latent factor (Podsakoff et al., 2003) into composite values where no common method bias was present.

Third, our data lacks objective performance data. Knowing we were going to conduct an individual level anonymous survey, we chose to include the Team Potency variable in lieu of performance data that we would not be able to obtain and match to team performance. As highlighted earlier, we accepted team potency as a reasonable alternative to performance data given its validated capability to predict team performance (Collins & Parker, 2010; Gully et al., 2002).

Fourth, our model lacks a feedback loop representative of the dynamic nature of teams. This would have required either a longitudinal study or a different type of analysis than we were prepared to perform at this time.

Given the limitations offered, some considerations for future research would be to escalate this study to a team level, incorporate objective team performance data, and identify optimal mechanisms to incorporate the effects of feedback to teams. In addition to objective team performance data such as progress toward goals, it would be valuable to include external stakeholder analyses. Such inclusions would help to minimize common method bias (Podsakoff et al., 2003) and would more holistically represent a team's performance and successful embedding into an organization. Accordingly, an additional study component that could add a depth of understanding to a team's connections in the organization would be to conduct a social network analysis. Furthermore, Joshi (2009) recommended a combination of a qualitative study with network analysis as most studies have conducted one or the other. Whatever additions or alterations are included, it would be our recommendation to continue to evaluate the combination of internal and external forces of teams, given their feedback effects on each other (Choi, 2002), in order to better

understand how the two can most effectively be combined to successfully embed a team in an organization to benefit both the team's and the larger system's goals and priorities.

Conclusions

The way groups embed into larger systems is a complex set of interconnections and interdependencies. According to individual participant perceptions, teams are confident and predictive of success when they are aligned with the organization and are connected to and relate well to others in the organization. This also facilitates the ability to manage changing priorities, tasks, and goals. Imposing a strong and positive effect on potency is, again, important because of its predictive capabilities regarding performance (Collins & Parker, 2010; Gully et al., 2002).

The model we offer represents the importance of valuing and supporting the complexity of interconnections and interdependencies of strategies, tasks, and relationships. From a scholarly perspective, our work offers a contribution to and building block for the theories of teams as a set of interrelated cognitions, behaviors, and attitudes (Ilgen et al., 2005; Salas et al., 2008). It also offers one approach to understanding teams as complex systems "existing in larger systemic contexts" (Ilgen et al., 2005: 519) to successfully embed a team in the organization such that it has the potential to benefit the larger system. If we are to consider teams as complex adaptive systems, embedding into larger organizations, it is important to be able to understand the factors that support such complexity and begin the path for measurement. For practitioners, we recommend those who are accountable for team performance encourage and facilitate teams to understand the system's priorities as well as connect to others outside the team to support adaptability and performance confidence.

CHAPTER 5: TEAM ADAPTATION AND MINDFUL BOUNDARY MANAGEMENT: HOW COGNITIVE AND AFFECTIVE ADAPTATION PATTERNS ARE DYNAMICALLY DEVELOPED THROUGH STRUCTURAL, COGNITIVE, AND BEHAVIORAL MECHANISMS

Introduction

Teams⁵ are necessary to operate successfully in complex and changing business landscapes (Burke et al., 2006); pursue continuous improvement and gain competitive advantage (Bell & Kozlowski, 2010; Kozlowski et al., 1999; Wiedow & Konradt, 2011); and are necessary to implement large organizational initiatives and create adaptive processes (Cooney & Sohal, 2004; Douglas & Judge Jr, 2001; Eisenhardt & Tabrizi, 1995; Hackman & Wageman, 1995). Such demands require internal awareness and sensitivity to potential needs and impact on people outside the team (Schippers et al., 2007) as well as openness (LePine, 2003) and responsiveness to boundary feedback internally (Ancona & Caldwell, 1992a; Schippers et al., 2007) in order to adapt team interactions and goals. Team adaptation is an important topic in organizational change and strategy literatures. It is also a complex topic as it calls for balancing internal and external dynamics (Coghlan, 1994: 16).

At the same time, team adaptation and how it influences team performance has received little empirical attention. So far team adaptation has been largely studied

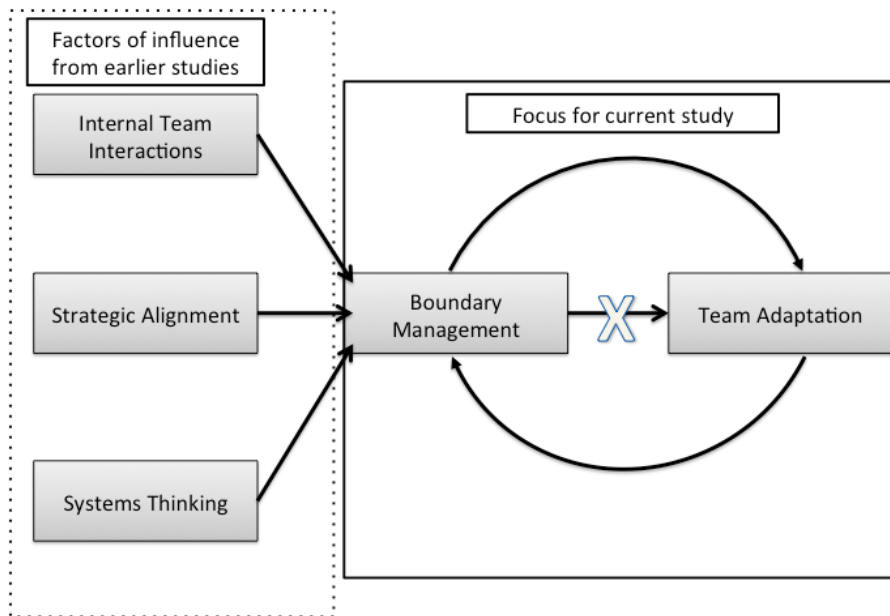
⁵ The words team and group are often used interchangeably in the literature and will also be used as such here Ancona, D. G. 1990. Outward Bound: Strategies for Team Survival in an Organization. *The Academy of Management Journal*, 33(2): 334-365. Teams and groups refer to organizational members who see themselves and are seen by others as a group who interact dynamically and interdependently to carry out an organizational task and/or work toward a valued goal Ancona, D. G. 1990. Outward Bound: Strategies for Team Survival in an Organization. *The Academy of Management Journal*, 33(2): 334-365, Kozlowski, S. W., Watola, D. J., Jensen, J. M., Kim, B. H., & Botero, I. C. 2009. Developing adaptive teams: a theory of dynamic team leadership. In E. Salas, G. F. Goodwin, & C. S. Burke (Eds.), *Team effectiveness in complex organizations: Cross-disciplinary perspectives and approaches*: 113-155. New York: Taylor & Francis Group, LLC..

conceptually. Several models of adaptation covering team characteristics and cycles of assessment, planning, execution, and learning as well as emergent states of shared mental models and situation awareness have been proposed (Burke et al., 2006; Klein & Pierce, 2001; Kozlowski et al., 1999; Kozlowski & Bell, 2008; Marks et al., 2001; Salas et al., 2009). Some empirical studies have been conducted, but they have primarily been lab-based experiments (Entin & Serfaty, 1999; LePine, 2003, 2005; Marks et al., 2000; Woolley, 2009) rather than focusing on complexity of team behaviors in *situ*. These studies offer a variety of mechanisms of adaptation, but “solid conceptual and empirical work on these themes are sparse” (Kozlowski & Bell, 2008: 16). Furthermore, the extent to which adaptation is directly linked to interactions occurring outside the team is still poorly understood. Teams that perform well have been identified as those who effectively manage their boundaries (Ancona, 1990; Ancona & Caldwell, 1988; Marrone, 2010) through functions such as coordinating complex tasks for product or service development (Ancona, 1990; Ancona & Caldwell, 1990; Marrone et al., 2007; Sundstrom et al., 1990); obtaining resources (Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Joshi et al., 2009); and exchanging information (Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Joshi et al., 2009; Marrone, 2010). The way in which team adaptation emerges in relation to such boundary management especially for teams in the field, however, is under-evaluated.

In two studies that led to this current inquiry, we found that boundary management facilitates adaptation and does so in a mediating way (Grooms & Lingham, 2013; Grooms & Lyytinen, 2014). By this we mean boundary spanning does not stand alone in driving adaptation rather it transforms internal and external interactions into the

ability to adapt. Our phase 1 qualitative study revealed combinations of systems-thinking, strategic alignment, boundary spanning, and motivation render explanatory influence on the emergence of team adaptation (Grooms & Lyytinen, 2014). Our phase 2 quantitative study demonstrated teams are confident and adaptable when they are aligned with the organization and are connected to and relate well to others in the organization (Grooms & Lingham, 2013). That is, measurable factors of alignment with organizational strategic priorities, connections outside the team, and collaboration with outsiders influenced adaptation. What we want to understand in greater detail, now, is what the distinct mechanisms are that teams use to manage their boundaries that allow patterns of adaptation to emerge and what those adaptation patterns are. We anticipate boundary management and adaptation iteratively influence each other, rather than act linearly, given that teams intentionally interact with other parts of the organization, the environment of which “reacts to the team and then both the team and context influence each other” (Ancona, 1990: 357) promoting adaptation. We represent our previous findings and topic of interest for this study in Figure 8.

Figure 8: Factors Influencing the Development of Boundary Management and Adaptation



The context in which teams interact with the boundary and its effect on adaptation is important (Bell & Kozlowski, 2008). The types of teams evaluated in the context of adaptation has generally covered team settings such as knowledge teams (Woolley, 2009), action teams (LePine, 2003, 2005; Marks et al., 2000) and broadly generalized teams (Schippers et al., 2007; Wiedow & Konradt, 2011). When industries are represented (i.e. the few studies not conducted in a lab-setting), they mainly focus on manufacturing, service industries, and public administration (such as education / social work) (Schippers et al., 2007; Wiedow & Konradt, 2011). Boundary spanning studies have primarily examined product development teams in manufacturing and/or high-tech industries (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Carlile, 2002, 2004; Faraj & Yan, 2009). Studies of teams that orient towards continuous organizational improvement in highly institutionalized and regulated settings are missing. Our context of

interest is Quality Improvement (QI) multidisciplinary project teams in hospital settings, which work together for a short time and pursue well-defined outcomes for improvement that could include process or clinical improvements often spanning multiple sub-units of the organization. In contrast to project teams, new product development teams work together over extended periods of time (Brown & Eisenhardt, 1995; Edmondson & Nembhard, 2009) and benefit when they are protected from other day-to-day routines, where increasing levels of separation are needed the more innovative the product (Barczak & Wilemon, 1989; Ernst, 2002). We would expect that both interactions with the external environment and internal team adaptations due to the heterogeneity of the teams in such settings are much more pronounced and complicated and worthy of study.

To understand boundary management and adaptation in our context of interest, our research question asks: how does team adaptation emerge through boundary management? Furthermore, what are the mechanisms of boundary management that facilitate gaining support from outside the team in settings that are highly institutionalized and regulated? To answer these questions, we conducted a qualitative study motivated by an extensive review of the literature on team adaptation and boundary management. Our analytic approach was founded on thematic analysis to address these questions broadly. A qualitative study using thematic analysis is deemed appropriate when a scholar wants to “discover rather than test variables” or conditions (Corbin & Strauss, 2008: 12). For us this study offers a way of seeing or making sense of seemingly unrelated material related to team interactions by analyzing and systematically observing interactions (Boyatzis, 1998). Following a review of the literature and a detailed description of our analysis, we finish with a report of our findings and conclude with a

discussion of the critical results as well as identify scholarly and practical contributions of the study.

Theoretical Framework

Team Adaptation

Team adaptation has been defined as “a change in team performance in response to a salient cue or cue stream, that leads to a functional outcome for the entire team” (Burke et al., 2006: 1190); “cognitive, affective, motivational and behavioral modifications made in response to the demands of a new or changing environment or situational demands” (Baard et al., 2014: 50); and as both a dynamic change occurring over time (LePine, 2005) as well as an outcome that includes post-change decision-making (LePine, 2003). Most conceptions of team adaptation consider the contextual nature of teams spanning organizational levels and needing to strike a balance between internal and external demands of dynamic and structurally complex environments (Ilgen, 1999). We adopt a definition of team adaptation as the extent to which a team generates a functional outcome by purposeful patterning of change through cognitive, affective, and/or behavioral changes between the team and its embedding context (Grooms & Lyytinen, 2014). Teams are triggered to adapt by a variety of demands or changes which can be expected or unexpected (Burke et al., 2006; Kozlowski & Bell, 2008) as well as internally or externally driven (Entin & Serfaty, 1999; Kozlowski et al., 1999).

Balancing Internal and External Factors of Adaptation

Teams experience both internal and external influences that trigger adaptation needs. Adaptive teams modify internal processes proactively and flexibly (Klein & Pierce, 2001) whereby members change cognitive states or goal-driven actions and

segment process-related tasks to address demands (Burke et al., 2006; Woolley, 2009). Interpersonal processes serve as both influences on and reflections of adaptation. Such processes include sense-making where members assign meaning to member actions (Salas et al., 2009); conflict management, confidence building, and affect management within the team (Ilgen et al., 2005); the development of shared mental models or cognitive states (Burke et al., 2006; Marks et al., 2000); and team reflection (Schippers et al., 2007; Wiedow & Konradt, 2011). While adaptation occurs within teams, external influences interest us as well, because teams are called upon to design and implement changes affecting multiple individuals or groups in the organization in a turbulent environment (Douglas & Judge Jr, 2001; Hackman & Wageman, 1995; Kozlowski & Ilgen, 2006).

External influences inviting adaptation emerge from the environment in which teams are engaged. Adaptive teams are able to adjust in response to barriers or constraints from the environment (Burke et al., 2006). Some examples of external influences include shifts in the external task or communication environment (Kozlowski et al., 1999; LePine, 2005; Marks et al., 2000); a new outcomes focus (Woolley, 2009); increasingly uncertain, ambiguous, or time pressured situations (Entin & Serfaty, 1999; Klein & Pierce, 2001); and changes in organizational systems such as structure (Kozlowski & Bell, 2008). The external perspective is an important one as teams often function in complex environments and their efforts affect others (Ilgen et al., 2005; Kozlowski et al., 2009; Schippers et al., 2007). Additionally, where teams are dependent on external interactions with the environment, external activities have been found as better predictors of team performance than internal processes (Ancona, 1990; Kozlowski et al., 2009). Yet,

no observational studies exist to explain the mechanisms of adaptation in regard to interactions with swaths of the organization.

Mechanisms of Adaptation

Adaptation is a dynamic phenomenon that has been described as arising from cognitive, affective, or behavioral characteristics, structures, or actions in response to expected or unexpected changes, as well as contribute to outcomes as indicators of performance (Burke et al., 2006; Kozlowski et al., 1999; Kozlowski et al., 2009; Kozlowski & Bell, 2008). As an emergent team-level phenomenon, adaptation properties “emerge from the perceptions, knowledge, feelings, actions, and interactions among individuals” and a phenomenon is emergent when it “originates in cognition, affect, behaviors or other characteristics of individuals...is amplified by interactions...and manifests as a higher level, collective phenomenon” (Kozlowski & Bell, 2008: 17-18). Few definitions have been offered for cognitive, affective, or behavioral adaptation, but limited descriptions offered include:

- Cognition: team observation of itself or other recognition of cues indicating a need for change leading to iterations of recognition, meaning-making, and mental model development (Burke et al., 2006; Klein & Pierce, 2001)
- Behavior: members working together to share knowledge and develop systems of communication (LePine, 2003).
- Affect: feeling that the plan is not going well or positive feelings of change when new opportunities are available (Klein & Pierce, 2001)

Mechanisms of adaptation have not been categorized specifically as cognitive, affective, or behavioral – just generally described as such (Kozlowski et al., 1999; Marks et al.,

2001; Salas et al., 2009) – but we purport this as a meaningful way to understand adaptation.

In order to attempt to categorize mechanisms of adaptation provided in the extant literature according to cognition, behavior, and affect, we first establish basic definitions of the concepts since no clear consensus exists for the concepts relevant to team adaptation. Merriam-Webster (2015) describes cognition as mental activities that include thinking, understanding, learning, and remembering as well as processes of perceiving, recognizing, and reasoning all of which are distinguished from experiences of feeling or willing. Behavior is a manner of conducting oneself or responding to the environment as individuals or groups (Merriam-Webster, 2015). Affect is distinguished from cognition and behavior as being based in feelings or emotion, but it also includes manifestations of subjectively experienced emotions (Merriam-Webster, 2015). Utilizing the few descriptions available in the team adaptation literature and their contextual associations as well as the dictionary definitions of cognition, affect and behavior, we explore what is known and what gaps exist in describing mechanisms of adaptation according to cognition, affect and behavior. For a summary table of the literature synthesis, please refer to Appendix D.

A variety of manifestations of cognitive, behavioral, and affective characteristics or mechanisms of adaptation are available. Cognition consists of mental activities among team members across categories of recognition, shared understanding, and goal or outcome orientations. Recognition includes a dynamic and recursive awareness of environmental cues or observation of the team from within for the need for change (Burke et al., 2006; Klein & Pierce, 2001). Knowledge, skills, attitudes, and capacity for

learning need to be present for pattern recognition and awareness of needs for network linkages in order to pursue continuous team improvement (Kozlowski et al., 1999; Kozlowski & Bell, 2008; LePine, 2003, 2005; Salas et al., 2009). Shared understanding includes knowledge structure development through learning as well as flexible shifting among task related contingencies (Burke et al., 2006; Entin & Serfaty, 1999; Kozlowski et al., 1999; Kozlowski & Bell, 2008; LePine, 2005; Marks et al., 2000; Schippers et al., 2007). Sense-making or assigning meaning to actions also contributes to shared understanding (Klein & Pierce, 2001; Salas et al., 2009). Teams having a goal orientation or outcome focus also need to employ learning and recognition in order to adapt (LePine, 2005; Woolley, 2009). We combine these descriptions to offer a definition of cognitive adaptation as mental activities of thinking, learning, and understanding as well as processes of perceiving, recognizing, and reasoning manifested through knowledge, skills, and attitudes supporting shared mental models contributing to continuous improvement, flexibility, and contingency management. Opportunity is available to contribute to a better understanding of what cognitive adaptation is and how it emerges as only team orientation – including goal and outcome focuses – as well as the value of developing a shared mental model have empirical support (Entin & Serfaty, 1999; LePine, 2003, 2005; Marks et al., 2000; Woolley, 2009).

Affect as a component of adaptation, yet weakly described in the extant literature other than the implied understanding of affect, is feelings and related beliefs. The few relevant descriptions can be synthesized as feelings about team progress and environmental feedback manifested in openness to experience, desire for collective cohesion, and beliefs about effectiveness (Entin & Serfaty, 1999; Klein & Pierce, 2001;

LePine, 2003; Marks et al., 2001; Salas et al., 2009; Schippers et al., 2007). The role affect plays in adaptation is a gap as we propose that a team's ability to perceive and connect with the environment emotionally or through beliefs about interactions will be an important facet of adaptation.

Behavioral adaptation is also challenging to define as compared to cognition because the behaviors described are wide-ranging and difficult to categorize in smaller groupings. Our review of the various descriptions resulted in the following definition: Response to the environment manifested through feedback seeking behaviors to assess gaps between actual and desired states; enacting and executing processes relevant to tasks and communication; interpersonal processes to share knowledge, demonstrate dependability, and modify network configurations (Burke et al., 2006; Ilgen et al., 2005; Klein & Pierce, 2001; Kozlowski et al., 1999; Kozlowski et al., 2009; Kozlowski & Bell, 2008; LePine, 2003, 2005; Schippers et al., 2007; Wiedow & Konradt, 2011; Woolley, 2009). Here again, empirical studies are limited, but where they have been conducted, conscientiousness in achievement and dependability in pursuit of change and task completion (LePine, 2003, 2005; Schippers et al., 2007; Wiedow & Konradt, 2011), working together to develop systems of communication (LePine, 2003), and reflection (Schippers et al., 2007; Wiedow & Konradt, 2011) are noted. The behaviors suggested in theoretical studies vary including problem solving (Klein & Pierce, 2001) and enactment or execution of processes for issues such as conflict management, motivation, back-up behaviors, coordination, and application of resources (Burke et al., 2006; Ilgen et al., 2005; Klein & Pierce, 2001). Such variation in behaviors leads us to suggest that behavioral adaptation does not fit in a box of singular action types. Rather it is a

dynamically active response pattern that emerges from cognitive and affective adaptation. Behaviors may be represented in a variety of ways but we submit most importantly include actions or responses executed according to interaction with the environment and the understanding and beliefs that emerge from those interactions.

Adaptation as a phenomenon has been considered too generic to explain performance change (Baard et al., 2014). Much of the team adaptation extant literature is theoretical or conceptual (Burke et al., 2006; Klein & Pierce, 2001; Kozlowski et al., 1999; Kozlowski et al., 2009; Kozlowski & Bell, 2008; Salas et al., 2009) and provides models to consider for triggers and mechanisms of adaptation, but they lack the power of observational or other empirical studies. Where empirical studies have been conducted, most are laboratory-based and emphasize action types of teams, which typically have specialized roles for their members cooperating in short-term events requiring improvisation and uncertain outcomes, (Entin & Serfaty, 1999; LePine, 2003, 2005; Marks et al., 2000; Sundstrom et al., 1990; Woolley, 2009) lacking the unpredictable contextual factors that can influence teams “in the wild” (Salas et al., 2008: 544) over the extended course of a project team’s mission. Our study offers to fill the gap of lacking empirical studies of teams in organizations as well as provide new insights to the internal and external balancing during adaptation and associated mechanisms. We also provide detailed attention to how adaptation behaviors are manifested.

Team Boundary Management

Because external influences invite adaptation and adaptive teams are responsive to their environments (Burke et al., 2006), we also explore interactions teams have with those beyond the team. Teams do not function in isolation (Ancona & Caldwell, 1992a;

Joshi et al., 2009). To be successful, they have a purpose that extends beyond the setting of the team itself (Ancona, 1990) and they must interact and coordinate with functions, groups, and individuals outside of the team (Ancona & Caldwell, 1990; Marrone et al., 2007). The team serves as the boundary and interactions with anyone beyond the team indicates a boundary activity. The terms boundary activity, boundary spanning, and boundary management are used throughout the literature, often interchangeably (Ancona & Caldwell, 1990; Choi, 2002; Faraj & Yan, 2009; Joshi et al., 2009). These terms are used to describe teams reaching beyond their boundaries, inside and outside of firms, to provide closer coupling or establish relationships and interactions with external actors as well as between functional units enabling the team to meet its goals (Ancona, 1990; Choi, 2002; Joshi et al., 2009; Marrone, 2010).

Beyond establishing relationships or interaction capabilities, teams span boundaries to achieve success. Boundary spanning enables teams to obtain information, resources, and support to create products and services making teams dependent on outside sources for high performance suggesting the importance of managing relationships (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Faraj & Yan, 2009; Marrone, 2010); therefore, we consider and adopt boundary management as the most descriptive term. To fully understand boundary management, it is important to explore what has been revealed about the mechanisms of boundary management and its effect on teams.

Boundary Management Influences, Mechanisms, and Context

How boundary management takes place suggests specific activities or processes constitute a part of boundary management. These processes are antecedents and influence

boundary spanning by directing boundary management activities. While our study is primarily interested in boundary management activities and underlying mechanisms, we include a consideration of antecedents to boundary management offered in the extant literature, because of our interest in contextual influence on complex interactions of boundary management (Joshi et al., 2009). The context promotes boundary management needs and adaptability as to respond to the environment (Marrone, 2010).

Antecedents to boundary management are described at the individual, team, and organizational levels. Individual characteristics include member skills and personalities (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988); “affective, cognitive, and behavioral responses to the environment” (Joshi et al., 2009: 734); and member roles and self-efficacy (Marrone et al., 2007). Team-level antecedents include a wide variety of characteristics including the nature of tasks (Ancona & Caldwell, 1988); team level attitudes, behaviors, and perceptions (Joshi et al., 2009); motivation and competency (Marrone, 2010); agreement about strategic and purposeful management of the environment (Marrone et al., 2007); and composition and development (Choi, 2002; Marrone et al., 2007). Organizational level characteristics denote organizational structure and climate (Joshi et al., 2009) and organizational tendencies for individuals to simultaneously serve on multiple teams contributing to boundary activity (Ancona & Caldwell, 1988). We propose that there may also be structural, affective, cognitive, and/or behavioral dimensions at the team level for boundary management as teams choose with whom to interact, why they engage others, and how those interactions occur.

An additional influence on team boundary management includes the task environment such as the extent to which the team or organization is experienced in the

goal being pursued (Ancona & Caldwell, 1990) or the team tasks required and their contingencies (Joshi et al., 2009) triggering external interactions. We would consider task based antecedents as most closely related to why a team would choose to or need to engage others beyond the team, whereas, the individual, team, and organizational level characteristics are those which may influence the extent to which they are skilled at or enabled to interact with others. Why a team chooses with whom to interact is an aspect of boundary management significance for this study as we seek to clarify why particular people beyond the team are deemed as those who can add value to the team. We propose there is more to explain about why particular individuals or groups are needed by the team or will be affected by the team.

Examining the mechanisms of boundary management begins to inform an understanding of why teams span and manage their boundaries. Teams often need resources, information, and support in order to accomplish their aims. The extant literature has repeatedly supported this as a reason for boundary management as well as the specific description of the mechanisms or activities undertaken as in:

- *Persuading* others, often senior leaders, to support the team through representation of the team's work (Ancona, 1990; Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003; Faraj & Yan, 2009; Marrone et al., 2007);
- *Coordination* of task activities across the organization (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Marrone, 2010);

- *Obtaining information* from various sources across the organization (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Marrone, 2010; Marrone et al., 2007);
- *Protecting the team* from being overloaded (Faraj & Yan, 2009; Marrone et al., 2007).

Given that most work in boundary spanning literature has focused on categories of information gathering, representation and coordination (Marrone, 2010), there is more to be learned about how teams directly interact with others in order to successfully achieve their goals.

The type of team and the organizational setting in which teams are active may further inform boundary management mechanisms. This aspect of context provides additional opportunity to contribute to a fuller understanding of boundary management. Where empirical studies of boundary management have been conducted, most include new product development teams in high technology industries where the teams' time, function, and responsibilities are specific to and primarily dedicated to that product (Ancona & Caldwell, 1990; Ancona & Caldwell, 1988; Ancona & Caldwell, 1992a; Faraj & Yan, 2009). Our study includes teams comprised of multi-disciplinary members who have other primary responsibilities in the organization and the team's purpose requires them to add functions and responsibilities above and beyond their typical day-to-day activities in healthcare organizations. The closest empirical representation we found to this was in consulting teams in educational settings in two studies (Ancona, 1990; Marrone et al., 2007), but only one utilized organizational teams (Ancona, 1990), while the other utilized MBA student teams (Marrone et al., 2007). We submit our team context

is underrepresented in the literature as many organizations utilize “cross-functional teams” to accomplish organizational needs and tasks and “have members who have concurrent commitments elsewhere” (Ancona, 1990: 334).

Effect and Results of Team Boundary Management

Boundary management suggests an internal and external balance because teams who do not engage in external activities can become too insular, yet continuous scouting for information can produce lower internal dynamics (Choi, 2002; Druskat & Wheeler, 2003). Our interest lies in the external activities of teams, but we must consider what effect it has on teams. External activity is associated with time spent with outsiders including understanding their needs and experimenting with solutions, frequent communication, and coordination across vertical and horizontal hierarchies (Ancona, 1990; Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a). Teams benefit from such boundary management activities by demonstrating increased team effectiveness, reducing overload on individual members (Marrone et al., 2007) and improving team psychological safety (Faraj & Yan, 2009). Because boundary management means teams cross levels in the organization, others benefit as well, such as increasing knowledge transfer within organizations and improving organizational learning (Marrone et al., 2007) and ingraining shared values across organizational units (Choi, 2002). The result of teams demonstrating increased boundary management activities is higher ratings of performance and enhanced viability (Ancona, 1990; Marrone et al., 2007). The reason is outsiders rate teams higher when teams depend on the outsiders and when they emphasize external interactions in order to diagnose outsiders’ needs and experiment with solutions (Ancona, 1990). What is unclear in boundary management studies is how

teams actively engage outsiders to willingly experiment with solutions. Our study seeks to answer the specific external activities and types of interactions that create coordination and cooperation as well as persuade support.

Boundary management has been established as predictor of positive team performance (Ancona, 1990; Ancona & Caldwell, 1988; Marrone, 2010). Team performance, however, is also influenced by iteratively interpreting environmental meanings, strategizing actions, and executing plans (Burke et al., 2006; Marks et al., 2000) while continuously scanning and engaging various levels of the organization (Druskat & Wheeler, 2003; Joshi et al., 2009; Marrone, 2010) as to accomplish team goals. This boundary management requires a great deal of effort, persistence (Marrone et al., 2007) and adaptability pointing to the interdependent nature of boundary management and adaptation, yet the two have not been studied together. We propose that boundary management behaviors serve as both a trigger for team adaptation as well as the way in which adaptation is functionally exhibited.

Methods

Design

Our research asks: 1) How does team adaptation emerge through boundary management? 2) What are the mechanisms of boundary management that facilitate support from outside the team? Our research study is motivated by these two questions, the lack of empirical studies combining team adaptation and boundary management, and the lack of team studies focused on the context of quality improvement. To answer our questions, we conducted a qualitative study using thematic analysis to reveal interrelationships of adaptation and boundary management that promote external support

and participation resulting in successful projects. A qualitative study founded on thematic analysis was deemed appropriate as we wanted to “discover rather than test variables” or conditions (Corbin & Strauss, 2008). This method offers a way of seeing or making sense of seemingly unrelated material by analyzing and systematically observing interactions (Boyatzis, 1998). A study examining team adaptation through boundary management may be considered on the continuum between nascent and intermediate theory as we attempt to answer how and why these constructs are interrelated for which qualitative data are appropriate (Edmondson & Mcmanus, 2007). Thematic analysis was selected, because it is a flexible tool with the potential to provide “a rich and detailed, yet complex, account of data” (Braun & Clarke, 2006: 78). Because this study drew upon two prior related studies (Grooms & Lingham, 2013; Grooms & Lyytinen, 2014) which indicated that boundary spanning served as an input to positive adaptation, we used theoretical thematic analysis coding to address the research questions (Braun & Clarke, 2006).

Sample

Our sample consists of interview data from 13 informants describing 15 different projects from 15 different hospitals. We sought informants who served as team leaders or other types of facilitators (e.g. project managers, quality consultants) because we wanted individuals who would be well informed about and could provide a description of team activities from the individual and the group perspective. Participants were drawn from hospitals in urban and suburban settings in the Midwest and were sourced through personal contacts from the interviewer’s current and previous working relationships in healthcare.

The 15 project teams represented in our findings include a common variety of hospital-based types of improvement initiatives including processes, clinical outcomes, patient / family experience, and growth. Our sample of 15 project teams is considered sufficient for analysis as it provides a mixture of high and low performing teams and aligns with Onwuegbuzie and Collins (2007) who recommend a sample size of 12 for semi-structured interviews, as conducted in our study. There was overlap in three project types – fall reduction, throughput, and growth – but each was conducted in different teams and hospitals. One project spanned an entire hospital system. There was also variation in team membership demographics – all were multi-disciplinary and 13 of 15 had a mix of men and women (two teams were comprised of all women). Team informants had roles as team leaders or facilitators and were generally either managers or project management professionals, except for one physician leader. Lengths of projects spanned six months to two years. There were seven teams that represented high performance, four that were intermediate, and four that were low performing. Teams were divided into these groups based on results as compared to their starting point and/or their adjusted project aims as well as the informants' perception of success. Table 11 provides an overview of each of these components.

Table 11: Project Summaries

Project Name	Performance at Start of Project	Resulting Performance	Length of Project	# Team Members	Composition of Team	Composition of Women, Men (General Percentages Given)	Improvement Methodology	Interview Participant Position / QI Credentials	High / Intermediate / Low Performance
Clinic Development and Growth	26%	(Desired direction down) 0.3%	9 months	10 to 12	Surgeons, Lab, Pharmacy, Physician Office staff	75% women, 25% men	Not very specific. Depended more on change management techniques than improvement methodology	Director of Clinic	High
Surgical Patient Continuum of Care Cycle Time	7.12 days	(Desired direction down) 5 days	6 months	11	Schedulers, nurses, midlevels (no surgeons). Champion of the team was Chair of Cardiac Surgery.	50% women, 50% men	Lean	Associate Director of Executive Education; Lean Six Sigma Black Belt	High
Division Growth Strategy Initiative	In previous years had grown at 10% rate, more recently 1-2%	Turned into a support plan - they needed to have an adequate support plan in place for the growth to occur.	11 months (to date)	7	VP of Division, 6 Directors or Managers responsible for various departments	40% women, 60% men	Established value proposition then looked at the critical success factors necessary to achieve that	Vice President for Division	High
Emergency Department Flow (Hospital 1)	375 minutes	(Desired direction down) 237 minutes	6 months	20	Emergency Department (ED), Radiology, Inpatient Unit nurses, Physicians, Hospitalists, Chief of Staff from ED, Nurses from ED.	60% women, 40% men	Lean - Kaizen	Corporate Director for Performance Excellence; Master Black Belt	High
Complication Rate Reduction	Originally planned as length of stay reduction but changed to complication rate reduction. Upon project plan change, baseline was 17-18 cases between surgical intervention need	(Desired direction up) 150+ cases between surgical intervention need	12 months	8 in core group. Up to 25 with ad hoc contribution.	Physicians, Nursing, Nutrition	50% each	Small tests of change	Six Sigma Black Belt	High

Charge Capture Improvement	Originally planned to improve completed and submitted charge tickets; Changed measure to days to charge capture = 6 days	(Desired direction down) 2 days	11 months (to date)	4	Grants, Operations, Finance, IT (mostly senior leaders - Directors and VPs)	25% women, 75% men	Process mapping. No other clear improvement methodology.	Director	High
Pain (PCA) Pump Activation	9-10/12 patients each month with 10/10 pain	(Desired Direction Down) 2 patients in 8 months with 10/10 pain	5+ months	6	5 staff nurses +Clinical Director	All Women	No specific methods	Clinical Director	High (their perspective - not as clear through objective information)
Fall Reduction (Single Hospital)	Did not provide exact baseline. Reported no significant change by end of project.	(Desired Direction Down) 3.7 falls /1000 patient days	~6 months	12 to 15	Nurses (including bedside staff), Physical Therapists, Educators, Leadership	80% women, 20% men	Model for Improvement - specifically Plan-Do-Study-Act cycles, Control Charts	Chief Nursing Officer	Intermediate
Patient Throughput and Capacity Improvement	Admission: Order to Bed in 165 minutes Discharges: 11-12% discharged before noon	(Desired Direction Up) Admission (measure changed): % of Order to Bed in 60 minutes = 6% Discharges: 21% discharged before noon	12 months (to date)	6-8 Steering plus put together additional specialty teams of 10-12 members	Various clinical and administrative staff	Unable to define due to too many subspecialty teams included	Lean - Value Stream Mapping	Lean Leader	Intermediate
Transfer Reduction	116 for the year	(Desired Direction Down) Approximately 5 per month (Most of gain was through "forcing" one physician to change practice)	1.5 years	7	Department head physician, Service Line Director, Clinical Manager, Lactation, 2 nursing staff members, improvement specialist	All women	Research to establish guidelines. No other specific improvement methods employed	Director of (specific area) Clinical and all Surgical Services.	Intermediate
Readmission Reduction	5.7 / 1000 births	(Desired Direction Down) 4 / 1000 births	6 months	7	Physician leader, Nurse manager, 2 Charge nurses, 2 Physician practice support nurses, Clinical operations person	85% women, 15% men	Model for Improvement	Physician / Medical Director	Intermediate (Approached goal only by not including holidays in data. Leader not satisfied with result.)
Emergency Department Flow (Hospital 2)	Did not state baseline but noted no change from baseline occurred	(Desired Direction Down) 335-347 minutes	10 months, ongoing	Number not provided	ED Chief, C-Suite, ED Director	Not specified	None used. Were guided to use methods, but team rejected it.	Corporate Director for Performance Excellence; Master Black Belt	Low

Fall Reduction - System-wide	Did not want to give measure	Rate is above 'where it should be'	1.5-2 years (interviewees term - team had been going on longer)	20	Physical Therapy, Occupational Therapy, Housekeeping, Maintenance, Nursing, Transportation, Patient Safety	66% women, 33% men	Inspection	Quality Improvement Manager	Low
Operating Room Efficiency and Throughput	<50% operating room utilization 50% First case On-Time Start 50% employee engagement	(Desired Direction Down) Reports metrics have not budged since starting	1 year (to date)	6	Surgery Chair, Director of Operating Room (OR), 2 OR managers, 2 Clinical Coordinators	66% women, 33% men	Kaizen - Lean. Small tests of change	Director of (specific area) Clinical and all Surgical Services	Low
Patient / Family Experience	Does not recall exact numbers.	Doesn't recall exact numbers but noted it met the 'failure' definition in the organization (less than 75% of goal)	2 years	8	2 Physician leads, 2 nurse leads. Staff nurses for remainder.	80% women, 20% men	Model for Improvement	Quality Improvement Consultant	Low

Data Collection

Data collection occurred from August 2014 to September 2014. Semi-structured interviews were utilized to obtain rich, lived experiences. Each interview lasted approximately 60 minutes and was guided by an interview protocol (Appendix E). The majority of interviews occurred face-to-face, except in instances of geographical barriers, in which case phone interviews were employed. The interviews were conducted in the environment of choice for the participant. All interviews were recorded with a digital recording device. The participants were advised of the recording and they were also assured of the confidentiality of the interview. They were offered the right to stop the interview at any time.

In order to understand adaptation and boundary management, we asked our informants to describe a single project – success or failure – of their choice. All were told we were *“interested in how the quality improvement project team engaged people outside the team to enhance project performance. [We are] particularly interested in three dimensions of team boundary management — who (roles, names are not needed) was identified as meaningful to the team’s ability to succeed, why they were meaningful, and how you engaged them in your efforts, or if you did not engage them, why.”* Informants were first allowed to provide an overarching description of their project. The interviewer then followed up with probing and clarifying questions to reveal who teams reached out to beyond the team and why; what ways and means they utilized to interact with boundary members; and what changes they found themselves making during the project.

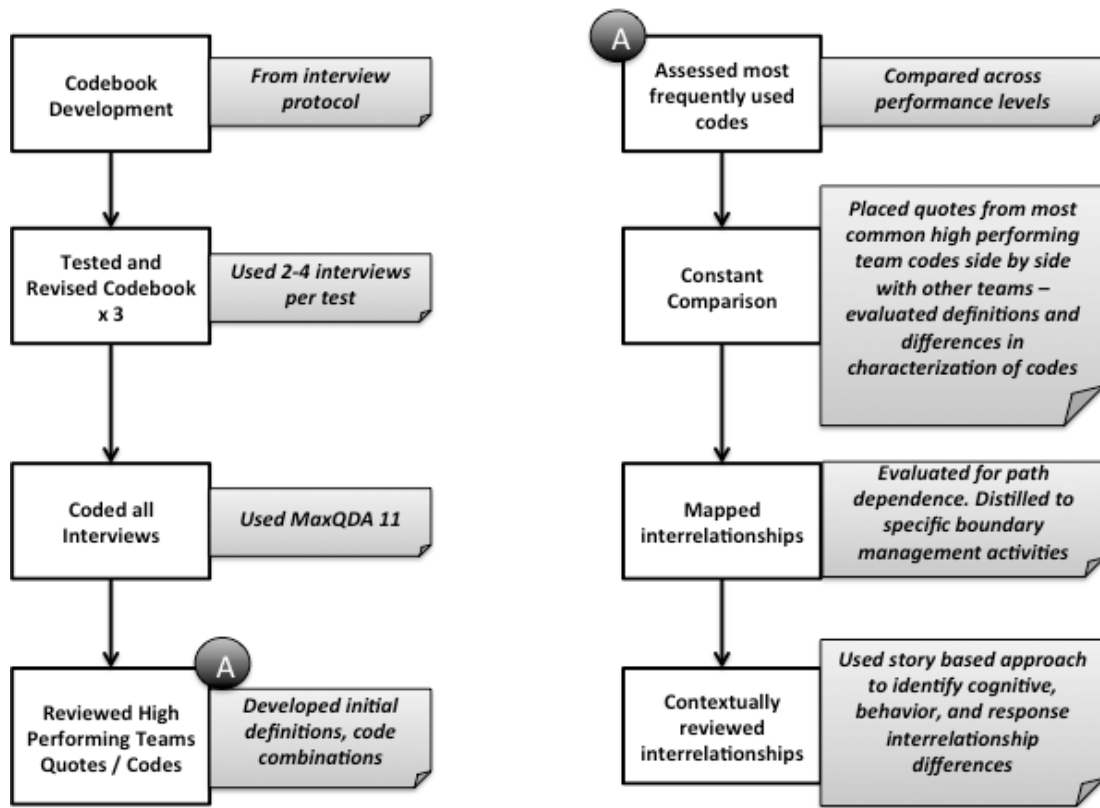
The recordings were provided to a single reputable, professional transcription service for anonymized, verbatim transcription resulting in 256 pages of transcripts.

Transcripts were validated for accuracy and reliability in two steps. The first was through the transcriptionist highlighting, in the transcript, unfamiliar terms and/or phrases or those that were not clearly understood. The second validation step was through the researcher, who also conducted the interviews, listening to the recordings to fill in gaps highlighted by the transcriptionist as well as comparing the transcript text for accuracy upon receipt of the transcription. Furthermore, due to the interviewer's professional knowledge of healthcare-based quality improvement methods and project management, the interviewer asked informants to explicitly describe various terms, relationships, and interactions to minimize assumptions about the project or methods used based on professional experience.

Data Analysis

We conducted a thematic analysis utilizing a codebook, which was developed prior to in-depth analysis (Fereday & Muir-Cochrane, 2008). The codebook or template was generated from the interview protocol, which was drawn on academic literature, the researcher's own experience, and exploratory research (King, Cassell, & Symon, 2004) and from two earlier studies (Grooms & Lingham, 2013; Grooms & Lyytinen, 2014). Figure 9 summarizes the analysis steps utilized beginning with the codebook development.

Figure 9: Data Analysis Map



The codebook was tested and revised three times before applying to all interviews (Initial and Final Codebooks, with notes about sources for codes, are available for review in Appendix F). After completing the coding, we closely examined all codes for the High Performing teams and continued to refine definitions as well as combine codes where categories overlapped. We also assessed the most frequently assigned codes across the different teams to begin a distillation process of the most common categories within themes. Once the most common categories were identified, the quotes were compared across all High, Intermediate, and Low performing teams to determine if the characteristics for the codes differed and to assess whether the Intermediate group was more similar to the high or low performing groups.

After code definitions were established, we conducted system dynamic modeling (Forrester, 1987; Luna-Reyes & Andersen, 2003; Wolstenholme & Coyle, 1983) for each interview and the selected categories to assess how categories and themes were interrelated.

Examples of the dynamic modeling conducted are available in Appendix G. Because we could not determine repetitive path combinations that were clearly always present or never present, we reviewed the interviews in a story format to assess how the themes were interrelated and how adaptation was revealed. Probing questions led to parts of the story being broken apart and supplemented with detail as the interview progressed. By reviewing the overarching description and the later probing questions, while utilizing the primary themes that emerged from our refined thematic analysis (Braun & Clarke, 2006; Fereday & Muir-Cochrane, 2008; King et al., 2004), team adaptation through interrelated boundary management mechanisms and the progressive development of engagement among high performing teams were revealed. These were then compared to intermediate and low performing teams revealing that characteristics of theme categories did not remain the same from high to low. Interviews were finally reviewed as stories and codified for unique instances of categories – rather than inadvertently treating them as multiple instances because details were added later in the interview – enabling us to assess adaptation as a process through recursive boundary management activities occurring in response to the extent of engagement or resistance encountered.

Findings

Cycle of Adaptation and Boundary Management

This study examined how team adaptation occurs in quality improvement projects through boundary management. We focused via thematic analysis on key themes and mechanisms of boundary management and how they influence team adaptation. Before describing the interrelated nature of adaptation and boundary management, we first provide context through descriptive characteristics of the theme categories used to analyze project cases. We embarked on allowing for three levels of team performance based on the extent to which they objectively and subjectively (per the informant) met their goals. The analysis was also used to determine if the intermediate teams more appropriately aligned with high performing teams or low performing teams. Intermediate team resulting definitions aligned directly with low performing teams and for that reason, in Table 12, we present definitions and associated evidence for theme categories for High and Low performing teams.

Table 12: Theme Category Definitions and Evidence

Theme / Category	Performance Level	Definition	Evidence
Structure (Who):	High and Low	Includes any person or group the team chooses to interact with for the purpose of advancing the project	• “Colleagues” of team members • “I had to go to the order set girl, bring her to the table” • “Another boundary was administration”
Cognition (Why): <u>Inclusiveness</u>	High	Large number of groups should be aware of or invited to the project. Awareness many groups need to know about or be included in the team’s pursuits to avoid failure and/or improve communication	• “I always believe that it’s better to have everybody at the table than not at the table because if somebody feels slighted, I don’t want them to...I don’t want the people shooting me in the foot” • “....because if you’re talking about growing [SERVICE], you’re impacting all these other areas and you’re gonna completely fail unless you include those people in the discussion”.
	Low	Inclusiveness not utilized	
Cognition (Why): <u>Competence</u>	High	Inclusion of knowledge holders from whom team can learn and/or add credibility to enhance the project. It involves a search outside the team for expertise from knowledge holders or the credibility to influence others, particularly when tapping into Power sources.	• <i>Knowledge Holders</i>: “....a problem was identified or understood that was not on our radar before in our discussion with her and then we tackled those, whoever it needed to be – the lower we got, [the] easier to fix.” • <i>Power sources</i>: “....in discussing with the patient access manager, we learned the minute details of working in their world, what kind of things that happened and then how things can go wrong.
	Low	Usefulness of boundary member is location driven and task-oriented.	• The team sought “....individuals [who] are cross-trained across the entire organization. So that made a good fit as well, because we could plug those individuals in to help with our pilot” • Nurses were directed by the team “....to change their practice, and they actually then started to keep [the product] on hand at all times for these weekend or holiday readmissions that we were trying to prevent.”
Behavior (How): <u>Use of Evidence</u>	High	Mindful and rich use of data, literature, and outcomes information. Teams made efforts to provide the type of evidence that would be most meaningful to the groups or representatives identified.	• <i>Literature-based support</i>: “I always brought the literature with me because I’m not special or different, why would they listen to me. So I always had best practice guidelines to back it up, Cochrane reviews, things like that. I said I’m not asking you to do something that’s not been tried and true, and by any means, you can always [do your usual procedure], but you can’t take [the procedure] back and the complications that ensue.” • <i>Data – actual vs. desired</i>: “So what we had to do, in communicating with the surgeons to make changes, was to collect data, trend the data, and display the data in order to validate why we were doing what we were doing”.

			• <i>Data – new process performance</i>: “What we did was we kept showing them the results. Especially after we got an outcome result associated with the change in the protocol.”
	Low	Low touch push of audits, data posting, information created by the team.	• <i>Audits</i> – “They like their paper. So they sorta used the [NAME] scale... They just kept saying we don't know how to connect with the staff until we get a piece of paper”. • <i>Results sharing</i> – “[Interviewer asked if they discussed results with units] No, we sent it out, but they were used to getting the reports sent out. And so we would review it in our meeting, and then it would get distributed to the individual units for them to look at. It was grid. Here's all the units, and here's how they're all doing on everything, and so they could see it was all public, transparent.”
Behavior: (How): <u>Partnership</u>	High	Acting on beliefs that feedback and collaboration are valuable from boundary members. Team actively seeks input from boundary members about how they want to see the processes or practices change	• <i>Systems feedback</i> – “We've been charged with doing it. We've made the initial efforts. We've talked to our key stakeholders outside of the department. What we wanna do is give you a framework of what we thought was important, but really turn it over to you guys and you come back to us and tell us where we missed the boat and what we need to delete and what you want us to add to it.” • <i>Testing feedback</i> – “what worked, what didn't work...Is there anything that we can tweak? If we're gonna tweak something, who do we need to tell that we're gonna tweak this particular task, and then continue, and then get some more results...”.
	Low	Strategy or test of change implementation site selected, imposed by team	• <i>Testing push</i> – “....who could we plug in to ensure that we carry it out in the way we need to carry it out. And we used the staff that were cross-trained across the organization...” • <i>Implementation push</i> – “Most other people were doing our work; meaning the clinical testing, getting the blanket, doing the lab follow-up. So we didn't work on that; we never did the actual groundwork.”
Behavior (How): <u>Process / Systems Management</u>	High	Process / Systems Clarity Seeking: Deference to outside expertise and knowledge exchange between team and boundary. Teams seek expertise from process owners regarding the current system and exchange knowledge and information between the team and the boundary to more clearly understand how changes could impact the system.	• <i>Deference to expertise</i> – “....they mention well, this happens and that happens. So understanding the process from people who are in the trenches was very critical” • <i>Knowledge Exchange</i> – “Once they had time to have some discussion and sort of play catch with options and concerns with them, then the second step was to go in to a second staff meeting where there was representatives from all groups then. So there was some more conversation... about well, we do it this way at our two hospitals, another group would do it a little bit different at their hospital, talk through the logistics of trying to make that change work for everybody...”
	Low	Process / Systems Implementation: Plans identified by team and presented as plan to implement, adopt	• Making “contact and explaining the intervention; what was gonna change; why it was gonna change, showing them how to do that. And then also we ran – we rolled it out and said here's the change...”

Behavior (How): <u>Power</u>	High	Inclusion of leaders who can lend credibility to, reinforce importance of project	• “...the chief nursing officer and the administrator...[brought] all the nurses together and walk[ed] through what we were doing, how we were doing it, and then basically explain[ed] and manage[d] those relationships with the nurses so that the nurses would continue to engage in the improvement activity;”
	Low	Leadership intervention to alter course of project	• “...we did have to do some contract addendums [from Leadership Office] with some of the penalty criteria that we put in there.” • [Because of a lack of accountability] “a lot of that pressure needs to come from our department of surgery chair.”
Engagement	High and Low	Boundary member willingness (verbally articulated or demonstrated through actions) to participate as requested by a team	• When the Chief Nursing Officer asked people to support the project it made “them puff our their chest and say, yeah I’m ready to go with this...” • “They understood things were coming. They would even say oh, that’s because of that [departmental] Kaizen they just had. They were prepared to look for changes...”
Resistance	High and Low	Boundary member lack of willingness (verbally articulated or demonstrated through actions) to participate as requested by a team	• “So there were some really famous doctors, they wouldn’t change their way of doing things.” • “...as they started looking more deeply into that protocol as to how much different it was than what they were used to do, I think that’s when a lot of the questions started happening. So the first indication of that was that people just weren’t complying.” • “I think the area that we still struggle with and we’re still trying to figure out how to embrace are the physicians ‘cause they are a group that just – they don’t see this as their responsibility.”

In summary, the table represents how Boundary Management Structure, Cognition, Behaviors, and responses to the teams (Engagement, Resistance) are defined. Who is important to the team's success provides structure for connections that need to be made as best exemplified by one team recognizing "...*this team...is the core team, but as you can see we're gonna have to go outside these walls or these boundaries and involve the people that you support in the organization doing and managing the research, so basically your customer.*" Inextricably linked with the structural aspect of identifying who from among the boundary is meaningful to the team, is the cognitive aspect of understanding and/or declaring why those groups or individuals influence team activities or outcomes. To actively engage boundary members in project efforts, teams use a variety of ways and means to interact with their boundaries. We defined means as the technology or vehicle for interactions and ways as the actual interaction behaviors. Most salient to this study is the arrays of ways teams engage boundary members identified for their varying reasons as noted in Table 12. First, we present the most common means utilized by teams in table 13, then we will describe the ways teams interacted with boundaries, through boundary management and adaptation, to create a willingness to participate in an issue of the project.

Table 13: Means of Boundary Management

HOW - MEANS		
Means (Top 3)	Frequency of Examples and Number of Teams utilizing example out of total in performance group	Examples of Means in Use
Meetings	• <u>High Performing Groups</u>: 16 examples from 7 of 7 groups • <u>Intermediate Performing Groups</u>: 6 examples from 2 of 4 groups • <u>Low Performing Groups</u>: 5 examples from 3 of 4 groups	• <u>High Performing</u>: Invitation to boundary members to attend project team meetings to exchange information; Presentation of project information in staff meetings or inservices • <u>Intermediate and Low Performing</u>: Use of "Summit", "Council" and Staff or Unit meetings to present information about the project
Improvement Methods	• <u>High Performing Groups</u>: 18 examples from 6 of 7 teams • <u>Intermediate Performing Groups</u>: 3 examples from 2 of 4 teams • <u>Low Performing Groups</u>: 5 examples from 3 of 4 teams	• <u>High Performing</u>: Variety of models and methods employed among teams including Kaizen events, Flow-charting, Value-Stream Mapping, Time Studies, Process Mapping, Small Tests of Change (e.g. Plan-Do-Study-Act), Brainstorming, Project Management based on Critical Success Factors, Lean, Control Charts • <u>Intermediate Performing</u>: Kaizen event, Value-Stream Analysis, Audits for compliance, Plan-Do-Study-Act cycles; Control Charts • <u>Low Performing</u>: Brainstorming, Small tests of change, Key Driver Diagram, Run Charts, and Inspection
Documents	• <u>High Performing Groups</u>: 13 examples from 4 of 7 teams • <u>Intermediate Performing Groups</u>: 2 examples from 2 of 4 teams • <u>Low Performing Groups</u>: 2 0 examples	• <u>High Performing Groups</u>: Process maps, educational documents, performance comparisons in charts, planned tests on poster boards, invitations, contracts, RAIL (rolling action item list) document • <u>Intermediate Performing Groups</u>: Poster board presentations, audits

Models of Boundary Management and Adaptation

Though we initially divided teams into three groups – High, Intermediate, and Low performing, we found through our analyses that Intermediate and Low performing teams' boundary management cognition and behaviors were highly similar in their characteristics – they had more in common with one another than with high performing teams. They also exhibited similar frequency and diversity of boundary management mechanisms utilized. Initial coding frequencies and distributions were not utilized to

develop models of Adaptation through Boundary Management as those codes were many times, used repetitively for the same unique event that was described in more detail later in the interview due to probing questions. Table 14 provides the final distribution results of unique events of our Boundary Management Behavior themes and categories of interest as well as occurrences of Engagement and Resistance Responses. Because Intermediate and Low Performing teams depended solely on Competence for Boundary Management Cognition, we have not included counts and distribution of Cognition categories. We do however, note that six of seven high performing teams used both Inclusiveness and Competence and the one team who used only one cognitive mechanism utilized competence for the purpose of connecting with knowledge holders outside the team.

Table 14: Frequency Distribution of Boundary Management Behaviors and Responses

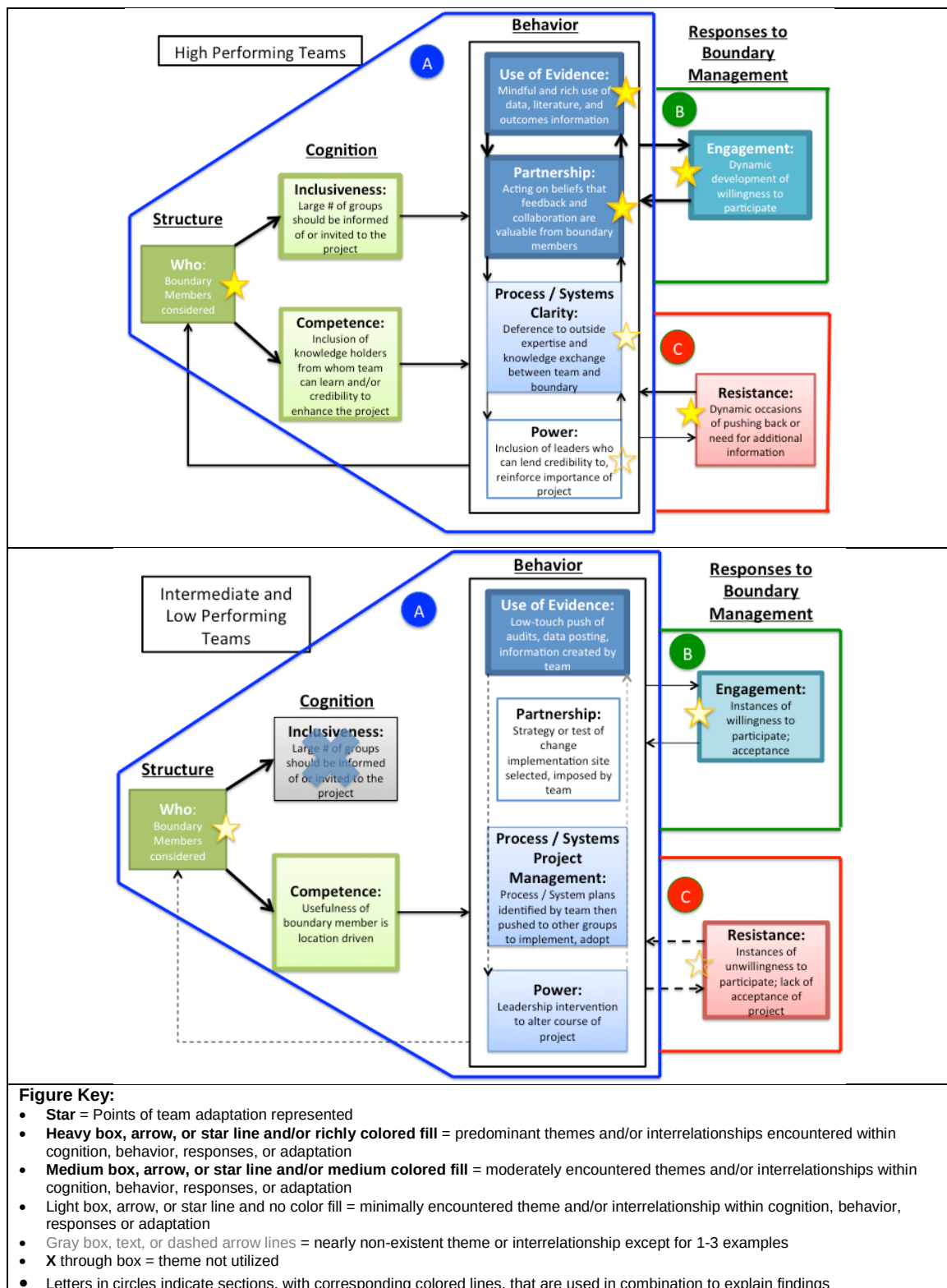
	Boundary Management Behaviors				Responses	
	Use of Evidence	Partnership	Process / Systems	Power	Engagement	Resistance
<i>High Performing Teams</i>						
# of Teams Represented (of 7)	7	7	7	5	7	4
Sum of Unique Events* (Total = 89)	19	19	13	7	22	9
% Distribution	21%	21%	15%	8%	25%	10%
<i>Low and Intermediate Performing Teams (Combined)</i>						
# of Teams Utilizing (of 8)	8	4	8	6	7 (weak)	8
Sum of Unique Events* (Total = 80.5)	21	4	14	11	10.5	20
% Distribution	26%	5%	17%	14%	13%	25%

* Where a portion of a point exists, credit was given for partial engagement achieved and reported.

Teams clearly demonstrated a process of adapting and employing boundary management elements of structure, cognition, and behaviors. There were, however, no

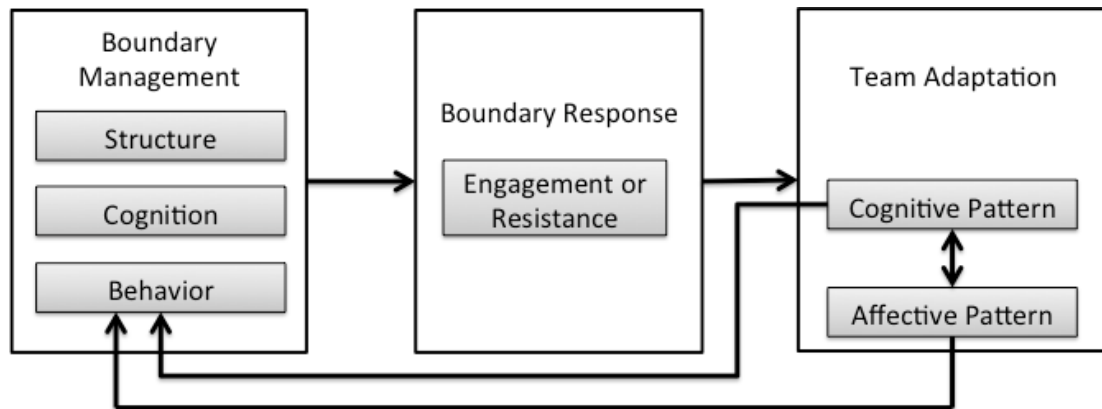
distinct consistent paths across the elements that could be drawn as a dynamic process of how this happened except to demonstrate that Boundary Management Structure, Cognition, and Behavior elements were consistently used in combination resulting in specific adaptive responses. We saw a variety of combined behaviors, especially among high performing teams demonstrating adaptation in approach as well as in response to the process of engagement or mitigating resistance. As seen in the distribution table, some boundary management behaviors were utilized more than others among each group. By analyzing the interviews for unique examples of the various conditions of Boundary Management and Responses, we found low performing groups utilized *fewer combinations of behaviors and fewer mitigating strategies* for resistance indicating less diversity in adaptation in their overall approaches. We provide the models, in Figure 10, below to distinguish the differences among the groups.

Figure 10: Dynamic Models of Boundary Management, Responses, and Adaptation



As we further analyzed the data to understand points of adaptation, we began to see patterns of cognition and affect among boundary responses. By patterns we mean the team reflected on and intellectually and emotionally responded as a configurable and meaningful whole to the feedback they received from the boundary. There was no difference whether it was engagement or resistance at the boundary. These responses generally occurred together – serially or in parallel, both styles were present – and gave rise to a set of choices for actions. The patterns reflected and drew upon available boundary management elements. At the same time we conjectured that adaptation was a state of cognitive acknowledgement of who was the salient element and why particular boundary members could assist the team in developing engagement. In a sense, the pattern embodied a belief system about the boundary’s responses and how the team should proceed to interact with them. This belief system was put into action through boundary management behaviors. In this way, a feedback system was erected over time of when to initiate boundary management as to move the project forward in the field (i.e. change the boundary), the boundary’s response of engagement or resistance, and team adaptation driven by cognitive and affective patterns, then led back to revise boundary management behaviors as to further develop engagement. This common pattern is depicted in Figure 11 and explored through detailed evidence in the sections that follow.

Figure 11: Process of Team's Boundary Adaptation



Boundary Management Mechanisms

We start with examples demonstrating that boundary management mechanisms are consistently present in QI teams, but the quality of how the mechanisms are used differs. The teams combined structure, cognition, and behaviors by identifying with whom they needed or wanted to interact, the reason(s) why these people were important to them, and how they would interact with them. The types of structures, cognitions, and behaviors found are represented in Figure 10 models' solid blue line hexagon labeled A. A representative high performing team, working on growth, demonstrates the interconnected nature of Boundary Management in the following story.

The group specifically identified other groups and representatives that would be meaningful (**Structure: Who**). *"We're gonna have to go and once in a while include a person from information technology. We're gonna once in a while have to include a person from the development office. We're gonna have to include legal and finance and kind of actually broaden the scope because if you're talking about [growth], you're impacting all these other areas and you're gonna completely fail unless you include those people in the discussion."* **Structure** and **Cognition** are inextricably linked as they

demonstrated awareness about why these boundary members were important through **Inclusiveness (Cognition)** as seen through “*impacting all these other areas*” and “*you’ll fail unless you include*” them. The primary **Behavior** employed with these structural representatives was **Partnership** as they invited people to meetings to ask for insights and feedback on what they were trying to pursue asking, “*...can you give us your perspective of how this would impact you in your role if we were successful in what we’re proposing here? And this just completely changed the whole dynamic of the thing.*”

Through their active boundary management, they also realized they had been putting together critical success factors, but they hadn’t connected with the people who do that work everyday (**Structure: Who**). An ad hoc member of the team asked them, demonstrating **Inclusiveness (Cognition)**, “*What if you rolled out these critical success factors and the people on the front lines didn’t agree with them...?*” The team then went group by group in the division to present the ideas to date and said “*We’ve been charged with [increasing growth]. We’ve made the initial efforts. We’ve talked to our key stakeholders outside of the department. What we wanna do is give you a framework of what we thought was important, but really turn it over to you guys and you come back to us and tell us where we missed the boat and what we need to delete and what you want us to add to it*” (**Behavior: Partnership and System Clarity**). The interest in hearing where the team’s plans might have gaps represents seeking **Competence (Cognition)** from the boundary members, their actions demonstrated behaviors meant to learn from the boundary and to bring them into partnership. The team used this as an opportunity to develop “*next generation leaders*” by asking for insights and guidance on the Critical Success Factors plan demonstrating **Partnership (Behavior)** with this new group. “We

created sort of the next generation of leaders in our own department, and put them together in a group that we called an affinity group for the growth plan. And we said you guys are now in charge of this for the next quarter or two...” The team sought additional **Process and System Clarity (Behavior)** by deferring to the ‘affinity’ group’s expertise and exchanging knowledge between groups – *“And I charged them in return with looking at...the action steps and the critical success factors, revising, adding, deleting any of those. And they tried generally to in some sense order the tasks at least to which ones they knew had to come before other ones... So they did all that.” “...they were still actively involved, still keeping me on a time line, still providing me with the right way to give progress reports to the president of the hospital... We fine-tuned it with them and then we rolled out sort of a final plan.”*

Low performing groups, on the other hand, differed in boundary management characteristics as represented in the Intermediate / Low performing team model’s solid blue line hexagon labeled A, Figure 10. For example, surgeons were identified as a critical group in an access and efficiency project, but the team had difficulty engaging them in change efforts. *“How do we have that accountability for the surgeons (Structure: Who) to say if your case is at 7:30, we also need to be in the room at 7:30? And a lot of that pressure (Behavior: Power) needs to come from our department of surgery chair (Cognition: Competence that is location driven).* The team also looked at an additional process, the “block schedule” they considered *“low hanging fruit...that [would] get immediate impact by implementing a policy change (Behavior: Process / System Push).* They knew conversations about this with several physicians would be difficult and even reported *“if you hear explosions happening on the east side of town,*

you'll know that was the day we had the discussions." At no point in that story was there an indication of considering alternative approaches in order encourage positive responses or engagement from the boundary. As in this example, low performing teams use Power as a leadership intervention **(Behavior)** to push a course alteration on others. Process and systems considerations are generally about creating plans for implementation rather than through deference to expertise and knowledge exchange. In this case, there was a plan to develop a policy to be approved by another committee then applied to the surgeons. They expected a negative response, but did not have a plan to adapt their approach.

Team Adaptation to Boundary Responses and Enacted Boundary Management Behaviors

High performing teams dynamically adapted in order to progressively engage groups outside the team they deemed important to the project's success. In Figure 10, we see engagement influences adaptation as high performing teams demonstrate cognitive and affective patterns of considering how to respond prior to and as they take action to develop engagement. As they move aspects of the project into the field, they encounter engagement and resistance from the boundary, stimulating adaptation patterns reflecting and leading to associated boundary management behaviors evocative of a feedback system. Because adaptation patterns and boundary management mechanisms are closely linked – patterns lead to action – we explored this feedback system as a whole. For orientation to the patterns, a table with associated labels and definitions is offered.

Table 15: Cognitive and Affective Patterns of Adaptation

Patterns of Adaptation Informed by and Promoting Boundary Management Mechanisms	
Cognitive	Affective
Who: Understanding of who comprises the boundary	Evidence-based dialogue: A desire to create mindful dialogue as part of understanding the boundary's real or potential responses.
Inclusiveness: Multiple targeted groups and individuals are considered important and valuable for interactions related to the project	Partnership Pull: Feelings of understanding or empathy for real or potential responses and values of boundary members sought for participation
Competence: Expertise and credibility beyond the team can contribute to the team's success	System Conscientiousness: Feelings or beliefs that 1) impact on the boundary from system changes need to be considered and/or 2) people beyond the team have the best information regarding current systems and potential impacts from changes.
System Clarity: Includes consideration for knowledge exchange and deference to expertise beyond the team, which can inform and advance the team's project.	

In a representative example from a high performing Throughput Efficiency project team, one of the structural groups needed ongoing interactions and responses from the team in order to enhance engagement. This example makes evident the ongoing nature of developing engagement by employing adaptation patterns and enacting boundary management behaviors according to the structure's needs.

In this high performing team, we see a starting point where the structural, cognitive and behavior aspects of boundary management initiate interactions. The primary group of interest was surgeons (**Structure – Who**), because the patient group for whom the team was trying to change process were surgical patients. They also identified the need to incorporate other physicians who managed the *continuum for aftercare* (or after surgery) to get their input on strategies (**Structure – Who; Cognition – Competence**). The team then examined current high level process steps to better understand boundary member activities (**Behavior – System Clarity**). To mitigate

resistance and progressively develop engagement with these boundary members, we saw adaptation patterns of cognition and affect, which guided their behaviors and interactions.

When asked how the physicians responded to potential changes in their processes, the informant reported there was some resistance and boundary members asked for data. The team recognized they could “*learn a lot about the culture of the organization by those reactions*” (**Affective Adaptation: Partnership Pull**) and that they would need to “*engage the physicians in a purposeful manner*” according to their data requests (**Affective Adaptation: Evidence-based dialogue**). In response, the team “*had to collect data and do historical data collection and then in-process data collection and posting of that data*” (**Behavior – Use of Evidence**). The physicians were still skeptical of the underlying purpose for the project so the team sought input from the chair of the department (**Cognitive Adaptation – Competence**) to “*determine what were the benefits of doing this project, and not just reduction. We know that reduction is good, but why do we want to have a reduction in the average length of stay? And it was better outcomes.*” The team needed to understand and develop a connection to what the physicians valued in order to provide the information the physicians needed with consideration to their values (**Affective Adaptation – Partnership Pull**). The team presented predictions of how process changes would result in capacity for an additional “*35 people that you could see, help, and touch and heal, [which] really resonated with those outside the team*” (**Engagement**). So it wasn’t just an arbitrary, ‘*Let’s just do more, be more,*’ it went back to the altruistic aspect of working in a non-profit hospital” (**Affective Adaptation – Partnership Pull**). Through a moment of reflection, our informant shared a valuable perspective on how boundary responses to boundary management behaviors influenced

adaptation patterns allowing engagement to develop. “*So there was a lot of managing of relationships and organizational dynamics with this particular project* (**Boundary Management**). *I think that’s something that was outside the scope* (**Recognition of need to adapt**). *You couldn’t just go in, modify the product, collect the data, and make the changes* (**Affective Adaptation – System Conscientiousness**). *There was a ripple effect with the project that required consistent, like, a feedback loop of seeing what the changes were, how they were affecting the dynamic, and then making adjustments to managing the perception of the project and the changes that were happening* (**Cognitive Adaptation – System Clarity; Affective Adaptation – System Conscientiousness; Boundary Management Behaviors according to need**).” Managing boundaries is complex requiring sensitivity to people and systems in order to adapt behaviors that result in better outcomes and engaged boundary members. Not every adaptation pattern is at work in each situation. Here there was an emphasis on feeling what boundary members valued, the importance of mindfulness in evidence-based dialogue, and systems considerations. High performing teams use various combinations of adaptation patterns to guide behaviors.

Low performing teams, however, demonstrated fewer adaptive patterns and less collaborative mechanisms. Boundary management behaviors were present, but few if any discernable considerations for potential or real boundary responses and adaptations as to develop further engagement. A low performing team provided an opposing example related to lack of adaptation or sensitivity to the boundary’s potential for contribution or responses to project implementations. The team determined a primary focus for the project was there simply needed “*to be an organizational-wide commitment*” to the

project's improvement goal. "We had some very specific strategies that we wanted all employees to do, including physicians (**Structure – who**). And so that was one of our strategies was to do hospital-wide education to all people who come on the nursing units about what their role should be in preventing falls (**Behavior – Process / System Project Management**)." These statements suggest a lack of consideration for or sensitivity to potential or real boundary responses to the team's strategies not to mention the lack of interest in expertise that might exist on the topic beyond the team.

The team recognized their lack of success and noted "we could not create any reliability in the system." The reasoning behind this lack of reliability, though, was described as "I don't think we had the nurse managers' support across the organization (**Resistance**). My favorite saying is you inspect what you expect, and it was more, 'oh, I got another one of these reports and we're not doing so well, oh well' (**Behavior – Use of Evidence, audits and push of reports**). And then their directors weren't putting any pressure on them (**Behavior – [desire for] Power**). So we started out with, okay, first of all everybody needs to know how the system is supposed to work, what the expectations are, and all that kinda stuff. So we did all of that, the education. They had to come down, be educated (**Behavior – Process / System Project Management**)...[but] nobody seemed to get the reliability piece of it...people didn't even understand what reliability was. They didn't even understand the concept. And I'm not talking about just staff. I'm talking about managers. I'm talking about department heads. They don't understand that if you don't do it, it 'ain't gonna' work. And so that piece of it was probably the most disappointing...the lack of support – actually, we were told...to stop inspecting (**Resistance**). They didn't wanna know (**Lack of Adaptation**)." This example represents

a push of the team's strategies and plans onto the boundary and when resistance, or lack of follow-through, by the boundary occurred there were no patterns of adaptation. The team simply resorted to the belief that boundary members were at fault for not understanding reliability and they had no desire to know what errors were happening.

In summary, team adaptation is a dynamic process embedded in boundary management activity, which is intended to promote responses of engagement from the boundary that recursively stimulates further adaptation. Boundary management involves inherent interrelationship between structure, cognition, and behaviors within which high performing teams exercise a mixture of these mechanisms. The positive response of engagement develops over time and does not reach a singular endpoint. Finally, resistance happens for all teams, but high performing teams adapt their boundary management to mitigate such behaviors and to ensure progression toward engagement. Also, high performing teams, specifically, were adaptable through sensitivity to their impact on the boundary and the need for boundary engagement. By way of mindful boundary management they explore and practice multiple reasons and actions to obtain confederacy.

Our key findings suggest team adaptation is generated dynamically through cognitive and affective patterns and enacted through boundary management mechanisms when teams mindfully interact with the boundary. Teams adapt in response to the extent to which they are able to engage the boundary where projects are conducted and changes embedded. Through a cognitive pattern, teams are actively aware of who is meaningful to the team, why they will be affected, and what aspects of system functions and consequences need to be uncovered. In the affective pattern teams are sensitive to the

boundary and mindfully consider the type of interactions they desire to have and that will result in a spirit of cooperation from the boundary. These patterns are converted to action through Boundary Management Behaviors resulting in progressive development of positive engagement. In summary, the propositions to support this lens on adaptation include the following.

1. Boundary management mechanisms utilized by project teams elicit a response from the boundary, which can be positive (engagement) or negative (resistance).
2. High performing teams are sensitive to the response and process the response cognitively and/or affectively prior to embarking on the next application or trial of boundary management mechanisms or behaviors, in particular.
3. Low performing teams lack sensitivity to boundary responses and continue to apply behaviors aligned with the team's internal plans, strategies, and/or expectations for the boundary.
4. Team adaptation, in high performing teams, builds a feedback loop that seeks to implement change by interacting with the boundary, the boundary responding, and the team recognizing and feeling what the boundary's needs are prior to enacting new boundary management mechanisms. They are mindful about their interactions in order to achieve boundary engagement, which ultimately assists the team in achieving project goals.

Discussion

We began our study asking how adaptation emerges through boundary management. Furthermore, we were interested in understanding what mechanisms of

boundary management facilitate support from outside the team. We carried out our research through a qualitative study using thematic analysis to reveal interrelationships of adaptation and boundary management to promote external support and successful team performance. Our study showed why and how team adaptation develops is through affective and cognitive patterns expressed through boundary management mechanisms in high performing teams' response to feedback derived from the extent of engagement from the boundary.

QI teams are embedded in the larger organization, promoting the need for adaptation and boundary management as these teams intervene in and respond to a dynamic, complex environment (Ilgen, 1999). Working with people outside the group, though, may have a negative influence on adaptability (Alderfer & Smith, 1982) creating a need to understand how project teams achieve success. Boundary spanning through vertical and horizontal communication, such as in ambassador and task coordinator activities, has been associated with manager ratings of strong team performance (Ancona & Caldwell, 1992a). Additionally, teams that actively engage outsiders and promote the team's achievement within the organization, called probing teams, have been deemed higher performing teams (Ancona, 1990). Our study adds to the understanding of team adaptation and boundary management by demonstrating that adaptation is cognitively and affectively patterned as well as dynamically expressed through mindful boundary management mechanisms in response to boundary engagement.

Team Adaptation is Dynamically Generated

Team adaptation is a dynamic process expressed in mindful boundary management's structural, cognitive, and behavioral mechanisms and through progressive

development of engagement. It does not stand alone, rather adaptation is intertwined with sensitivity to the environment and mindful interactions. This aligns with Burke et al.'s (2006: 1192) premise that adaptation is a “phenomenon that emanates as team members and teams recursively display behavioral processes and draw on and update emergent cognitive states to engage in change.” Our high performing teams demonstrated active changes in reasons to engage particular boundary members and the way in which they interacted with them, which may be reflective of outcomes oriented knowledge work teams who have been found to be more adaptable in work processes (Woolley, 2009). Specifically, high performing teams in our study found it important to engage multiple targeted groups and individuals as well as to utilize the boundary's knowledge and credibility by drawing the boundary into the project through attentive dialogue regarding evidence, partnering with them, and deferring to the boundary's expertise to adequately evaluate current and future systems. Importantly, this engagement developed over the course of the project by continually adapting reasons for interaction and behaviors to which the boundary would positively respond. Low performing teams, alternatively, identified boundary members who were geographically and/or functionally positioned where the team planned to intervene and pushed implementations out to them without the benefit of collaborative behaviors.

Team adaptation is generally conceived as important to team performance (Burke et al., 2006; Kozlowski et al., 2009; LePine, 2003; Woolley, 2009). We found team adaptation, in high performing teams, occurs by employing multiple reasons for and ways of engaging the boundary; however, prior studies and theories are primarily internally focused on activities such as adjusting internal role networks and structures (Kozlowski

et al., 1999; LePine, 2005) and team process improvements (Klein & Pierce, 2001; Wiedow & Konradt, 2011). Burke et al. (2006) also discuss team adaptation from within the team – such as psychological safety, team back-up and coordination behaviors, leadership, and internal communication. Little insight has been offered, however, to how team interactions with the environment facilitate and reveal adaptation. Our study demonstrates boundary management provides feedback from the environment and is a primary group of behaviors emerging from cognitive and affective mechanisms of adaptation. Furthermore, adaptation has been described as cognitive, affective, and behavioral; however, empirical analysis has been lacking about how these elements are represented in teams interacting with their environments. Combinations of cognitive and affective mechanisms of adaptation give rise to specific types of behaviors carried out by teams. For quality improvement teams in healthcare, the overarching behavior that emerges within adaptation is mindful boundary management.

Boundary Management Mechanisms

Boundary management is attentive to structure and concurrently cognizant of who and why boundary members are meaningful to the team and combine this awareness with active interactions that seek engagement from boundary members. The scholarly foundation in boundary spanning has previously revealed processes including interacting with and gaining knowledge from the environment (Ancona, 1990; Druskat & Wheeler, 2003; Marrone, 2010); communication with the vertical hierarchy to both protect the team and obtain resources (Ancona & Caldwell, 1992a; Marrone, 2010); and lateral communication to coordinate issues with other groups (Ancona & Caldwell, 1992a; Ancona & Caldwell, 1992b; Druskat & Wheeler, 2003; Marrone, 2010; Druskat &

Wheeler, 2003; Marrone, 2010) lead to higher performing teams. Such activities have been treated as both an antecedent to team performance (Marrone et al., 2007) and behavioral outcomes of boundary spanning activity (Joshi et al., 2009); however, none have clearly identified the interactions of boundary management mechanisms that facilitate boundary engagement. A study of social capital at the business unit level offers some validation to our findings. Indeed, Tsai and Ghoshal (1998) found three aspects of social capital – 1) Structure which comprises social interactions blurring boundaries, 2) Cognitive or shared vision which embodies collective goals and aspirations of the organization, and 3) Relational or trust which acts as an antecedent to cooperation – in combination had significant effects on the extent of resource exchange associated with product innovation. We found the mechanisms comprising boundary management similarly included a combination of 1) Structure – who among the boundary is meaningful to the team; 2) Cognition – why various boundary members are needed by or will be affected by the team; 3) Behavior – specific interaction behaviors are utilized to actively engage the boundary.

In the past, boundary management has not been treated as an activity that is progressively developed to engage otherwise disinterested parties as most other studies utilize product development teams who have clear paths of coordination with other groups (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Faraj & Yan, 2009; Joshi et al., 2009; Marrone et al., 2007) working on individual aspects of ultimately a shared goal of producing a good or service. Our study covers healthcare quality improvement teams who are tasked with changing processes and practices that other people carry out and who, prior to the project, often have no knowledge that this change is sought and

considered a desirable pursuit. We contend describing the mechanisms of boundary management – structure, cognition, and behavior – similar to social capital, that facilitate engagement promoting team project success contributes an added layer of understanding to boundary management. Boundary management, though, is also a functional expression of team adaptation. Without team adaptation, recursive boundary management that progressively develops engagement does not occur. To examine team adaptation and mindful boundary management as concepts that iteratively influence each other, we examine organizing patterns that are cognitive and affective.

Cognitive and Affective Patterns of Team Adaptation

Common intergroup characteristics, regardless of context, include group boundaries delineating membership but also include: permeability for conducting transactions; affective patterns varying in the degree to which positive or negative feelings about other groups are generated and affect permeability; and cognitive formations influenced by boundaries and affective patterns contributing to the team's explanation of experiences and influences on other groups (Alderfer & Smith, 1982). Teams conducting QI projects pursue the design and implementation of changes that will reside in areas, among people, who may or may not share the same interest in or insight to the need for change. Because of this, teams face a challenge and need to engage boundary members in various activities to build commitment and achieve success (Costa, 2003). We found responsiveness from the boundary activates team adaptation arising from cognitive and affective patterns, in reaction to the extent to which they are able to engage the boundary.

Cognitive and affective patterns in teams have primarily been studied in terms of trust (Greenberg, Greenberg, & Antonucci, 2007; Kanawattanachai & Yoo, 2002; Schaubroeck, Lam, & Peng, 2011; Wilson, Straus, & McEvily, 2006). The extant literature on trust provides unexpected insight to why engagement may or may not occur and the patterns teams then demonstrate to adapt in response to the boundary. Interpersonal trust – between individuals or groups – is closely linked to the willingness to act cooperatively or to engage based on the interactions, words, or decisions of others (Costa, 2003; McAllister, 1995). Cognitive based trust includes beliefs about others' competence, reliability, professionalism, and dependability, whereas, affective based trust includes emotional bonds where there is care and concern for others and rapport based on shared regard for others (Kanawattanachai & Yoo, 2002; McAllister, 1995; Schaubroeck et al., 2011; Wilson et al., 2006). Trust develops, builds, declines and resurfaces (Rousseau, Sitkin, Burt, & Camerer, 1998); likewise, we found engagement develops as teams adapt in response to boundary feedback. Engagement is obtained through functional mechanisms of boundary management, but mindfully managing the boundary to obtain cooperation reveals cognitive and affective adaptive patterns.

We categorize boundary management mechanisms within cognitive patterns of adaption according to knowledge of who comprises the boundary, competence among boundary members, system clarity informing the project through knowledge exchange and deference to expertise, and inclusiveness. Again, we can draw parallels to trust. A study on distrust in the relevant context of implementing a Total Quality Management (TQM) program in an organization revealed task-ambiguous groups considered the managerial control system to be mismatched with the perceived nature of work and

viewed the TQM efforts as “unthinkingly transplanted solutions” (Sitkin & Stickel, 1996). Our low performing teams demonstrated a similar cognitive pattern where there was no effort given to inclusiveness – counter to the multiplex nature of trust crossing organizational levels (Rousseau et al., 1998) – and their interventions were simply pushed to location-based groups to implement. High performing teams, alternatively, broadly identified and included multiple targeted groups and individuals who could offer competence through expertise and credibility to the team, while seeking system clarity from knowledge holders and experts among the boundary to better inform the team’s project plans. They adapted and matched their interactions and interventions to the feedback received.

Affective patterns of adaptation are based on the beliefs that consideration needs to be given to the boundary in regard to the team’s implementation efforts. We saw this expressed through statements indicating consideration for the team’s affect on the boundary as well as in relational sentiments. One team understood their work on efficiency would result in more patient volume, which can sound like more work to others. However, they connected to the boundary by describing the additional “35 people that you could see, help, and touch and heal, really resonated with those outside the team”. Another team described their interactions became “based on a relationship between two people, not on what’s the formal protocol between the departments”.

The word pattern can be taken one step farther, though. We found that consideration for the boundary’s responses and the value the boundary could add was revealed through particular patterns of the 1) desire to partner with boundary members through collaborative behaviors; 2) a desire to enter into mindful evidence-based

dialogue in preparation or response to boundary feedback; 3) beliefs that people beyond the team possess the best information about current systems and potential impacts from changes; and 4) the consideration of and reflection on feedback from the boundary to guide adaptations in how the boundary is managed.

Cognitive and affective patterns of adaptation in conjunction with mechanisms of boundary management dynamically influence each other. High performing teams demonstrated enhanced cognitive and affective patterns as they increased the use of various boundary management mechanisms as revealed by one team leader noting “the only way I was able to be enlightened and realized that [the project] finally started to happen [was] because we took different approaches with people.” This experience is not unlike virtual teams who are challenged to develop trust because of limited direct interaction where trust is developed in the short-term through assessments of ability and integrity and over the long-term by assessments of benevolence and continued evaluation of integrity (Greenberg et al., 2007). We looked to the trust literature to assist understanding and explaining the cognitive and affective patterns of adaptation revealed in our study because the team adaptation literature and the boundary management literature are silent on cognitive and affective adaptive patterns promoting successful team outcomes through boundary management. Our study is offered as new light on this as it describes team adaptation occurring through combined cognitive and affective patterns. It is functionally expressed through boundary management mechanisms leading to boundary member engagement resulting in successful team performance.

Limitations

This study of team adaptation at the boundary combines the two concepts of adaptation and boundary management in new ways. It offers a contribution to the literature by organizing adaptation processes according to cognitive and affective patterns. We do, however, acknowledge limitations in this study. For one, as a qualitative study it provides a rich description of themes and how adaptation, boundary management, and engagement contribute to team success. But it stops short of providing a more generalizable validation. In this regard a qualitative comparative analysis (QCA) could validate the identified theme interactions – particularly those among the cognitive and affective patterns – that were present in high/low performing teams through verifying the combinations of causal conditions (combinations of themes) across cases (Ragin, 1987; Ragin, 1999). QCA accommodates small sample sizes (Rihoux & Ragin, 2009); however, the Boolean algebra used to identify logically possible combinations requires a large enough sample size to ensure more combinations of conditions (or themes) do not result than there are cases (Berg-Schlosser & De Meur, 2009). This would have occurred with 15 team projects. Future studies would benefit from including more projects in order to conduct a rigorous QCA.

Our study also depended on individual team informants who were in positions to provide insight to all team member activity. While our informants provided rich and comprehensive details about the projects and aggregate team activities, we may have gained more insight to the patterns enacted by teams if more team members would have been represented for each project. Furthermore, reasons for engagement or resistance and

the boundary's perception of how the teams interacted with them would have been improved upon by including interviews of boundary members.

Finally, our teams originated from a specific context of quality improvement project teams in hospitals. These are important groups to understand because of their potential impact on the need to continually improve the healthcare system, but their structure, approach, and relationships may not be generalizable to other contexts such as manufacturing and new product development teams. We do suggest, however, our findings are generalizable to other service industries or teams that conduct project work that is not necessarily central to their day-to-day roles, who meet infrequently (Higgins & Routhieaux, 1999), and who are tasked with implementing innovations at varying levels of the organization (Alexander et al., 2006; Lingham et al., 2009).

Implications for Future Research and Management

Research Implications

Adaptation through Boundary Management Mechanisms

Team adaptation has only lately begun to be more closely studied and defined, likely due in part to its complexity (Burke et al., 2006; Salas et al., 2009). It has been evaluated as a process that is influenced by context or situation, resources, problem solving, goal difficulty or orientation, ability to learn and improve internal team processes (Burke et al., 2006; Klein & Pierce, 2001; LePine, 2003; Wiedow & Konradt, 2011), but it has not been directly connected to boundary management. The influences on adaptation, such as context, situation, and learning indicate an interaction with the environment, but this is the first study of which we are aware clearly linking team

adaptation to boundary management and describing how adaptation dynamically evolves through boundary management mechanisms.

We further offer a lens on boundary management that defines mechanisms used by QI teams that both serve as a conduit for teams to recognize adaptation opportunities and as actions through which adaptation is functionally revealed. We would suggest, however, teams who do not recursively revisit boundary management mechanisms do not exemplify adaptive teams. We have not seen adaptation expressed as being revealed through boundary management, nor has boundary management been described in terms of a pattern of mechanisms. This may be due to the context in which boundary spanning / boundary management has been previously studied (mostly NPD teams in manufacturing environments).

Cognitive and Affective Patterns of Adaptation through Mindful Boundary Management and Responsiveness to Engagement

Team adaptation has been described as having cognitive processes, but primarily in terms of sense-making regarding member activity and developing a shared cognitive frame (Burke et al., 2006; Salas et al., 2009). We found no studies describing or defining affective patterns in adaptation. Our study found cognitive and affective patterns of adaptation are both revealed in and derived through boundary management mechanisms (cognitive) and mindful boundary management considerations (affective). Because of feedback from the boundary as seen in levels of engagement or resistance, teams adapted boundary management reasons for and methods of interaction behaviors. We offer this as a new addition to the literature as we have not seen boundary management described as mindful nor has team adaptation been empirically described through cognitive and affective patterns.

Insights into Quality Improvement Teams

An additional contribution this study offers is increasing understanding about team adaptation in quality improvement teams, particularly those found in healthcare. Current team literature on adaptation relies more on manufacturing and technology firms where teams are involved in innovative behaviors associated with new product development (NPD) (Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003; Hoegl & Weinkauff, 2005; Sydow & Staber, 2002; Van Der Vegt et al., 1999; Zeller, 2002). NPD teams conduct the majority of their work together, demonstrate increasing performance over time, and benefit when they are protected from day-to-day routines for product innovation (Barczak & Wilemon, 1989; Brown & Eisenhardt, 1995; Edmondson & Nemhard, 2009; Ernst, 2002). Less is known about adaptive behaviors among QI teams charged with designing and implementing changes promoting a need to engage people across varying levels of the organization while working in short-lived teams, meeting infrequently, and carrying out other full-time professional responsibilities yet managing competing priorities (Alexander et al., 2006; Batalden & Davidoff, 2007; Davidoff & Batalden, 2005; Higgins & Routhieaux, 1999; Leape et al., 2006). Our study offers insight to adaptation as a dynamic event with cognitive and affective patterns that allow high performing QI teams to better understand their impact on and potential value gained from the boundary as they utilize diverse mindful boundary management mechanisms to engage people beyond the team in their efforts. High performing teams effectively identify pertinent boundary members (individuals or groups), intellectually and emotionally understand the boundary's needs and/or concerns, select mechanisms for

engaging boundary members, and adapt both their understanding and their approach to working with the boundary based on feedback received.

Management Implications

Quality Improvement teams seeking to improve performance should consider the extent to which they are sensitive to the environment by cognitively and affectively adapting expressed through mindful boundary management. Starting with a cognitive pattern, they could ask themselves whether they have clearly identified all meaningful parties beyond the team and why they are influential to the team's success. For instance, are there a large number of individuals or groups who may affect the project's outcomes based on their collaboration or are there those will be affected by the team's efforts that should be given insight to the project? Perhaps, there are boundary members who add credibility to the project based on the nature of their role or expertise in the organization or who hold knowledge and expertise about the reality of the current system and the capabilities for changing it. Traversing the affective pattern should lead teams to mindfully consider the value they place on creating dialogue based on evidence about current and planned performance. They should further give consideration to beliefs about creating a spirit of collaboration through partnership and understand if they hope to pull people into engagement or simply apply an implementation onto an area of interest. Furthermore, they would likely benefit from also considering their beliefs about whether the best knowledge of the system is represented in the team or what additional insights can be gained from boundary members. The highest performing teams use a variety of these aspects of cognitive and affective patterns making adjustments and trying different paths based on the extent to which the boundary is willing to engage with them.

Conclusions

Team adaptation requires an internal and external balance in order for teams to achieve their goals. QI teams or those that are multi-disciplinary, work together temporarily and as an addition to their day to day work are often called upon to design and implement changes that affect multiple people across varying organizational levels impressing the need to span boundaries and engage others in order to bring about changes. We found high performing teams demonstrate cognitive and affective patterns of adaptation revealed in and derived from mindful boundary management mechanisms and behaviors. By this we mean high performing teams embark on purposeful interaction with people beyond the team eliciting a response to the team's efforts, which teams examine, reconcile, and rejoin according to the boundary's needs for engagement.

We propose that in order for QI and similar project teams to reach their potential, they must mindfully manage their boundaries, but do so by dynamically adapting to such interactions. This can be accomplished through clearly identifying who the team needs to engage, why they should be included as well as the value they add to the team, and how these individuals or groups will be best informed of or pulled into the team's efforts. Teams must be aware of the responses from the boundary – through cognitive knowledge and affective feelings and beliefs – and willing to use a repertoire of interactive behaviors lest their efforts be stymied by a lack of consideration for people affected by the team's work. Furthermore, we contend this line of research should be continued in teams in the field to further understand how context or team types (e.g. new product development teams, decision-making teams, external consulting teams, etc.) influence mindful boundary management and adaptation. Adaptation is complex, but we purport it is among

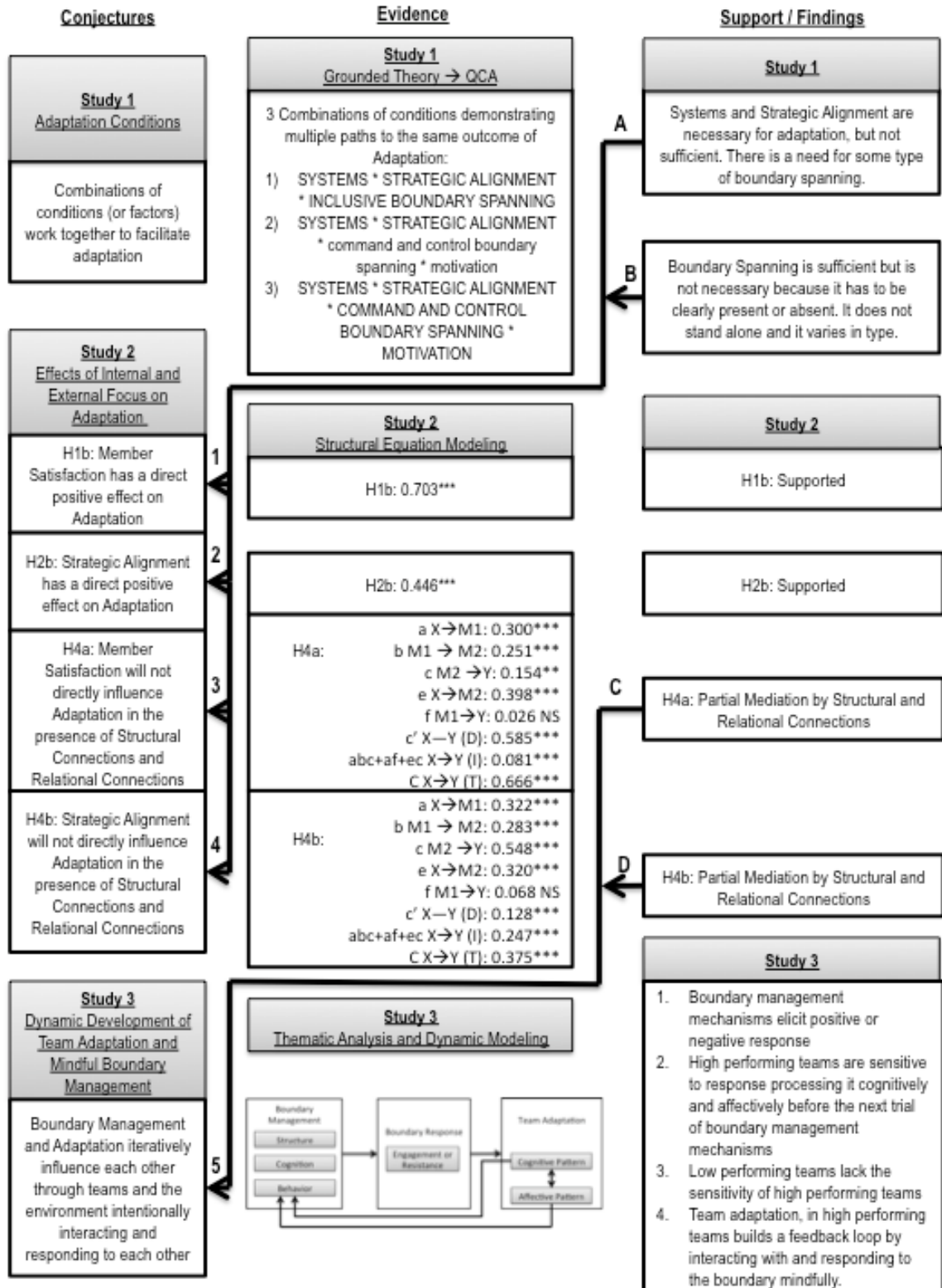
the most important skills a team needs in order to be successful and we encourage the scholarly community to continue to develop concrete descriptions teams in the field can actively utilize.

CHAPTER 6: INTEGRATED FINDINGS AND DISCUSSION

The first study conducted, which was a mixed methods approach, revealed that team adaptation results from teams identifying meaningful processes, management structures and measurement systems, and agreed upon organizational priorities, as well as through two types of boundary spanning – command and control or inclusive. Through Qualitative Comparative Analysis (QCA) (Ragin, 1987), three combinations of the factors (or conditions in QCA parlance) each led to adaptation, which is known as “multiple conjunctural causation” meaning more than one path leads to the same outcome, allowing for the complexity (Berg-Schlosser et al., 2009: 8) that exists in group efforts to effect changes in organizations. The second study, which was quantitative, used new scales developed from the phase I study factors as well as validated scales from literature to assess the effects of an internal vs. external focus of teams and the mediating effect boundary spanning has on carrying internal and external focuses to adaptation. Findings revealed that when team participants see the team as strategically aligned with the organization, Boundary Spanning partially carries the influence of Strategic Alignment on Adaptation, meaning knowing who to know and relating well are important influences in Adaptation. The last – qualitative – study assessed mechanisms of Boundary Management and patterns of Adaptation and found that team adaptation is generated dynamically through cognitive and affective patterns and enacted through boundary management mechanisms when teams mindfully interact with the boundary. Teams adapt in response to the extent to which they are able to engage the boundary where projects are conducted and changes embedded.

Findings from each study promoted conjectures of interest that informed the next study. In study 1, it became clear that multiple factors in combination play a role in Adaptation, but the combinations can vary. This facilitated an interest in trying to measure the impact of key factors from study 1 – in particular strategic alignment and boundary spanning activities – on Adaptation. In addition, I wanted to begin a more direct pursuit to understand the effect internal team relationships had on boundary spanning activities and adaptation in order to elicit clarity about the balance teams must strike between internal and external activities. The result of finding boundary spanning as a partial mediator in study 2 gave rise to a conjecture that perhaps boundary management mechanisms' effect on adaptation are not so linear and, therefore, should be evaluated for their dynamic interactions. Figure 12 represents the linkages between the studies' conjectures, evidence, and findings.

Figure 12: Linkages between Studies



The train of thought how findings influenced and generated conjectures for follow-on studies is outlined in Table 16 according to the Finding letter label and its connection to the Conjecture number label.

Table 16: Linkages between Findings and Follow-On Study Hypotheses

Finding → Conjecture	Reasons for conjecture generation
A → 1	Member satisfaction was not included in Study 1, but was selected in study 2 as an internal focus for teams to serve as a comparison point for how teams balance an internal and external focus and their effects on Adaptation.
A → 2	Interested in the potential to measure a direct effect of Strategic Alignment on Adaptation, as an external focus of teams, since it was found in Study 1 as a condition influencing Adaptation.
A → 3	Member Satisfaction was not included in Study 1, but Boundary Spanning was found as an integral component of generating Adaptation. While evaluating the effect of the internal focus of teams (Member Satisfaction), I wanted to assess whether boundary spanning would carry the effect of Member Satisfaction to Adaptation.
A → 4	Because Strategic Alignment was found to not be sufficient to stand alone and instead worked in coordination with some type of Boundary Spanning, I wanted to measure the extent to which activities of Boundary Spanning would carry the effect of Strategic Alignment.
B → 3 and 4	Because Boundary Spanning did not stand alone in study 1, I wanted to assess its ability to carry the effect of the internal or external focus of teams. Additionally different aspects of Boundary Spanning were included to assess their differing influences as mediators on Adaptation.
C and D → 5	Boundary Spanning constructs of Structural and Relational Connections only partially carried the effects of the internal and external focus of teams on Adaptation. Because Adaptation by definition is dynamic, I sought to describe how Adaptation and Boundary Management iteratively influence each other as well as distinguish differences in Adaptation patterns between high and low performing teams.

Study one findings guided the development of scales and a measurement model for Study two. The results from Study one and Study two gave rise to a detailed thematic analysis of the dynamic and interconnected nature of Boundary Management and Adaptation in Study three. The same foundational factors that influenced Adaptation across Studies one and two continued to be used to evaluate the internal and external balance of teams in Study three. At the same time, Study three compared high and low performing teams relevant to Adaptation rather than teams, in general, to discern better the dynamism of Adaptation. This chapter discusses interesting and unexpected findings

about boundary management and adaptation. Rather than an exhaustive review of the results of the three studies, the next section focuses on the most salient results. The discussions offered here represent important aspects and linkages between studies concerning: the role of strategic alignment and systems thinking; boundary management as a linchpin for adaptation; and the dynamic patterns of adaptation.

The research questions explored the factors of adaptation and how teams could achieve an effective internal and external balance while interacting with the larger organization to pursue project-based quality improvements in healthcare. The results of these studies, therefore, are specifically relevant to quality improvement teams in healthcare settings; however, they are likely generalizable to any project-based team that works together intermittently to embed change in the larger organization. Together, the results of the three studies provide insight to teams about the factors and their interactions that support and influence Adaptation in order to achieve a goal. This chapter discusses the impact of the integrated findings from the three studies.

Strategic Alignment and Systems Clarity

QI teams facilitate and experience adaptation through changing a portion of the organization's process or practices. They accomplish this by combining strategic alignment (i.e. alignment with organizational priorities) and systems clarity (i.e. processes, infrastructure, and measurement systems) with behaviors that engage people outside the team (Boundary Management) in their efforts. In Study one (QUAL → QUAN), QCA revealed that Systems Clarity and Strategic Alignment are necessary for Adaptation as conditions that are always present in Adaptation, but they also needed to work with some form of Boundary Spanning to result in Adaptation.

Study two (QUAN) examined Strategic Alignment as the external focus of teams and its direct and indirect effects on Adaptation finding that it does have a positive and significant direct effect on Adaptation, but it is also partially carried by Boundary Spanning activities to positively influence Adaptation. One counterintuitive finding in Study two demonstrated that Strategic Alignment influenced Adaptation differently between team members and team leaders. For team members, there was a positive and significant relationship whereas for leaders there was a negative and non-significant relationship between Strategic Alignment and Adaptation. While there were no further empirical analyses for this, I conjecture that this may suggest that leaders were already connected to the strategic priorities so it served no new purpose for them, but for members it may have made the project more meaningful influencing their readiness to adapt.

Finally, Study three focused more on Systems Clarity rather than Strategic Alignment and found that Systems Clarity is utilized as a behavioral mechanism for Boundary Management in high performing teams as it offers a means to defer to outside expertise, exchange knowledge, and engage others in a team's effort. As the boundary responds to the team's efforts, the team responds to the boundary through Adaptation and cycling back into Boundary Management activities when needed – including Systems Clarity – in order to achieve goals.

Strategic Alignment connects the team, both internally and externally, to the priorities of the organization. It supports seeing their work, and the resulting necessary interactions, as whole rather than disconnected parts (Senge, 2006). Systems Clarity provides teams with insight to those processes, management structures, and measurement

systems that are meaningful both internally and externally. This facilitates mindful dialogue and understanding about how their project influences the larger system. My research found that the factors of Strategic Alignment and Systems Clarity work in combination to influence Adaptation. Teams surface as well as develop processes and infrastructure systems and they both recognize and relate their work to organizational priorities in order to continuously adapt internally while bringing about change at a larger level of the organization.

Although prior studies have demonstrated goal orientation (individual vs. collective) and difficulty of goals promote team effectiveness and adaptation within teams (Knight et al., 2001; LePine, 2005; Porter, 2008), which is similar to the finding of strategic alignment influencing adaptation, this study may be among the first to specify alignment with organizational priorities as an input to adaptation. Additionally, the QM literature points to systems thinking characteristics of process management, data quality, and reporting to have a positive effect on performance (Kaynak, 2003; Nair, 2006; Sadikoglu & Zehir, 2010; Sousa & Voss, 2002). These studies, however, were primarily at the firm level and this literature stream does not consider the combination of systems clarity with strategic alignment or their effects on team adaptation. This research found that high performing teams deem it important to engage the boundary in the project through shared beliefs in organizational priorities and attentive dialogue regarding evidence. This involved deferring to the boundary's expertise to adequately evaluate current and future systems that promote change in the overall system and adaptation within the team. Strategic Alignment and Systems Clarity work in combination to support Adaptation, but they do so through Boundary Management, which leads to the next

important finding: the mechanisms of boundary management and their interconnected influence on Adaptation.

Boundary Management as a Linchpin to Adaptation

Boundary Management both informs and serves as a conduit for team adaptation. In the first two studies, Boundary Management activities were situated as inputs to Adaptation through collaborative and cooperative types of actions or through more directive styles. Because there were different styles recognized in Study one and different measurable degrees of influence between who to know (Structural Connections) and how to work with others (Relational Connections) in the second study, the third study explored the various mechanisms of Boundary Management and their reciprocal influence on Adaptation. Neither Boundary Management nor Adaptation are static events, rather each influences the other as a project develops.

First, a review of the types and styles of boundary spanning found in each study is offered. In Study one, boundary spanning was defined as two different styles. One style was Inclusive in which teams explored mutual interests with people outside the team and sought collaboration, ideas, and deference to expertise. The other style was called Command and Control because the team – primarily through the team leaders – directed people outside the team to certain actions rather than seeking a harmonious relationship of working together. In Study two, the exploratory factor analysis of Boundary Spanning items resulted in two factors – Relational Connections and Structural Connections. Relational Connections are oriented toward the types of relationships members have with people beyond the team such as influential, well-connected, trusted, and considerate of responses from the outside. Structural Connections are more action-oriented providing

structure for the connections created such as exchanging knowledge, collaborating, inviting others into the work, and proactively identifying the effect their work will have on others. Study three sought to more robustly describe the patterns and mechanisms of Boundary Management, which are richly described within Study three but summarized here as structural (who teams connect with), cognitive (why teams choose to make connections), and behavioral (the actions taken to engage the boundary). The third study clarified the more general descriptions of Boundary Spanning found in Studies one and two.

The important finding about Boundary Management is that it serves as a linchpin to Adaptation. In the first two studies, Boundary Management (or Boundary Spanning) served as a type of input to Adaptation. In Study one, there were two conditions of Boundary Management – Inclusive Boundary Spanning and Command and Control Boundary Spanning. In two of the three condition combinations leading to Adaptation, one type of Boundary Spanning had to be consistently present in the sample cases for QCA (Rihoux & Ragin, 2009) for Adaptation to occur. In the third combination, it was common for Inclusive Boundary Spanning to be present, but Command and Control Boundary Spanning had to be distinctly and consistently absent in order for Adaptation to occur. Boundary Management as a sufficient condition for Adaptation to emerge was met because it needed to be present in some form (i.e. Inclusive or Command and Control Boundary Spanning), but it could not stand alone to bring about Adaptation (Berg-Schlosser et al., 2009). Boundary management conditions had to work in conjunction with Strategic Alignment and Systems Clarity. This led to a decision to evaluate Boundary Spanning as a mediator in Study two to determine if factors of Boundary

Spanning could carry the influence of other inputs to Adaptation. The result was that Boundary Spanning did provide partial mediation for both the external focus of teams (Strategic Alignment) and the internal focus of teams (Member Satisfaction) to influence Adaptation. In Study three I was most interested in describing the patterns and mechanisms of how the external focus of teams and connection to the larger organization, through Boundary Management, facilitated Adaptation. Findings clarified multiple facets of Boundary Management, i.e. that there is a feedback loop with Adaptation. Boundary Management has mechanisms in which teams identify who from outside the team is important for connection, why those connections matter, and the behaviors that will elicit a response from the boundary. High performing teams, through Adaptation, are sensitive to boundary responses and process the responses cognitively and affectively prior to embarking on a new application of boundary management – which is the cycle of Boundary Management and Adaptation mindfully intertwined.

Boundary spanning activities that engage others beyond the team – vertically and horizontally – have previously demonstrated positive results in team and organizational performance as well as successful organizational change initiatives (Ancona, 1990; Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Marrone, 2010). Boundary Spanning studies, however, have not combined Systems perspectives and Strategic Alignment with Boundary Spanning as inputs or as behavioral mechanisms of Boundary Management. Nor has Boundary Management been directly tied to facilitating team Adaptation.

Dynamic Patterns of Adaptation

Adaptation acts as the primary outcome of interest for this dissertation.

Historically, it has been a challenging construct to study because Adaptation is inherently a change in the team and measuring or evaluating change over time is complex (Burke et al., 2006). This study is not longitudinal but it does evaluate whether teams experience change. In Studies one and three, Adaptation was defined as the extent to which a team generates a functional change by purposeful patterning of change through cognitive and behavioral changes between the team and its embedding context. Study two measured this functional change through a construct in which the reflective items were related to the team's ability to identify, coordinate, and direct tasks while also managing changing task demands. The first two studies treated Adaptation as a somewhat static outcome or result of the inputs. In Study three, however, more dynamic patterns were explored, defined, and produced findings.

Patterns of Adaptation emerged by way of teams reflecting on and intellectually as well as emotionally responding to the feedback they receive from the boundary. Adaptation embodied a dynamic belief system about the boundary's responses and how the team should proceed to interact with them. The interactions were carried out through Boundary Management setting up the feedback system for further Adaptation to occur. By dynamically mapping Adaptation points into stories shared by study participants, patterns of cognitive and affective adaptation were revealed which are outlined in Table 17 and replicated from Study three.

Table 17: Cognitive and Affective Patterns of Adaptation

Patterns of Adaptation Informed by and Promoting Boundary Management Mechanisms	
Cognitive	Affective
Who: Understanding of who comprises the boundary	Evidence-based dialogue: A desire to create mindful dialogue as part of understanding the boundary's real or potential responses.
Inclusiveness: Multiple targeted groups and individuals are considered important and valuable for interactions related to the project	Partnership Pull: Feelings of understanding or empathy for real or potential responses and values of boundary members sought for participation
Competence: Expertise and credibility beyond the team can contribute to the team's success	System Conscientiousness: Feelings or beliefs that 1) impact on the boundary from system changes need to be considered and/or 2) people beyond the team have the best information regarding current systems and potential impacts from changes.
System Clarity: Includes consideration for knowledge exchange and deference to expertise beyond the team, which can inform and advance the team's project.	

A difference could be detected between high and low performing teams. While not every adaptation pattern is at work in each situation, I did find that high performing teams used a richer tapestry of Adaptation patterns to guide behaviors. Low performing teams, in contrast, demonstrated fewer adaptive patterns and less collaborative mechanisms. In each study, Adaptation results because of interactions with a team's context through boundary activities. The first two studies applied a more linear and static lens on inputs and the outcome of adaptation. Adaptation, however, is all about change and change is dynamic. The final study revealed team Adaptation is dynamically generated through a cognitive pattern in which teams are actively aware of who is meaningful to the team, why they will be affected, and what system functions and consequences need to be uncovered. Adaptation is also dynamically generated through affective patterns in which teams are sensitive to the boundary and mindfully consider the type of interactions desired and that will result in a spirit of cooperation. As noted earlier,

inputs and mechanisms of Systems Clarity, Strategic Alignment, and Boundary Management, in particular, have not been clearly included in studies of Adaptation. The third study in this dissertation specifically offers distinct descriptions of patterns of Adaptation, which heretofore, have only been described in the most basic and limited ways (Burke et al., 2006; Klein & Pierce, 2001; LePine, 2003).

CHAPTER 7: IMPLICATIONS, CONTRIBUTIONS, LIMITATIONS, AND FUTURE RESEARCH

Contributions and Implications

There are four key contributions to the scholarly literature highlighted here: 1) the use of mixed methods to empirically evaluate Adaptation and its antecedents 2) the use of teams *in situ* and the use of QI teams in a service industry; 3) a study of a combination of factors not previously evaluated together in team Adaptation contexts; 4) expansion of Boundary Management and Adaptation descriptions. The remainder of this section outlines what is new and interesting relevant to these contributions.

This dissertation utilized mixed methods research, the value of which is in its use of “complementary strengths” of both quantitative and qualitative research in which various methods, approaches, and strategies are put together in creative ways (Onwuegbuzie & Johnson, 2006: 51). The three studies examined Adaptation qualitatively through a grounded theory approach (Glaser & Strauss, 1967) and thematic analysis (Boyatzis, 1998; Braun & Clarke, 2006) as well as measured adaptation and its antecedents, quantitatively, through a structural equation model (SEM) (Hair et al., 2010) and through Qualitative Comparative Analysis (QCA) (Ragin, 1987; Rihoux & Ragin, 2009). By incorporating multiple qualitative and quantitative approaches, this research offers robust empirical support for how team Adaptation is defined, how it occurs, and what feeds it. Furthermore, the use of QCA to initially evaluate the combinations of factors that are influential on Adaptation is also new. As previously noted, team Adaptation literature is strongly oriented toward conceptual descriptions without the benefit of empirical evidence. This research fills the gap of empirical evidence concerning team Adaptation and its antecedents.

Teams *in situ* were used for all three studies. This is particularly important for team Adaptation given that most of the prior work is theoretical (Burke et al., 2006; Ilgen et al., 2005; Klein & Pierce, 2001; Kozlowski et al., 1999; Kozlowski & Bell, 2008; Marks et al., 2001; Salas et al., 2009). Team studies call for increased use of teams in the field to ensure the unpredictable contextual factors that can influence teams are accounted for and evaluated (Salas et al., 2008). With my research questions and attention turned toward how teams adapt and how they interact with people beyond the team through Boundary Management, without studying teams *in situ* I may not have adequately anticipated or predicted the attention teams gave to measuring and clarifying the systems in which they were intervening or the importance to team members, in particular, to be strategically aligned with larger organizational priorities. Additionally, this research focused on QI teams in institutionalized, hierarchical service-oriented organizations not NPD teams in manufacturing – the teams and context more commonly used in empirical team interaction behavior and outcome studies (Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003; Hoegl & Weinkauff, 2005; Sydow & Staber, 2002; Van Der Vegt et al., 1999; Zeller, 2002) – offering an expanded view on the importance and mechanisms of Adaptation as a result of interacting with the embedding context for teams pursuing projects that impact the internal organization.

Another contribution offered is in the group of factors combined in this research, which is different than has been studied together before in Team Adaptation. Whereas Systems Clarity, Strategic Alignment, and Boundary Spanning have individually been examined, they have mostly been studied in relation to team performance or organizational performance (Ancona, 1990; Ancona & Caldwell, 1990; Kaynak, 2003;

LePine, 2005; Marrone, 2010; Nair, 2006; Porter, 2008; Sadikoglu & Zehir, 2010; Sousa & Voss, 2002), but not in regard to ‘mediating mechanisms’ of Adaptation and definitely not as a feedback system as this research found. The organizational level literature does reflect similar combinations of factors relevant to systems thinking, strategic alignment, and boundary spanning within and across organizations (Benner & Tushman, 2003; Kaplan & Norton, 2001; Volkoff et al., 1999) but again only in relation to organizational performance; however, Adaptation is not considered here either. This research found and validated, through QCA and SEM, combinations of factors – specifically Strategic Alignment, Systems Clarity, and Boundary Management – work together interdependently to result in Adaptation allowing teams to facilitate change at varying levels. While the quality management literature calls on teams to implement improvements, it lacks clarity in how they do so (Cooney & Sohal, 2004; Daily & Bishop, 2003; Hackman & Wageman, 1995; Sousa & Voss, 2002). In this regard this research specifies teams as critical units facilitating desired outcomes through ongoing team Adaptation that arises from Systems Clarity, Strategic Alignment, and Mindful Boundary Management.

Finally, this research offers an expansion of Boundary Spanning and Adaptation descriptions. These are not new constructs, but there has not been a description of mechanisms of Boundary Management nor has there been a description of patterns of Adaptation, nor studies indicating an inextricable link between Boundary Management and Adaptation. In the past, Boundary Management has been conveyed as an activity to inform leadership of the team’s activities and/or to coordinate resources among groups who have established clear paths of interaction with other groups and work on a shared

goal of producing a good or service (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a; Faraj & Yan, 2009; Joshi et al., 2009; Marrone et al., 2007). This study, however, found that Boundary Management is a combination of mechanisms utilized to actively engage the boundary in the team's efforts including 1) Structure – who among the boundary is meaningful to the team; 2) Cognition – why various boundary members are needed or will be affected by the team; 3) Behavior – specific interaction behaviors are utilized to actively engage the boundary. Furthermore Boundary Management is a functional expression of team Adaptation, because without Adaptation, recurring and developmental Boundary Management resulting in engagement does not emerge. Team Adaptation occurs through cognitive and affective patterns. Through a cognitive pattern, teams are actively aware of who is meaningful to the team, why they will be affected, and what aspects of system functions and consequences need to be uncovered. In the affective pattern, teams are sensitive to the boundary and mindfully consider the types of interactions they desire to have and that will result in a spirit of cooperation from the boundary. This is a contribution to team Adaptation literature since the only stream of research that could be found to shed light on cognitive and affective patterns in teams centered on trust (Greenberg et al., 2007; Kanawattanachai & Yoo, 2002; Schaubroeck et al., 2011; Wilson et al., 2006). Boundary Management mechanisms as the behavioral expression of Adaptation and as a pattern of recurring mechanisms contributing to a feedback loop with Adaptation have also not been previously described in the literature allowing this research to add a new depth of understanding to these topics.

For management practitioners, this research offers practical and useful implications. First and foremost, Adaptation is a dynamic process that can be mindfully

guided through clarifying the systems that teams will intervene in, connecting the team to the organization's larger priorities, and supporting and developing active engagement with people beyond the boundary of the team. Teams can accomplish this by tying the change they are attempting in project work to an organizational goal or priority and by understanding the processes and infrastructures that are being affected. Teams should also actively seek input and cooperation from people beyond the team who will be affected by or responsible for sustaining changes. They can accomplish this by being sensitive to the environment through cognitive and affective Adaptation expressed through mindful Boundary Management. Practically speaking, this could look like teams asking themselves – using a cognitive pattern of Adaptation – whether they have clearly identified all meaningful parties beyond the team and why they are influential to the team's success. Applying an affective pattern of adaptation, teams can be more mindful and deliberate about their beliefs regarding the importance of creating dialogue based on evidence in relation to current and planned performance. Furthermore, they should give consideration to the beliefs and feelings they hold about creating a spirit of collaboration through partnership and where the best knowledge of the system is represented, whether inside or outside the team. The highest performing teams used a variety of cognitive and affective patterns of Adaptation intertwined with mindful Boundary Management mechanisms that ultimately resulted in a 'willing' boundary to engage with a team's goals.

Limitations

Constructs

The team Adaptation literature is highly conceptual as noted frequently in this dissertation. This allowed for broad opportunity to select and pursue various constructs for empirical evaluation. The constructs examined throughout the three studies originated from a grounded theory approach in the first study. Researchers and the participants are not empty receptacles untouched by their roles, experiences, knowledge, and purposes; in fact “qualitative research of all sorts relies on those who conduct it” (Charmaz, 2006: 15). Because of this the constructs that emerged may have been influenced by my years of experience and desire to explain why some QI projects succeed and others fail in the presence of using the same tools and methods across projects. Even so, there were multiple people interviewed in two different qualitative studies where the same types of constructs continued to be present and saturation of themes could be achieved (Glaser & Strauss, 1967). Because this was a developmental study, though, the constructs evolved too – with increased detail and richness – as the research progressed so they were not measured and described identically across the three studies. Finally, none of the studies included objective team performance data obtained separately from research participants, which would add validity to the outcomes.

Methods

This dissertation utilized a mixed methods approach with a design strategy that was continual development using the results from one study to inform and develop the next method for the next study in order to increase the validity of inquiry and related constructs (Greene et al., 1989). A challenge in mixed methods research is inference

transferability indicating the findings can be transferred to other entities or settings (O’Cathain, 2010). The qualitative studies of phases one and three provide rich descriptions of themes and how Boundary Management, Systems Clarity, and Strategic Alignment contribute to patterns of team Adaptation in quality improvement teams in healthcare, but may stop short of providing a more generalizable validation. The transparency provided about the sample descriptions, though, would allow other researchers to consider similarities and potential applicability to their team contexts (O’Cathain, 2010), which may mitigate the challenge of transferability.

Another challenge in mixed methods research is in analytic adequacy related to data quality (O’Cathain, 2010). The purely quantitative study, phase two, measured a small existence of common method bias, but given its nearness to the cutoff I accepted the risk of a small influence on the results rather than incorporate an unmeasured latent factor (Podsakoff et al., 2003) into composite values where no other common method bias was present. In this study, only one method of data collection was utilized – a single survey – contributing to the common method bias; however, all other indicators of data quality were met in the study including a) data transparency – the methods used were described in detail; b) data rigor – the methods used were evidence-based and follow validated analytic techniques; c) sampling adequacy – the sample was large enough for rigorous analytics to be conducted (O’Cathain, 2010).

Finally, interpretive rigor presents another challenge in mixed methods research meaning conclusions are based on the findings (O’Cathain, 2010). Validation of theme interactions is provided through QCA in the phase one study and SEM in the phase two study; however the lack of a similar quantitative approach in the final qualitative phase

three study allows for questions to remain about theme interactions contributing to cognitive and affective patterns and their differences among high and low performing teams. While the sample was large enough in that final study for an appropriate qualitative evaluation, a larger sample would have been needed to conduct QCA. The risk of not meeting interpretive rigor on inferences made on the combined studies is mitigated by offering interpretive transparency and consistency by demonstrating which findings emerged from which methods (such as found in Figure 12 and Table 16) and the inferences were consistent with the objective findings (O’Cathain, 2010). Additionally, theoretical consistency was maintained by tying inferences and findings to current knowledge and theory (O’Cathain, 2010). It is possible, however, that by not including a quantitative evaluation of combinations of conditions in Study three that others could reach a different conclusion about key interactions. I would submit, though, that conducting dynamic mapping of the interactions of mechanisms of Boundary Management and patterns of Adaptation minimized my subjective interpretation of theme interactions.

Sample

Each of the studies included team representatives, but none of the studies included multiple members from teams. Team studies are more robust, especially in quantitative studies, when multiple members of the same team are included in the sample in order to obtain a team level measure. This dissertation included a significant amount of qualitative research so to mitigate the lack of multiple members per team in the samples, team leaders or project facilitators served as participants in order to obtain the broadest view possible of overall team activities. In the first study, most of these team leaders were also

organizational level leaders. By including those in a senior leadership role, however, findings in the first study may not be generalizable to other teams in which a directive style of Boundary Spanning was found. This may be why the third study revealed less about directive styles and more about leaning on people who offer the team credibility – often due to the position such people held in the organization – as a means of creating engagement. Additionally, the context for this study was in institutionalized, hierarchical organizations – almost exclusively in hospitals – which may prohibit the findings from being fully generalizable to flatter project-based organizations such as manufacturing organizations and new product development teams.

Future Research and Final Conclusions

This research has sought to explain team Adaptation and identify factors that contribute to a team's internal and external balance, which helps a QI project achieve success. The result was that a combination of factors including clarifying the Systems being affected, Strategic Alignment with organization's larger priorities, and mindful Boundary Management to engage others in the team's work contribute to, facilitate, and reflect team Adaptation. There are, however, still opportunities to further validate these findings and expand our understanding of the complexity of the feedback that takes place in Adaptation.

To expand and further validate the findings from this dissertation, a valuable addition to the research would be to include multiple members from the teams to obtain a more comprehensive and validated perspective on the teams' experiences. Additionally, including more teams and projects to the research in order to conduct a robust QCA and/or SEM would increase clarity about what combinations of factors influence the

patterns of Adaptation and which factors have the most influence. Furthermore, including objective team performance data as well as stakeholder (people outside the team) analyses would create a more robust distinction between high and low performing teams to better analyze the mechanisms of Boundary Management that facilitate Adaptation. Finally, considering the feedback loop between Boundary Management mechanisms and Adaptation patterns revealed in the phase three study, a longitudinal study using a mixed methods approach where teams can be both observed and measured over time would incorporate and validate how time and multiple interactions influence team Adaptation.

Ultimately, team Adaptation is reflected in the team's the ability to balance internal and external influences. QI teams are charged with embedding changes and elevating performance at levels across the organization; therefore, they must be well-versed in the processes and infrastructure that will be affected with their planned changes, aligned with organizational priorities, and competent in the Boundary Management mechanisms that will facilitate Adaptation and eventually lead to positive project performance. Team Adaptation is complex, but I purport it is a critically important skill teams need to be successful. As part of the scholarly community, there is opportunity to continue to develop concrete descriptions and measures of mindful Boundary Management interacting with cognitive and affective Adaptation. As a member of the practitioner community, it is also important to continue to study teams in the field and to consider applying intervention studies in order to directly assist teams in developing their capabilities to balance internal and external interactions for Boundary Management and Adaptation to lead to successful outcomes.

Appendix A: Interview Protocol

1. People often talk about cost and quality as polarized topics. I would like to know how you think about them as part of your work. Can you give me an example of how you have considered cost and quality in your work?
Note - Examples might include:
 - a. Individual patients?
 - b. Institutional level?
 - c. Population health?
 - d. Interactions that influence decisions?
2. Physicians and administrators want to do the best possible for all patients. During individual treatment decisions:
 - Can you give me an example of a decision that was easy to resolve when considering the impact of cost?
 - Can you give me an example of a decision that was difficult to resolve when considering the impact of cost?
 - Other probes might include:
 - Is consideration given to any potential impact on costs that treatment choices may have down the road?
 - If so, what do you take into consideration?
 - What or who influences your decisions?
 - Are others involved in treatment decisions and if so, who and what is their involvement and your interaction with them? (Provide an example)
3. Most areas of medicine have associated quality measures, which often have required treatments associated with specific diagnoses in order to be compliant with the measure. Clinical judgment may not recognize the required components as important / impactful / optimal. Can you give me an example of how you approached that type of situation?
 - a. Examples of sources for quality measures if needed: The Joint Commission Core Measures, CMS quality metrics, Commercial Insurance metrics for quality rankings, or a Professional or Regional Collaborative Quality Initiative
4. Decisions about services and/or treatment decisions can be influenced by issues other than evidence based care or cost such as family desires, one's own ethics, or legalities. Can you give me an example of when you have had to resolve such issues and decisions?
5. Optional: Physicians are frequently exposed to the latest and greatest treatment modalities / equipment / implants. Can you give me an example of when one of these prompted a request for purchase? What were (and who was included in) the conversations that occurred in the decision to purchase or not

Appendix B: Quantitative Phase 2 Tables and Figures

Table B1: Model Constructs and Scale Development

1. Name of Scale: <u>Member Satisfaction</u>	Referenced Citation: Oetzel, J. G. 2001. 'Self-Construals, Communication Processes, and Group Outcomes in Homogeneous and Heterogeneous Groups.' <i>Small Group Research</i> , 32:1, 19	Cronbach's Alpha: 0.930
Description: - Member satisfaction relates to processes of group dynamics and the resulting outcome of their experience of working with that team. “Satisfaction is a positive or negative affective response to the environment (Jablin, 1987) and an important consideration for members of small groups. Satisfaction can affect future participation of group members and thus indirectly affect group performance (Gouran, 1973; Jurma, 1978). In addition, member satisfaction has implications for cohesiveness and group climate (Downs and Hazen, 1977).” (Oetzel 2001, 29)		Items: a. I am satisfied with the group’s outcomes to date b. I am confident that our performance during this activity is satisfactory c. I am happy with the way our group interacts d. I have confidence in the members of my group e. All of our members are prepared f. I am happy with the way our group works during this activity g. My personal level of satisfaction with the group is high h. I would like to work with this group again
2. Name of Scale: <u>Strategic Alignment</u>	Referenced Citation: Newly developed scale	Cronbach’s Alpha: 0.846
Description: - Teams as systems embedded in larger organizations are both physical and social structures that include a network of interdependent and interrelated components working together to achieve an aim through a commitment to ideas, issues, goals, and management processes. Perceived achievement in teams embedding in organizations requires a whole picture approach in recognizing all the tasks, relationships, structures, boundaries, aims, and ideas. This scale and its items were developed from findings and related statements in the (Grooms & Lyytinen, 2014) qualitative study with refinements with support from the following: (Ilgen, 1999) Multi-level nature of teams and organizations as social systems in which teams are embedded. Time component of teamwork. (Manning, 2008) Time components of teamwork. Systems have boundaries guiding actions. (Sydow & Staber, 2002) Team members draw on structures and structural properties for task development and social interaction		Items: a. Members align the team’s efforts with organizational routines b. Members recognize underlying organizational structures that can affect the team’s work c. Members pursue shared goals with organizational leadership d. Members are aware of the influence our team has on the use of organizational resources e. Members consider the organization’s strategic plan when designing changes f. Members consider the effect our team has on the organization’s strategic priorities g. Members identify multiple levels of the organization needed for team success

• (Deming, 1994) Network of interdependent components that work together to accomplish an aim of the organization • (Senge, 2006) Recognition of unseen forces and underlying structures that change behavior. Shared commitment across groups to ideas, issues, goals, management processes. Systems thinking includes seeing the whole - not tasks or relationships alone but putting the whole picture together, seeing all the interrelationships • (Arshinov & Fuchs, 2003) Emergence as a quality of self-organizing systems - one component is systemness: "Self-organization takes place in a system i.e. in a coherent whole that has parts, interactions, structural relationships, behavior, state, and a border that delimits it from its environment."	h. Members utilize organizational structures (such as data systems, processes, improvement practices) that support our work	
3. Name of Scale: <u>Embedded Interconnections:</u> <u>Structural Connections and Relational Connections</u>	Referenced Citation: Newly developed scale	Cronbach's Alpha: Structural Connections = 0.732 Relational Connections = 0.796
Description: • This scale was originally conceptualized as one factor reflective of the interconnections among actors that support embedding teams into larger systems. In the process of analysis, it became clear that the items measured two constructs – connections that are structural in nature and those that are relational. This scale and its items were developed from findings and related statements in the (Grooms & Lyytinen, 2014) qualitative study with refinements with support from the following: - o (Dacin et al., 1999) Social structure constrains and generates "configurations and forms of actors" (Dacin et al., 1999). Market exchange is embedded in and defined by social processes. Also, there are "communicative and interactional linkages within an arena of institutional life" (Dacin et al., 1999) . Additionally, there are direct, indirect network linkages, local and non-local ties, vertical and horizontal flows of materials and resources and linkages between social actors. - o (Dacin et al., 1999) Useful for the importance of understanding embedding into organizations: "Hence, understanding these elements is essential for explanation and for practice - so that managers can take an active role in the management of interactor ties	Items: a. Members have relationships with others outside the team who can help influence change (Relational) b. Members are influential with their peers outside the team (Relational) c. Members are well connected to others in the organization (Relational) d. Members effectively exchange knowledge with others in the organization (Structural) e. Members collaborate with others in the organization to advance our team's efforts (Structural) f. Members consider the effect our work will have on others (Structural) g. Members consider how others will respond to our team's work (Relational) h. Members invite others in the organization to engage with our work (Structural) i. Members are trusted by others in the organization (Relational)	

and development of collaborative capabilities that recognize both opportunities and costs in embedded relations." - o (Sydow & Staber, 2002) Social capital can encourage collective capability or be exclusionary - leads to inertia or stagnation. "Institutional embeddedness...addresses the forms and processes by which meaning is constructed, trust is built (or destroyed), knowledge is exchanged..." (Sydow & Staber, 2002). - o (Ilgen, 1999) Consider team members' ability to work together; growth and well-being of team members. Ilgen notes these items as particularly important for teams embedded in organizations vs. studying teams in isolation. There is an exchange relationship in which members give their time and effort in exchange for personal return. Additionally, there is consideration for a shared mental model or common view of interaction structure among team members. - o (Manning, 2008) Social systems and social contexts are brought about by social practices. Embedding is conceptualized "as a continuous process which project participants and stakeholders actively engage in when they enact, transform and reproduce task, time and team features of particular projects and relate them to multiple systemic environments" (Manning, 2008).		
4. Name of Scale: <u>Adaptability</u>	Referenced Citation: Newly developed scale	Cronbach's Alpha: 0.888
Description: • Adaptability includes the concepts of task adaptation within the team in alignment with organizational needs, an eye toward goals, and change management as needed both to achieve adjust tasks and progress toward goals. The items were originally developed from statements related to key findings regarding recognized interdependencies in systems from the (Grooms & Lyytinen, 2014) qualitative study. Items were further refined with consideration given to such concepts and findings as the following: - o (Lenox et al., 2006) Interrelationships of activities. Adjustment of practices over time - o (Sydow et al., 2004) Tension of task demands - o (Van Der Vegt et al., 1999) Action-orientation - trying to get something accomplished. One's outcomes are affected by other's actions. Structural feature of relationships between team members –	Items: a. Members effectively identify activities to advance our work in the organization b. Members effectively coordinate tasks across the team c. Members align tasks with organizational priorities d. Members adjust activities to advance the team's goals e. Members are good at managing changing task demands f. Members direct tasks to accomplish a specific result	

this is a management of interfaces among team members - one's actions within a team affects other team member actions such as completing materials or communicating information - o (Hoegl & Weinkauff, 2005) Management of interfaces with others who are associated with the project (outside of the direct project team) - o (Rivkin & Siggelkow, 2003) The concept is organizational design but we considered the applicability to smaller systems – teams. Division of tasks into manageable parts yet coordinated to achieve harmonious action. - o (Huberman & Wilkinson, 2010) Coordination of tasks dynamically integrated into a whole. - o (Gully et al., 2002) Task interdependence = degree of task-driven interaction among members (referencing Shea and Guzzo). Goal interdependence guides efforts among members.		
5. Name of Scale: <u>Potency</u>	Referenced Citation: Collins, C. G. & Parker, S. K. 2010. 'Team capability beliefs over time: Distinguishing between team potency, team outcome efficacy, and team process efficacy.' <i>Journal of Occupational & Organizational Psychology</i>, 83:4, 1003-23.	Cronbach's Alpha: 0.904
Description: Team potency has a general emphasis and reflects a global evaluation of a team's confidence in its ability to succeed and general capabilities. Collins and Parker (2010) reference Guzzo et al's 1993 work in reporting team potency refers to "the collective belief in a team that it can be effective".	Items: a. This team believes it can become unusually good at producing high-quality work b. This team expects to be known as a high performing team c. My team feels it can solve any problem it encounters d. My team has confidence in itself e. My team believes it will get a lot done when it works hard f. No task is too tough for this team g. My team believes it can be very productive h. My team expects to have a lot of influence in the organization	

Figure B1: New Scales – Development Process

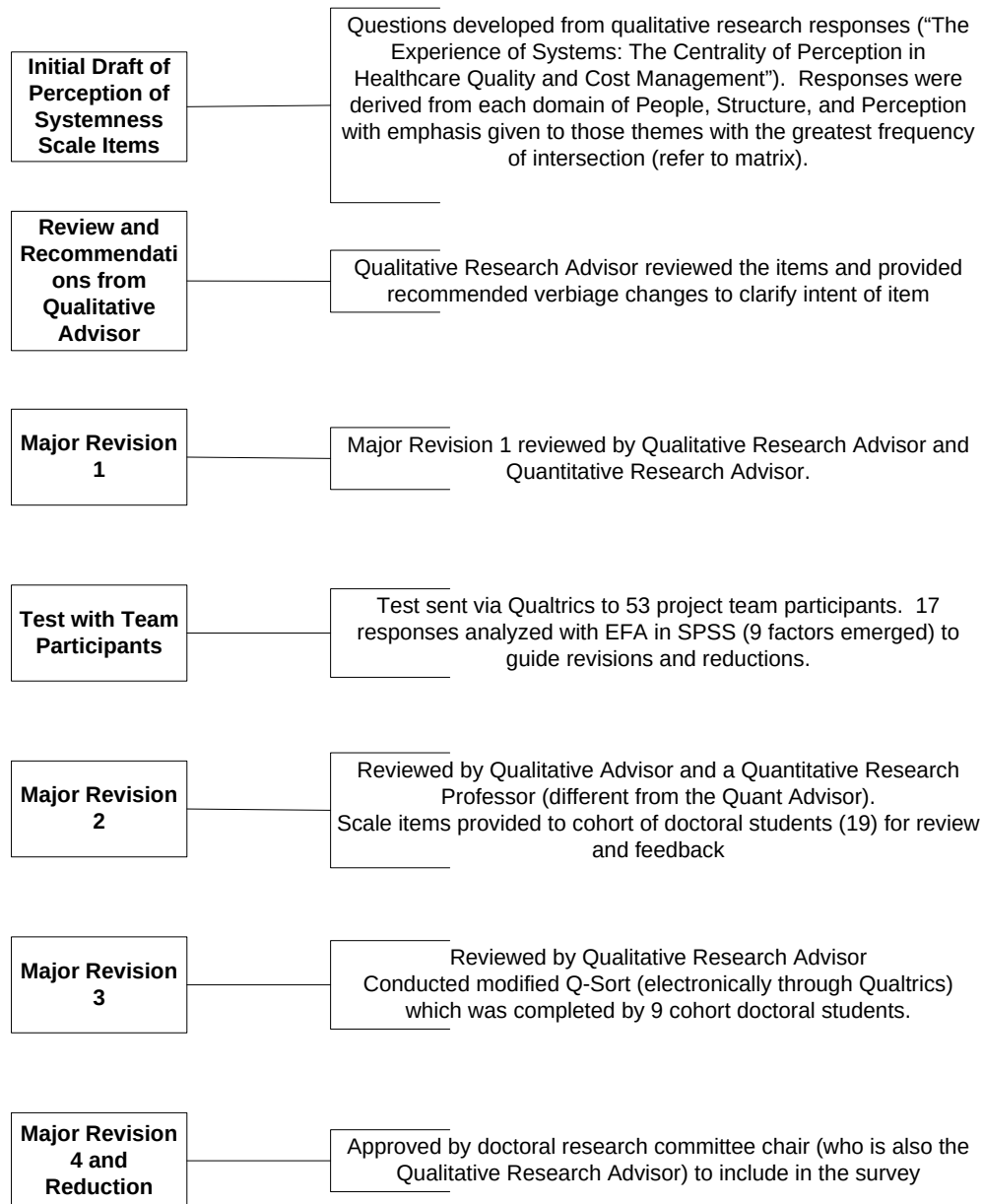


Table B2: Demographics

Demographic Item	Descriptor	Frequency	Percent
Team Type Count = 249 Missing = 1	Project Team	81	32.5
	Committee	17	6.8
	Task Force	19	7.6
	Cross Disciplinary Team	11	4.4
	Department	47	18.9
	Executive Team	17	6.8
	Work Team	23	9.2
	Consulting Team	14	5.6
	Ad Hoc Team	1	.4
	Cross Functional Team	11	4.4
	Other	8	3.2
Organization Type Count = 249 Missing = 1	Healthcare	143	57.4
	Education	41	16.5
	Manufacturing	16	6.4
	Consulting	11	4.4
	Marketing / Advertising	3	1.2
	Finance	11	4.4
	Transportation	1	.4
	Communications	3	1.2
	Hospitality	2	.8
	Other	18	7.2
Role on Team Count = 249 Missing = 1	Team leader	67	26.9
	Team coach or facilitator	29	11.6
	Team member	127	51.0
	Ad hoc member (e.g. providing specialty knowledge as needed)	6	2.4
	Project management professional	9	3.6
	Expert	7	2.8
	Other	4	1.6
Meeting Frequency Count = 244 Missing = 6	Daily	19	7.8
	Once a week	84	34.4
	Twice a month (or every other week)	53	21.7

	Once a month	53	21.7
	Once every other month	15	6.1
	Once a quarter	14	5.7
	Twice a year	6	2.5
Tenure of Team Count = 246 Missing = 4	1 - 6 months	76	30.9
	7 - 11 months	35	14.2
	1 to 3 years	70	28.5
	Greater than 3 years	65	26.4
Education Count = 249 Missing = 1	High School Graduate	1	.4
	Some College	12	4.8
	Undergraduate	71	28.5
	Graduate	129	51.8
	Doctorate	36	14.5

EFA:

Table B3: EFA Refinement Steps and Results

Step	Adequacy				Validity		Reliability			Decision
	<i>KMO / Bartlett</i>	<i>Communalities</i>	<i>Total Variance</i>	<i>Residuals</i>	<i>Convergent</i>	<i>Discriminant</i>	<i>Factor Label</i>	<i>Cronbach's Alpha</i>	<i>Specification</i>	
	(Hair: 2010, 104) ≥0.8 meritorious ≥0.7 middling ≥0.6 mediocre ≥0.5 miserable <0.5 unacceptable P<0.05 good	Loading <.50 have insufficient explanation (Hair : 2010, 119)	>60% sought (Hair, et al 2010)	Residuals <5% sought (Lyytinen & Gaskin, 2011)	Variable communality loads of >0.5 (Hair, et al, 2010)	Cross loadings <0.2 Rotated Factor Matrix <0.7 below diagonal	List factor labels	Report Cronbach's Alpha (lower limit should be 0.7)	Reflective vs. Formative	Note items being reduced or other decisions made (such as switch from Eigenvalues to Fixed Factors)
1 - Eigenvalues	0.933	12/39 <0.5; Lowest: ST2=0.305	58.895%, 7 factors	11%	Factors 1, 2, 4, 6, 7 (non cross loaded) items all >0.5	Some problematic cross loads; Factor 1 / 3 >0.7	TBD	TBD	Reflective	Delete ST2 - lowest communality, low load in pattern matrix
2 - Eigenvalues X ST 2	0.934	12/38 <0.5; Lowest: ST4=.296	59.619%, 7 factors	9%	Factors 1,2,3,6,7 (non cross loaded) items all >0.5	Some problematic cross loads; Factor 1 / 3 >0.7	TBD	TBD	Reflective	Delete ST4 - lowest communality, low load in pattern matrix
3 - Eigenvalues X ST4	0.934	12/37 <0.5	60.446%, 7 factors	8%	Factors 1,2,3,6,7 (non cross loaded) items all >0.5	Problematic cross loads: ITD3, ST7, ST8, ST1, Emb9; Factor 1/3 >0.7	TBD	TBD	Reflective	Delete Pot8 - 2nd to lowest communality, 2 cross loads
4 - Eigenvalues X Pot8	0.934	11/36 <0.5	61.042%, 7 factors	8%	Factors 1,2,3,6,7 (non cross loaded) items all >0.5	Problematic cross loads: ITD3, ST7, ST8, ST1, Emb8, Emb9; Factor 1/3 >0.7	TBD	TBD	Reflective	Delete Emb8 - 2 problematic cross load, communality <0.5

5 - Eigenvalues X Emb8	0.933	10/35 <0.05	61.414%, 7 factors	8%	Factors 1,2,3,6,7 (non cross loaded) items all >0.5	Problematic cross loads: ITD3, ST7, ST8, ST1, Emb9; Factor 1/3 >0.7	TBD	TBD	Reflective	Delete Emb7 - 1 problematic cross load, communality <0.5
6 - Eigenvalues X Emb7	0.934	12/34 <0.5	59.587%, 6 factors	10%	Factors 2,4,5,6 (non cross loaded) items all >0.5	Problematic cross loads Emb3, Emb2, ST3, Emb9; Factor 1/4 >0.7	TBD	TBD	Reflective	Delete Emb9 - problematic cross load, communality <0.5
6 - Eigenvalues X Emb9	0.932	10/33 <0.5	60.126%, 6 factors	9%	Factors 2,4,5,6 (non cross loaded) items all >0.5	Problematic cross loads Emb3, Emb2, ST3, Emb9; Factor 1/4 >0.7	TBD	TBD	Reflective	Delete ST8 - problematic cross load, communality <0.5
7 - Eigenvalues X ST8	0.932	9/32 <0.5	60.686%, 6 factors	8%	Factors 2,4,5,6 (non cross loaded) items all >0.5	Problematic cross loads Emb3, Emb2, ST3, ST1; Factor 1/4 >0.7	TBD	TBD	Reflective	Delete ST1 - problematic cross load, communality <0.5
8 - Eigenvalues X ST1	0.929	8/31 <0.5	61.245%, 6 factors	8%	Factors 2,3,5,6 (non cross loaded) items all >0.5	Problematic cross loads Emb3, Emb2, ST3; Factor 1/3 >0.7	TBD	TBD	Reflective	Delete ST3 - problematic cross load, communality <0.3
9 - Eigenvalues X ST3	0.927	7/30, <0.5	61.964%, 6 factors	8%	Factors 2,3,5,6 (non cross loaded) items all >0.5	Problematic cross loads Emb2, Emb3, ITD3; Factor 1/3 >0.7	TBD	TBD	Reflective	Delete MS2 - nearing a problematic cross load. Expect 6 factors, but Member Sat items should load together unless items are problematic - MS is from a validated and well used scale
10 - Eigenvalues X MS2	0.925	5/29 <0.5	62.583%, 6 factors	6%	Factors 1,2,3,5,6, (non cross loaded) items all >0.5	Problematic cross loads ITD3, ST7; Factor 1/3 >0.7	TBD	TBD	Reflective	Delete ST7 due to problematic cross loads, low communality
11 - Eigenvalues X ST7	0.926	6/28 <0.5	63.450%, 6 factors	6%	All factors (non cross loaded) items >0.5	Problematic cross loads ITD3, possibly MS1; Factor 1/3 >0.7				Delete ITD3

12 - Eigenvalues X ITD3	0.924	7/27 <0.5	63.910%, 6 factors	6%	All factors (non cross loaded) items >0.5 except one at 0.488). 2 Heywood cases	No problematic cross loads; Factor 1/3 >0.7 (.730)				Delete MS1 for same reason as MS2 and because it is the only cross loaded item. Also might be contributing to Heywood cases.
13 - Eigenvalues X MS1	0.922	7/26 <0.5	64.191%, 6 factors	5%	All factor (non cross loaded) items >0.5 except 1 (at 0.480). 1 Heywood case	No problematic cross loads; Factor 1/3 >0.7 (.723)	1. Potency 2. Mem Sat 3. Adapt 4. Rel Con 5. Struc Con 6. Strat Align	1. Potency=0.904 2. Mem Sat=0.930 3. Adapt=0.888 4. Rel Con=0.796 5. StrucCon=0.732 6. StratAlign=0.846 All items combined = 0.940	Reflective	Test with PAF to see if it improves the Heywood case on ST5. Previously tested in CFA and had no issues with this item. It works with PAF. Move on to CFA to refine any additional reductions

Table B4: EFA Pattern Matrix

	Factor					
	Potency	Mem Sat	Adapt	Rel Connect	StrucConnect	Strat Align
Pot_3_I	.858					
Pot_5_I	.828					
Pot_7_I	.784					
Pot_2_I	.762					
Pot_4_I	.703					
Pot_1_I	.687					
Pot_6_I	.654					
MS_3_I		.986				
MS_4_I		.883				
MS_6_I		.847				
MS_7_I		.740				
MS_8_I		.702				
MS_5_I		.659				
ITD_6_I			.885			
ITD_2_I			.775			
ITD_4_I			.772			
ITD_5_I			.709			
ITD_1_I			.681			
Emb_2_I				.869		
Emb_3_I				.799		
Emb_1_I				.545		
Emb_5_I					.989	
Emb_6_I					.553	
Emb_4_I					.480	
ST_6_I						1.007
ST_5_I						.648

Extraction – Maximum Likelihood; Rotation – Promax

CFA:

Table B5: CFA Refinement Steps and Results

Step	Component change carried out and evaluated	Chi Square CMIN / DF <3 good; <5 sometimes allowable P-Value >0.05	PCLOSE >0.05	CFI >0.95 great; >0.90 traditional; >0.80 sometimes allowable	TLI >0.95 is desired >0.92 OK with >30 items	GFI >0.95	NFI 0.90 - 0.95 is acceptable, >.95 is good	RMSEA <0.05 good; 0.05 - 0.10 moderate; >0.10 bad	SRMR <0.09	Decision for Next Step (e.g. Covary Errors; Item to Remove)
1	Base Model	1.73, p=0.0	0.195	0.949	0.942	0.874	0.889	0.054	0.0523	X MS7 (high MI issues)
2	X MS7	1.655, p=0.0	0.393	0.954	0.947	0.885	0.893	0.051	0.0527	X Pot2 (High MI issues)
3	X Pot 2	1.571, p=0.0	0.638	0.962	0.956	0.896	0.903	0.048	0.0507	X Pot3 (High MI Issues)
4	X Pot3	1.47, p=0.0	0.855	0.97	0.965	0.904	0.913	0.043	0.0471	Test deleting Emb 4 to improve convergent validity
5	X Emb4	1.382, p=0.0	0.948	0.977	0.973	0.914	0.923	0.039	0.0437	Move to CMB tests

Figure B2: Resulting Model Including Common Method Bias Testing with Unmeasured Latent Factor

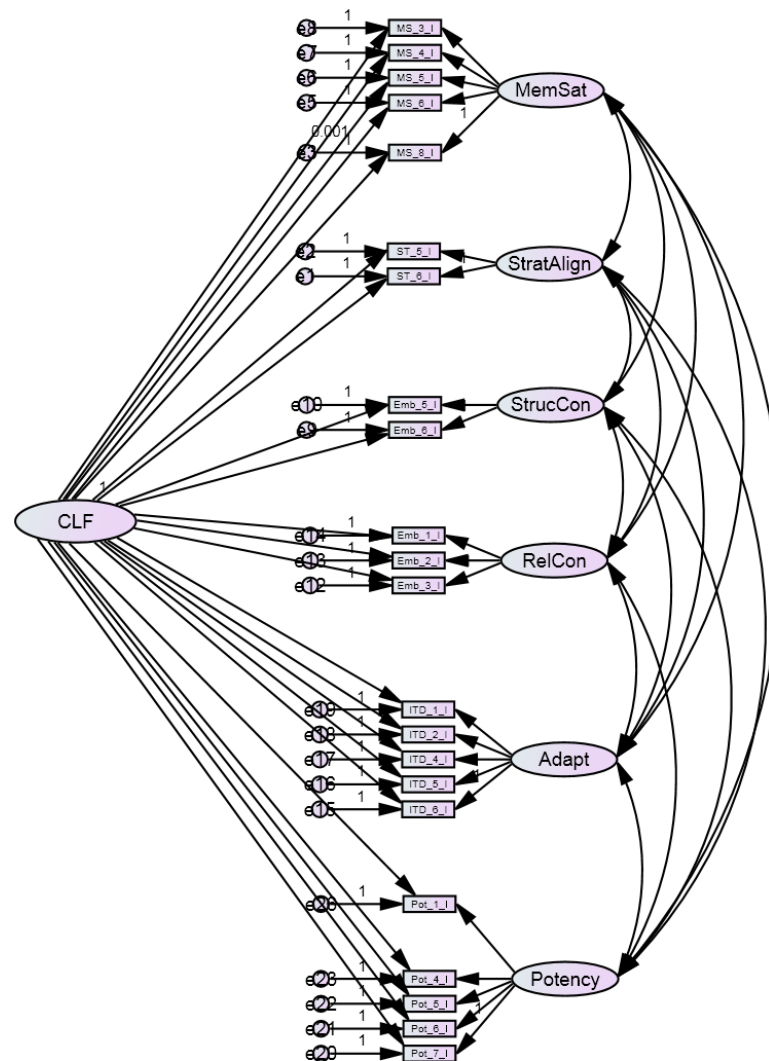


Table B6: Validity Measures

	AVE	MSV	ASV
Adapt	0.621	0.551	0.330
StratAlign	0.734	0.294	0.221
MemSat	0.684	0.551	0.331
StrucCon	0.595	0.172	0.133
RelCon	0.571	0.372	0.269
Potency	0.620	0.445	0.264

Table B7: Nomological Validity

	Mean	Std Dev	CR	AVE	Adapt	StratAlign	MemSat	StrucCon	RelCon	Potency
Adapt	3.779	0.687	0.891	0.621	0.788					
StratAlign	3.896	0.793	0.846	0.734	0.517	0.857				
MemSat	3.889	0.775	0.915	0.684	0.742	0.440	0.827			
StrucCon	3.914	0.624	0.742	0.595	0.343	0.415	0.306	0.771		
RelCon	3.958	0.618	0.799	0.571	0.563	0.542	0.610	0.415	0.756	
Potency	4.189	0.646	0.890	0.620	0.632	0.421	0.667	0.329	0.438	0.788

Table B8: Item-Level Correlation Matrix

	Pot_1_1	Pot_4_1	Pot_5_1	Pot_6_1	Pot_7_1	ITD_1_1	ITD_2_1	ITD_4_1	ITD_5_1	ITD_6_1	Emb_1_1	Emb_2_1	Emb_3_1	Emb_5_1	Emb_6_1	MS_3_1	MS_4_1	MS_5_1	MS_6_1	MS_8_1	ST_5_1	ST_6_1
Pot_1_1	1																					
Pot_4_1	0.552	1																				
Pot_5_1	0.591	0.688	1																			
Pot_6_1	0.481	0.573	0.617	1																		
Pot_7_1	0.555	0.684	0.762	0.614	1																	
ITD_1_1	0.333	0.394	0.386	0.371	0.415	1																
ITD_2_1	0.274	0.499	0.391	0.405	0.436	0.563	1															
ITD_4_1	0.336	0.406	0.384	0.393	0.406	0.554	0.65	1														
ITD_5_1	0.296	0.462	0.398	0.43	0.425	0.488	0.647	0.648	1													
ITD_6_1	0.356	0.464	0.428	0.411	0.464	0.564	0.683	0.689	0.668	1												
Emb_1_1	0.242	0.367	0.327	0.354	0.361	0.315	0.438	0.446	0.451	0.413	1											
Emb_2_1	0.117	0.221	0.182	0.207	0.266	0.209	0.343	0.33	0.314	0.277	0.514	1										
Emb_3_1	0.169	0.288	0.258	0.263	0.332	0.229	0.356	0.396	0.313	0.286	0.541	0.639	1									
Emb_5_1	0.129	0.221	0.219	0.099	0.197	0.15	0.161	0.147	0.228	0.193	0.165	0.225	0.203	1								
Emb_6_1	0.205	0.209	0.247	0.153	0.27	0.222	0.246	0.22	0.31	0.208	0.255	0.298	0.278	0.57	1							
MS_3_1	0.36	0.467	0.433	0.45	0.478	0.415	0.546	0.508	0.521	0.517	0.436	0.296	0.351	0.113	0.21	1						
MS_4_1	0.367	0.516	0.473	0.473	0.519	0.361	0.508	0.529	0.519	0.468	0.5	0.372	0.446	0.154	0.234	0.791	1					
MS_5_1	0.342	0.436	0.361	0.335	0.398	0.278	0.495	0.405	0.476	0.415	0.371	0.231	0.352	0.192	0.227	0.576	0.639	1				
MS_6_1	0.389	0.456	0.424	0.376	0.476	0.415	0.585	0.528	0.562	0.528	0.458	0.391	0.379	0.202	0.247	0.78	0.715	0.596	1			
MS_8_1	0.434	0.512	0.52	0.5	0.522	0.402	0.496	0.545	0.501	0.554	0.469	0.363	0.416	0.149	0.239	0.702	0.732	0.556	0.678	1		
ST_5_1	0.179	0.269	0.265	0.274	0.311	0.399	0.339	0.378	0.35	0.365	0.32	0.365	0.364	0.197	0.309	0.272	0.332	0.248	0.344	0.352	1	
ST_6_1	0.259	0.288	0.312	0.293	0.359	0.344	0.31	0.377	0.358	0.345	0.311	0.295	0.428	0.252	0.318	0.281	0.361	0.353	0.318	0.317	0.733	1

Table B9: Invariance Testing – Model Fit for Configural Invariance

Step	Invariance - Group Tested	Chi Square CMIN / DF <3 good; <5 sometimes allowable P-Value >0.05	PCLOSE >0.05	CFI >0.95 great; >0.90 traditional; >0.80 sometimes allowable	TLI >0.95 is desired >0.92 OK with >30 items	GFI >0.95	NFI 0.90 - 0.95 is acceptable, >.95 is good	RMSEA <0.05 good; 0.05 - 0.10 moderate; >0.10 bad	SRMR <0.09
1	Gender Majority	1.475, p=0.0	0.886	0.945	0.935	0.837	0.851	0.044	0.0704
3	Role (Leader, Member, Other)	1.428, p=0.0	0.986	0.929	0.916	0.787	0.803	0.042	0.0624

SEM:

Figure B3: Final Model Retained

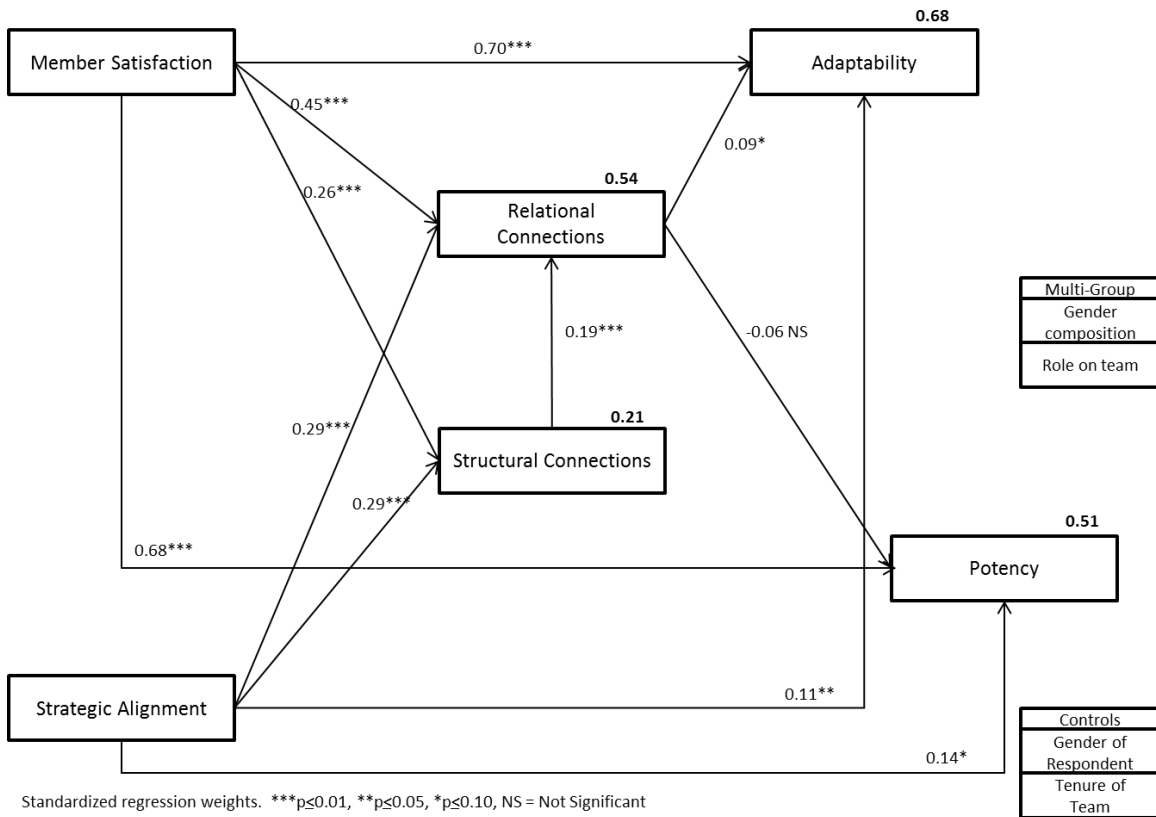
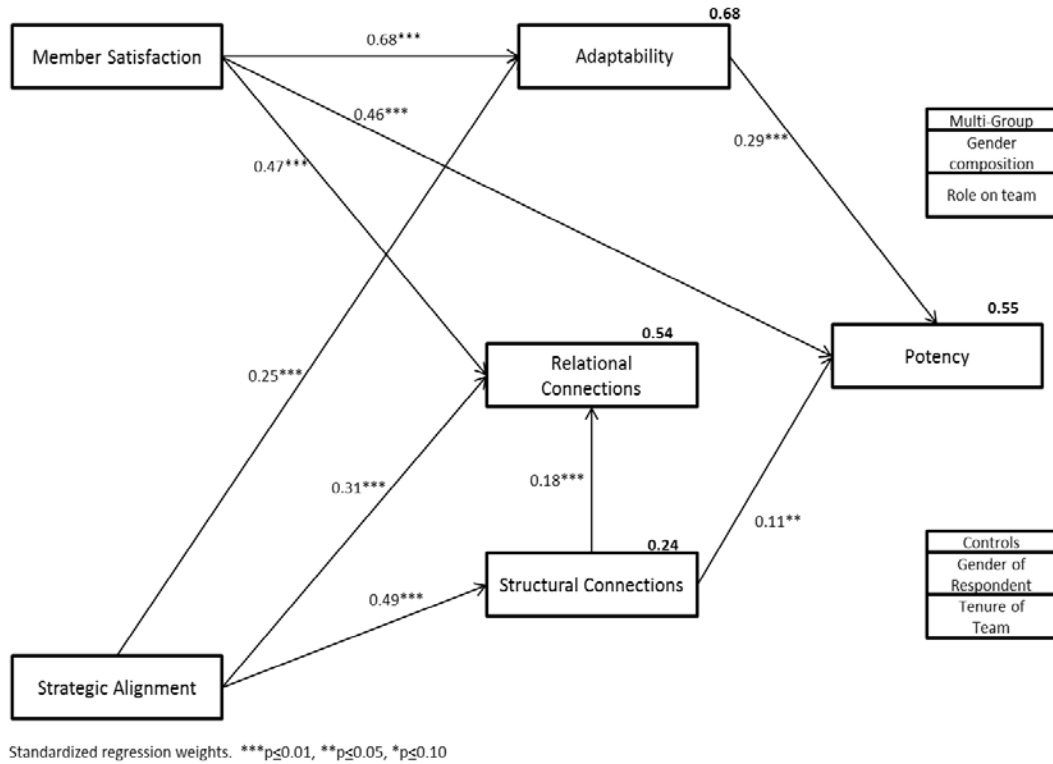


Figure B4: Alternate Model Evaluated

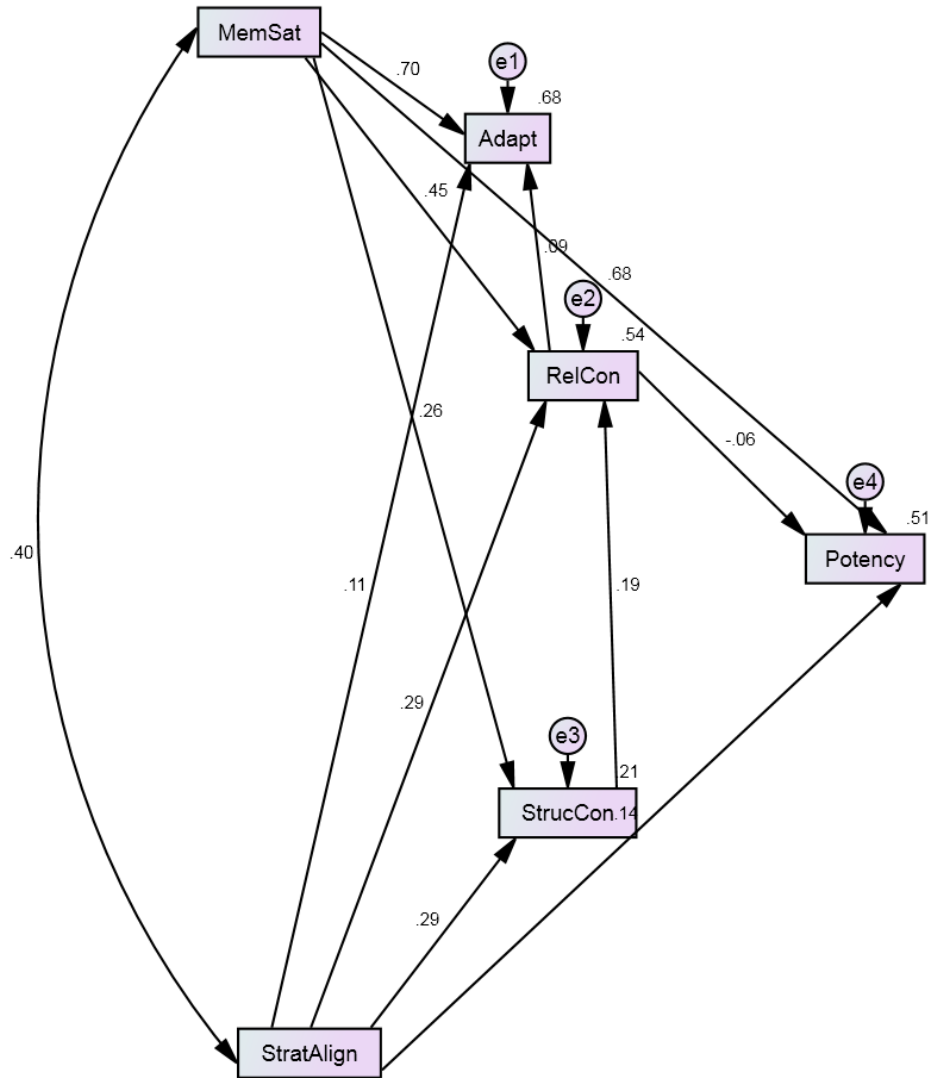


Rejected Model

Table B10: SEM Model Fit – Interaction Testing

Step	Action	Chi Square CMIN / DF <3 good; <5 sometimes allowable P-Value >0.05	PCLOSE >0.05	CFI >0.95 great; >0.90 traditional; >0.80 sometimes allowable	TLI >0.95 is desired >0.92 OK with >30 items	GFI >0.95	NFI 0.90 - 0.95 is acceptable, >.95 is good	RMSEA <0.05 good; 0.05 - 0.10 moderate; >0.10 bad	SRMR <0.09	Decision for Next Step (e.g. Covary Errors; Item to Remove)
1	Base Model	1.231, p=.297	0.509	0.999	0.992	0.995	0.994	0.033	0.0123	X MSxSA- >RC
2	X MSxSA- >RC	0.923, p=0.449	0.682	1.0	1.003	0.995	0.994	0.0	0.0123	X MSxSA- >SC
3	X MSxSA- >SC	0.739, p=0.594	0.809	1.0	1.009	0.995	0.994	0.0	0.0123	X MSxSA- >Pot
4	X MSxSA- >Pot	0.696, p=0.694	0.882	1.0	1.012	0.995	0.994	0.0	0.0128	X MSxSA- >Adapt
5	X MSxSA- >Adapt	0.618, p=0.742	0.915	1.0	1.013	0.994	0.993	0.0	0.0136	X MSxSA
6	X MSxSA	1.23, p=0.297	0.51	0.999	0.995	0.994	0.994	0.033	0.0142	Accept

Figure B5: Final Model



Appendix C: Mediation Testing

According to the Shrout and Bolger (2002), when mediation effects are theorized as distal, it may not be necessary to first test the $X \rightarrow Y$ direct effects. In particular, it may be “unwise” to “defer considering mediation” until the direct effects are established because the $X - Y$ association may be more “powerful when mediation is taken into account” (Shrout & Bolger, 2002). Also, it is assumed that there is theoretical justification for the mediator to precede the dependent variable in time (Carlo et al., 2012; Shrout & Bolger, 2002). Testing additionally included bootstrapping and bias corrected estimates for testing mediation effects and significance as it provides more accurate interpretation for small and medium sized studies (Carlo et al., 2012; Mathieu & Taylor, 2006; Shrout & Bolger, 2002).

In this paper, we follow the Shrout and Bolger (Shrout & Bolger, 2002) method referring to Indirect Effects as Distal Mediation (Carlo et al., 2012). Below outlines how results are categorized according to distal mediation, partial mediation, or complete mediation.

- **Distal Mediation** = X does not have a significant effect on Y (Total Effect C = not significant); Indirect Effect = $a \times b$ is significant when the mediator is present or $X \rightarrow Y$ under Indirect Effects is significant
 - Following Carlo, Lyytinen, and Rose (2012) who follow Shrout and Bolger (2002) – When $X \rightarrow Y$ (Total Effect C) is not significant, but the Total Indirect Effect ($abc + af + ec$) is, this is Distal Mediation.
 - If $X \rightarrow Y$ (Total Effect C) is significant AND the Total Indirect Effect ($abc + af + ec$) is significant, the effect is either partial or complete mediation.

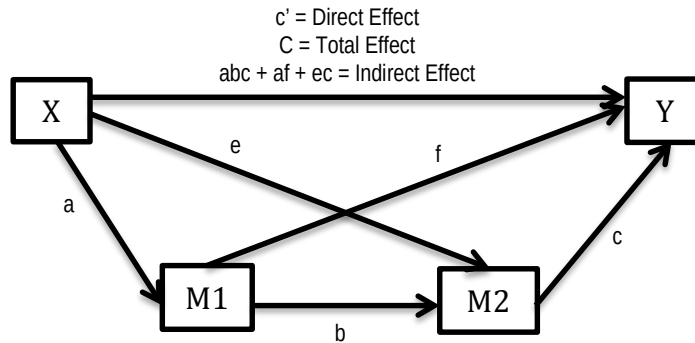
- **Partial Mediation** = $X \rightarrow Y$ is significant and remains so when the mediator (c') is introduced. The ratio of $(a \times b/C) < 100\%$ indicating only a portion of the effect is explained through the mediator.
 - If e , f , and c' are significant, it suggests partial mediation
- **Complete Mediation** = $X \rightarrow Y$ is significant (direct effect), but significance is eliminated in the presence of a mediator (c') indicating all the influence of X to Y is mediated through the mediator.
 - If e , f , and c' are each non-significant, then it suggests complete mediation

Our mediation paths include two mediators in a chain. Following Carlo, Lyytinen, and Rose (2012), Fletcher (2006), and Shrout and Bolger (2002), we tested the following procedures:

1. Total Effect: $X \rightarrow Y = C$
2. Direct Effects:
 - a. $X \rightarrow M1 = a$
 - b. $M1 \rightarrow M2 = b$
 - c. $M2 \rightarrow Y = c$
 - d. $X \rightarrow M2 = e$
 - e. $M1 \rightarrow Y = f$
 - f. $X \rightarrow Y = c'$
3. Estimate Indirect Effect $X \rightarrow M1 \rightarrow M2 \rightarrow Y$ path ($abc + af + ec$). This is actually $X \rightarrow Y$ under Indirect Effects in the AMOS output
4. Direct effect $X \rightarrow Y$ with $M1$ and $M2$ included in the model and paths connected (c')

5. Estimate the strength of mediation $(abc + af + ec)/C$ – reporting Complete, Partial, or Distal Mediation.

Graphically, this is represented as:



Below is our detailed results table for our Mediation analyses:

H	Mediation Path		Paths	Bootstrap		Bias Corrected	
				Unstd. Est.	S.E.	95% CI	Signifi- cance
H3a	Member Satisfaction → Structural Connections → Relational Connections → Potency	a	Member Satisfaction → Structural Connections	0.300	0.048	0.207, 0.395	0.001
		b	Structural Connections → Relational Connections	0.251	0.047	0.152, 0.339	0.001
		c	Relational Connections → Potency	-0.037	0.077	-0.186, 0.110	0.652
	Direct	e	Member Satisfaction → Relational Connections	0.398	0.043	0.312, 0.478	0.001
		f	Structural Connections → Potency	0.109	0.060	-0.010, 0.228	0.063
		c'	(Direct) Member Satisfaction → Potency	0.545	0.045	0.455, 0.629	0.001
		abc+af+ec	(Indirect) Member Satisfaction → Potency	0.015	0.035	-0.055, 0.084	0.651
		C	(Total Effect) Member Satisfaction → Potency	0.560	0.034	0.491, 0.626	0.001
H3b	Strategic Alignment → Structural Connections → Relational Connections → Potency	a	Strategic Alignment → Structural Connections	0.322	0.056	0.212, 0.435	0.001
		b	Structural Connections → Relational Connections	0.283	0.050	0.180, 0.377	0.001
		c	Relational Connections → Potency	0.320	0.079	0.161, 0.473	0.001
	Partial Mediation	e	Strategic Alignment → Relational Connections	0.320	0.044	0.229, 0.404	0.001
		f	Structural Alignment → Potency	0.144	0.068	0.007, 0.279	0.037

		c' $X \rightarrow Y (D)$	(Direct) Strategic Alignment $\rightarrow$ Potency	0.136	0.063	0.021, 0.266	0.019
		$abc+af+ec$ $X \rightarrow Y (I)$	(Indirect) Strategic Alignment $\rightarrow$ Potency	0.178	0.038	0.108, 0.257	0.001
		C $X \rightarrow Y (T)$	(Total) Strategic Alignment $\rightarrow$ Potency	0.314	0.053	0.212, 0.419	0.001
H4a	Member Satisfaction $\rightarrow$ Structural Connections $\rightarrow$ Relational Connections $\rightarrow$ Adaptability Partial Mediation	a $X \rightarrow M1$	Member Satisfaction $\rightarrow$ Structural Connections	0.300	0.048	0.207, 0.395	0.001
		b $M1 \rightarrow M2$	Structural Connections $\rightarrow$ Relational Connections	0.251	0.047	0.152, 0.339	0.001
		c $M2 \rightarrow Y$	Relational Connections $\rightarrow$ Adaptability	0.154	0.065	0.030, 0.280	0.011
		e $X \rightarrow M2$	Member Satisfaction $\rightarrow$ Relational Connections	0.398	0.043	0.312, 0.478	0.001
		f $M1 \rightarrow Y$	Structural Connections $\rightarrow$ Adaptability	0.026	0.049	-0.069, 0.121	0.623
		c' $X \rightarrow Y (D)$	(Direct) Member Satisfaction $\rightarrow$ Adaptability	0.585	0.048	0.489, 0.680	0.001
		$abc+af+ec$ $X \rightarrow Y (I)$	(Indirect) Member Satisfaction $\rightarrow$ Adaptability	0.081	0.029	0.025, 0.137	0.003
		C $X \rightarrow Y (T)$	(Total) Member Satisfaction $\rightarrow$ Adaptability	0.666	0.036	0.598, 0.739	0.001
H4b	Strategic Alignment $\rightarrow$ Structural Connections $\rightarrow$ Relational Connections $\rightarrow$ Adaptability Partial Mediation	a $X \rightarrow M1$	Strategic Alignment $\rightarrow$ Structural Connections	0.300	0.056	0.212, 0.435	0.001
		b $M1 \rightarrow M2$	Structural Connections $\rightarrow$ Relational Connections	0.251	0.050	0.180, 0.377	0.001
		c $M2 \rightarrow Y$	Relational Connections $\rightarrow$ Adaptability	0.154	0.078	0.395, 0.697	0.001
		e $X \rightarrow M2$	Strategic Alignment $\rightarrow$ Relational Connections	0.320	0.044	0.229, 0.404	0.001
		f $M1 \rightarrow Y$	Structural Connections $\rightarrow$ Adaptability	0.068	0.071	-0.066, 0.210	0.317
		c' $X \rightarrow Y (D)$	(Direct) Strategic Alignment $\rightarrow$ Adaptability	0.128	0.059	0.016, 0.247	0.029
		$abc+af+ec$ $X \rightarrow Y (I)$	(Indirect) Strategic Alignment $\rightarrow$ Adaptability	0.247	0.040	0.174, 0.330	0.001
		C $X \rightarrow Y (T)$	(Total) Strategic Alignment $\rightarrow$ Adaptability	0.375	0.053	0.266, 0.477	0.001

Appendix D: Table of Cognitive, Behavioral, and Affective Mechanisms of Adaptation

Concept	Description	Source
<i>Cognition</i>		
Detection and framing of cues	Promoting dynamical and recursive development of shared cognition	(Burke et al., 2006)
Situation assessment and awareness	Team has a shared perspective of environmental elements of importance to team goals	(Burke et al., 2006)
Learning	Facilitating development of knowledge and collective understanding of a situation	(Burke et al., 2006; Kozlowski & Bell, 2008; LePine, 2005; Schippers et al., 2007)
Outcome focus	Employed in order to identify multiple approaches to accomplish goals	(Woolley, 2009)
Shared mental models	Having similar knowledge structures that are shifted flexibly and contingencies that connect team networks to tasks	(Entin & Serfaty, 1999; Kozlowski et al., 1999; Marks et al., 2000)
Cognitive ability	Members' individual differences in capacity to process information and learn	(LePine, 2003, 2005)
Team orientation	Orientation toward performance vs. learning. A learning orientation better facilitates adaptation with difficult goals	(LePine, 2005)
Metacognition	The team's observation of itself and whether it needs to develop new or adjust existing routines	(Klein & Pierce, 2001)
Knowledge, skills, and attitudes (KSA)	KSAs to recognize patterns and needs for role or network linkages and team compilation to contribute to continuous improvement	(Kozlowski et al., 1999; Kozlowski & Bell, 2008; Salas et al., 2009)
Sensemaking	Assigning meaning to actions	(Klein & Pierce, 2001; Salas et al., 2009)
<i>Behavior</i>		
Enacting and executing processes	Includes plan execution representing actions of back-up behaviors, communication, coordination, and application of resources	(Burke et al., 2006; Klein & Pierce, 2001)
Task dynamics and segmentation	These emerge from a process focus and, with repetition, providing experiences for learning and skill acquisition	(Kozlowski et al., 2009; Kozlowski & Bell, 2008; Woolley, 2009)
Interpersonal processes	Includes conflict management, motivating, confidence building, and affect management	(Marks et al., 2001)
Conscientiousness	Included achievement and dependability	(LePine, 2003)
Working together, sharing knowledge	For the purpose of developing systems of communication	(LePine, 2003)
Modifying the configuration or network of team roles	For purposes of task execution and exploration of transaction alternatives to meet internal or external contingencies	(Kozlowski et al., 1999; LePine, 2005)
Team reflection	Behaviors fostering insights about processes and performance such as examining tasks, methods, resources as well as questioning, planning, analysis, and use of knowledge	(Schippers et al., 2007; Wiedow & Konradt, 2011)
Living up to agreements	For the purpose of production of change such as making decisions and initiating changes	(Wiedow & Konradt, 2011)

Feedback-seeking behavior	To obtain information on the gap between actual and desired states relevant to goals or to assess outcomes and processes	(Klein & Pierce, 2001; Schippers et al., 2007)
Problem Solving	Includes the behavior of parsing	(Klein & Pierce, 2001)
<i>Affect</i>		
Emergent states	Reflected in collective efficacy, potency, and cohesion	(Marks et al., 2001)
Openness to experience	Included imaginativeness, curiosity, originality, broad-mindedness	(LePine, 2003; Salas et al., 2009)
Proactive personalities	Meaning those that are more likely to challenge norms and values	(Schippers et al., 2007)
Belief	In regard to the situation such as whether the plan is going well or not	(Klein & Pierce, 2001)
Stress	May be induced by uncertainty or ambiguity	(Entin & Serfaty, 1999)

Appendix E: Interview Protocol and Questions

Team Boundary Management

Step 1: Introduction and Explanation

Introduction (Interviewer): *“Hi (name). I just want to thank you for taking the time to meet with me today. I appreciate the time and attention. Before getting started, there are a couple of things I would like to cover up front about why we are doing this interview.”*

Purpose and Format for the Interview (Interviewer): *“As you know from the letter you have received from me, I am interested in how quality improvement project teams in hospitals manage their boundaries. To make sure we are on the same page, I am not interested in any proprietary information about your institution. I am more interested in how the quality improvement project team engaged people outside the team to enhance project performance. I am particularly interested in three dimensions of team boundary management — who (roles, names are not needed) was identified as meaningful to the team’s ability to succeed, why they were meaningful, and how you engaged them in your efforts, or if you did not engage them, why. That is really the focus of what we are going to talk about today”*

Confidentiality (Interviewer): *“Everything you share in this interview will be kept in the strictest confidence, and your comments will be transcribed anonymously – omitting your name, anyone else you refer to in this interview, as well as the name of your institution and any identifiable information of you or your institution. Your interview responses will be included with all the other interviews I conduct.”*

Audio Taping (Interviewer): *“To help me capture your responses accurately and without being overly distracting by taking notes, I would like to record our conversation with your permission. Again, your response will be kept confidential. If at any time, you are uncomfortable with this interview, please let me know and I will turn the recorder off.”*

“Any questions before we begin?”

Step 2: Opening Icebreaker Question

Interviewer: *“(Name) tell me about yourself and your role here at (institution name).”*

Probing questions to ask only if the respondent does not provide responses to:

Interviewer:

- *“Where are you from?”*
- *“How long have you been with this institution?”*
- *“How long have you been in this role?”*
- *“What are the major areas of responsibility in your role?”*
- *“How did you get to this position?”*
- *“What is your educational background?”*
- *“Tell me more about this,”*

Step 3: Experiential Questions on Institutional Business Model changes over the last 5 years

Introduction: *“I am going to ask you a few questions regarding your experience with your quality improvement project and team at (institution name). Please be as specific as you can and provide me with as much detail as you can remember. If I have a clarifying question, I will ask you.”*

Interviewer Q1: *Tell me about a Quality Improvement project you led and completed within the last 24 months. If you need to go back farther for a project you’d like to discuss, please select a project completed no longer than 48 months ago. First, I would specifically like to know the following background information:*

(NOTE: The lines following the bullet point items are for researcher notes, they are not intended for interviewee to complete).

- o **Project Name:** _____
- o **Project Aim / Goal:** _____
 - Baseline measurement: _____
 - Project completion measurement: _____
- o **Length of Project:** _____
- o **Team:**
 - Number of members: _____
 - Composition – Professions represented: _____
 - Men / Women: _____
 - Selection of Members (how, why): _____
 - Member turnover? _____

Interviewer Q2: ***Who (individuals / groups / organizations) did your team identify as potentially having an influence on your success?***

Clarifying questions that will be asked if needed to engage specifics from the respondent, or to add detail to their story:

- o *At what point in the project were these influencers identified?*
- o *What brought them to your attention?*
- o *Describe why they were considered meaningful or relevant to your success (or lack)*
- o *Did your team formally identify these influencers or did it happen another way?*

Interviewer Q3: ***Who did you connect with outside the team?***

Clarifying questions that will be asked if needed to engage specifics from the respondent, or to add detail to their story:

- o *Names are not needed, just professional / work roles*
- o *Were there people / groups you identified as influencers on the project that you didn't contact? If so, why?*
- o *When, in the project, did you identify who might influence the project (and in what way)?*

Interviewer Q4: ***Why did you connect with those individuals or groups?***

Clarifying questions that will be asked if needed to engage specifics from the respondent, or to add detail to their story:

- o *What role did they have relevant to the team's success or lack of success?*
- o *How did they help or hinder the project's progress?*
- o *Possible additional detail probes related to whether they provided or were connected through power, identity, efficiency, competence, other*

Interviewer Q5: ***Who reached out to people outside the team (e.g. team leader, team members, certain titles or roles)?***

Clarifying questions that will be asked if needed to engage specifics from the respondent, or to add detail to their story:

- o *How did people outside the team respond to your requests, needs?*

Interviewer Q6: ***What styles or tactics (that is, ways) of engagement did your team use to manage its boundaries for the purpose of advancing the project?***

Clarifying types of styles and tactics will be provided if needed to engage specifics from the respondent, or to add detail to their story:

- o *Dependence on existing relationships*
- o *Deference to expertise*
- o *Negotiation for resources or testing*
- o *Promotion of team activities*
- o *Aligned incentives*
- o *Develop trust*
- o *Offer a benefit*
- o *Collaboration / Cooperation*
- o *Expectation setting*

Interviewer Q7: ***What instruments or methods (i.e. means) did your team use to manage its boundaries (connections with people outside the team)?***

Clarifying questions and/or types of instruments, methods, or practices will be provided, as examples, if needed to engage specifics from the respondent, or to add detail to their story:

- o *Specific change management practices*
- o *Specific improvement methodology practices*
- o *Communication tools*
- o *Communication styles*
- o *How often did you connect with selected people outside the team to support your project?*

Interviewer Q8: ***Did the team ever have to change the ways or means (above) you were utilizing in order advance the project?***

Clarifying questions that will be asked if needed to engage specifics from the respondent, or to add detail to their story:

- o *Did the team ever have to change from a collaborative approach to a directive approach or vice versa?*
- o *Did you have to adjust instruments (e.g. change from e-mail to face-to-face updates), methods, or practices?*
- o *If you had to make adjustments, how did you know an adjustment was needed and why did you select a particular alternative?*

Interviewer Q9: ***What effect – positive or negative – did your team’s boundary management have on:***

- o *Team tasks*
- o *Team confidence in success*
- o *Project outcome*

Step 4: Closing

Interviewer: *“That concludes our interview. Do you have any additional comment you would like to make regarding quality improvement project team boundary management in hospitals?”*

“Thanks very much for your time and for sharing your experiences with me. In the event that I have a need to clarify some of your responses, would it be ok for me to give you a short follow up phone call? Thank you again.

Appendix F: Codebook Development for Thematic Analysis

Initial Codebook	Final Codebook
<u>Project / Team Information</u> • Project Name: • Project Aim / Goal: - o Baseline measurement: - o Project completion measurement: • Project Length: • Team Description: - o Number of members: - o Composition – Professions represented: - o Men / Women: - o Selection of Members (how, why): - o Member turnover? • (Intra) Team Dynamic: <i>(decided to add based on a couple of the interviews)</i> - o Familiarity / Comfort with team members - o Cooperative / Agreeable - o Contentious <u>(Who) External Connections:</u> • Unit level - o Individual - o Group - o Organization • Role - o Related professionals (e.g. physicians / nurses on team connected with other external physicians and nurses) - o Non-represented roles (e.g. no IT representation on team, but connected with IT externally) • Timing of Connection - o Early Project - o Mid Project - o Late Project • Identification Trigger (for +/- Influencers) - o From within team - o From outside the team - o Mechanism for identification • Connectors from team - o Team leader - o Team members <u>(Why) Connections Made:</u>	<u>Project / Team Information</u> <i>(header for reference, not a code)</i> • Project Name: - o Project Purpose: • Project Aim / Goal: - o Baseline measurement: - o Project completion measurement: • Project Length: • Team Description: - o Number of members: - o Composition – Professions represented: - o Men / Women: - o Selection of Members (how, why): - o Member turnover? • (Intra) Team Dynamic: <i>(decided to add based on a couple of the interviews)</i> - o Familiarity / Comfort with team members - o Cooperative / Agreeable - o Contentious <u>(Who) External Connections:</u> <i>(header for reference, not a code)</i> • Unit level <i>(header for reference, not a code)</i> - o Who-Individual - o Who-Group - o Who-Organization • Role <i>(header for reference, not a code)</i> - o Who Role-Related professionals (e.g. physicians / nurses on team connected with other external physicians and nurses) - o Who Role-Non-represented roles (e.g. no IT representation on team, but connected with IT externally) • Timing of Connection <i>(header for reference, not a code)</i> - o Who Timing-Early Project - o Who Timing-Mid Project - o Who-Late Project • Identification Trigger (for +/- Influencers) <i>(header for reference, not a code)</i> - o Who IDs-From within team - o Who IDs-From outside the team - o Who IDs-Mechanism for identification • Connectors from team <i>(header for reference, not a code)</i> - o Who Connects-Team leader - o Who Connects-Team members • Who Other

• Meaningfulness to team (role in success) - o Power (Santos & Eisenhardt, 2005) - o Identity (Santos & Eisenhardt, 2005) - o Efficiency (Santos & Eisenhardt, 2005) - o Competence (Santos & Eisenhardt, 2005) - o Resource needs (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a) (Ancona and Caldwell) - o Coordination of tasks (Ancona & Caldwell, 1990; Ancona & Caldwell, 1992a) - o Culture (Schein, 2010) - o Rapport (<i>word used in a few interviews</i>) - o Pragmatics (<i>saw instances of not purposely selecting people to work with, it was more of a matter of who was available – usually for testing changes</i>) • Type of Influence on team - o Barrier - o Partner / Collaborator - o Supporter • (Why) Incomplete connection - o Elected to not connect - o Rejected by connection <u>(How) External Engagement:</u> • Ways (styles, tactics) of engagement - o Dependence on existing relationships (Santos & Eisenhardt, 2009) - o Power (Santos & Eisenhardt, 2005) (<i>included Power thru Competence here in addition to meaningfulness to team based on read-through of a couple of interviews</i>) ▪ Empowerment (<i>heard this in a couple of interviews</i>) - o Identity (Santos & Eisenhardt, 2005) - o Efficiency (Santos & Eisenhardt, 2005) - o Competence (Santos & Eisenhardt, 2005) - o Deference to expertise (Weick & Sutcliffe, 2007) - o Negotiation for resources or testing (Ancona & Caldwell, 1992a; Druskat & Wheeler, 2003) - o Promotion of team activities (Ancona, 1990; Ancona & Caldwell, 1992a) - o Aligned incentives (Present in Qual 1) - o Develop trust (Personal experience) - o Offer a benefit (Personal experience) - o Collaboration / Cooperation (Ancona, 1990) - o Expectation setting (Present in Qual 1) - o Use of evidence (Personal experience) • Means (instruments, methods) of engagement (<i>each of the items below could easily lead to another level of codes like Lean, Model for Improvement, Innovation Adoption, Technology vs. In-person communication, etc.</i>) - o Change management practices - o Improvement methodology practices - o Communication tools	<u>(Why) Connections Made:</u> (<i>header for reference, not a code</i>) • Why-Value from Boundary • Why-Team Effect on Boundary • Why-Boundary Effect on Team • Why-Culture • Why-Competence • Why-Cooperation • Why-Inclusiveness / Inclusion • Why-Pragmatism • Why No Connection – Team Choice • Why No Connection – Rejection by Boundary • Why Other <u>(How) External Engagement:</u> (<i>header for reference, not a code</i>) • Ways (styles, tactics) of engagement (<i>header for reference, not a code</i>) - o How Ways-Dependence on network relationships (Santos & Eisenhardt, 2009) - o How Ways-Power (Santos & Eisenhardt, 2005) (<i>included Power thru Competence here in addition to meaningfulness to team based on read-through of a couple of interviews</i>) - o How Ways-Empowerment (Spreitzer, 1996) - o How Ways-Identity (Santos & Eisenhardt, 2005) (Note – peer identity) - o How Ways-Efficiency (Santos & Eisenhardt, 2005) - o How Ways-Deference to expertise (Weick & Sutcliffe, 2007) - o How Ways-Shared Vision (Present in Qual 1 Paper (Grooms & Lyytinen, 2014)) - o How Ways-Trust (Personal experience) - o How Ways-Reward, Benefit (Personal experience) - o How Ways-Partnership (Note – <i>include these in code definition – they are synonyms: Collaboration / Cooperation</i> (Ancona, 1990) (Present in Qual 1)) - o How Ways-Use of evidence (Personal experience) - o How Ways-Development - o How Ways-Other • Means (instruments, methods) of engagement (<i>each of the items below could easily lead to another level of codes like Lean, Model for Improvement, Innovation Adoption, Technology vs. In-person communication, etc.</i>) (<i>header for reference, not a code</i>) - o How Means-Improvement methodology practices - o How Means-Other Technologies - o How Means-E-mail - o How Means-Face-to-face - o How Means-Phone - o How Means-Documents - o How Means-Meetings - o How Means-Project Management documents or tools
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- o Communication styles - o Development (e.g. Individual employee development, next level leadership development) (<i>this came from a couple of interview read-throughs; NOTE – this might cross over into 'why a positive influence'</i>) - o Reasons for selecting means • Frequency of engagement • Change in ways or means - o How team knew to make change - o Type of change selected - o Why the change was selected • External Response - o Helped advance project - o Hindered project <u>Effect on Team</u> • Potency (Gully et al., 2002; Guzzo & Shea, 1992) • Internal Adaptation (Burke et al., 2008; Burke et al., 2006; Kozlowski et al., 1999; Kozlowski et al., 2009; LePine, 2005; Wiedow & Konradt, 2011) • Psychological Safety (Edmondson, 1999; Faraj & Yan, 2009) • Learning (Edmondson & Roloff, 2009; Edmondson & Nemphard, 2009; Kozlowski & Bell, 2008; Wong, 2004) • Network Development (Ancona & Caldwell, 1988; Brass, 1984; Marrone, 2010; Mørk, Hoholm, Maaninen-Olsson, & Aanestad, 2012) <u>Why a Positive Influence</u> • External “Buy-In” (<i>most of these ideas emerged from personal experience or interviews</i>) - o Connecting to value (e.g. personal or organizational values; value / worthiness of project) (<i>personal experience, interviews</i>) - o Improves external boundary's day-to-day activities (WIIFM) (<i>personal experience, interviews</i>) - o Supports micro or meso-system need (<i>personal experience</i>) - o Reduces fear of change (might also need to add the flip side – develops trust) (<i>interviews</i>) - o Accountability (e.g. Creation of, 1-way or 2-way) (<i>interviews</i>) - o Relationship development (<i>interviews</i>) - o Provides a forum for idea sharing (<i>interviews</i>) • Organizational Support - o Encourages fulfillment of resource needs for team - o Team helps meet organizational goals	- o Development (e.g. Individual employee development, next level leadership development) (<i>this came from a couple of interview read-throughs; NOTE – this might cross over into 'why a positive influence'</i>) Moved to How Ways - o Reasons for selecting means • Frequency of engagement (<i>header for reference, not a code</i>) - o How Frequency-1 x Only - o How Frequency-Daily - o How Frequency-Weekly - o How Frequency-Monthly - o How Frequency-Quarterly - o How Frequency-As needed • Change in ways or means (<i>header for reference, not a code</i>) - o How Changes-How team knew to make change - o How Changes-Type of change selected - o How Changes-Why the change was selected • External Response (<i>header for reference, not a code</i>) - o Results-Advanced project ▪ BM Results Changed-Advanced to Resistance - o Results-Resistance ▪ BM Results Changed-Resistance to Advanced - o Results-Equivalent to, Exceeded Goal - o Results-Approached Goal (not met) - o Results-Failed Goal (far from) <u>Effect on Team</u> • Potency (Gully et al., 2002; Guzzo & Shea, 1992) • Internal Adaptation (Burke et al., 2008; Burke et al., 2006; Kozlowski et al., 1999; Kozlowski et al., 2009; LePine, 2005; Wiedow & Konradt, 2011) • Psychological Safety (Edmondson, 1999; Faraj & Yan, 2009) • Learning (Edmondson & Roloff, 2009; Edmondson & Nemphard, 2009; Kozlowski & Bell, 2008; Wong, 2004) • Network Development (Ancona & Caldwell, 1988; Brass, 1984; Marrone, 2010; Mørk et al., 2012) <u>Why Does BM Matter</u> • External “Buy-In” (<i>most of these ideas emerged from personal experience or interviews</i>) - o Matters: Value connection (e.g. personal or organizational values; value / worthiness of project) (<i>personal experience, interviews</i>) - o Matters: WIIFM -Improves external boundary's day-to-day activities (<i>personal experience, interviews</i>) - o Matters-Shared Vision, Belief (Supports micro or meso-system need <i>personal experience</i>) - o Matters-Reduces fear of change (might also need to add the flip side – develops trust) (<i>interviews</i>) - o Matters-Builds Trust
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	- o Matters-Accountability (e.g. Creation of, 1-way or 2-way) (<i>interviews</i>) - o Matters-Relationship development (<i>interviews</i>) - o Matters-Boundary Voice (<i>interviews</i>) - o Matters-Encourages fulfillment of resource needs for team - o Matters-Team helps meet organizational goals - o Matters-Active boundary participation - o Matters-Other
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Appendix G: System Dynamic Modeling

Figure G1: High Performing Team – System Dynamic Modeling of Boundary Management Interrelationships and Adaptation



Figure G2: Low Performing Team – System Dynamic Modeling of Boundary Management Interrelationships and Adaptation

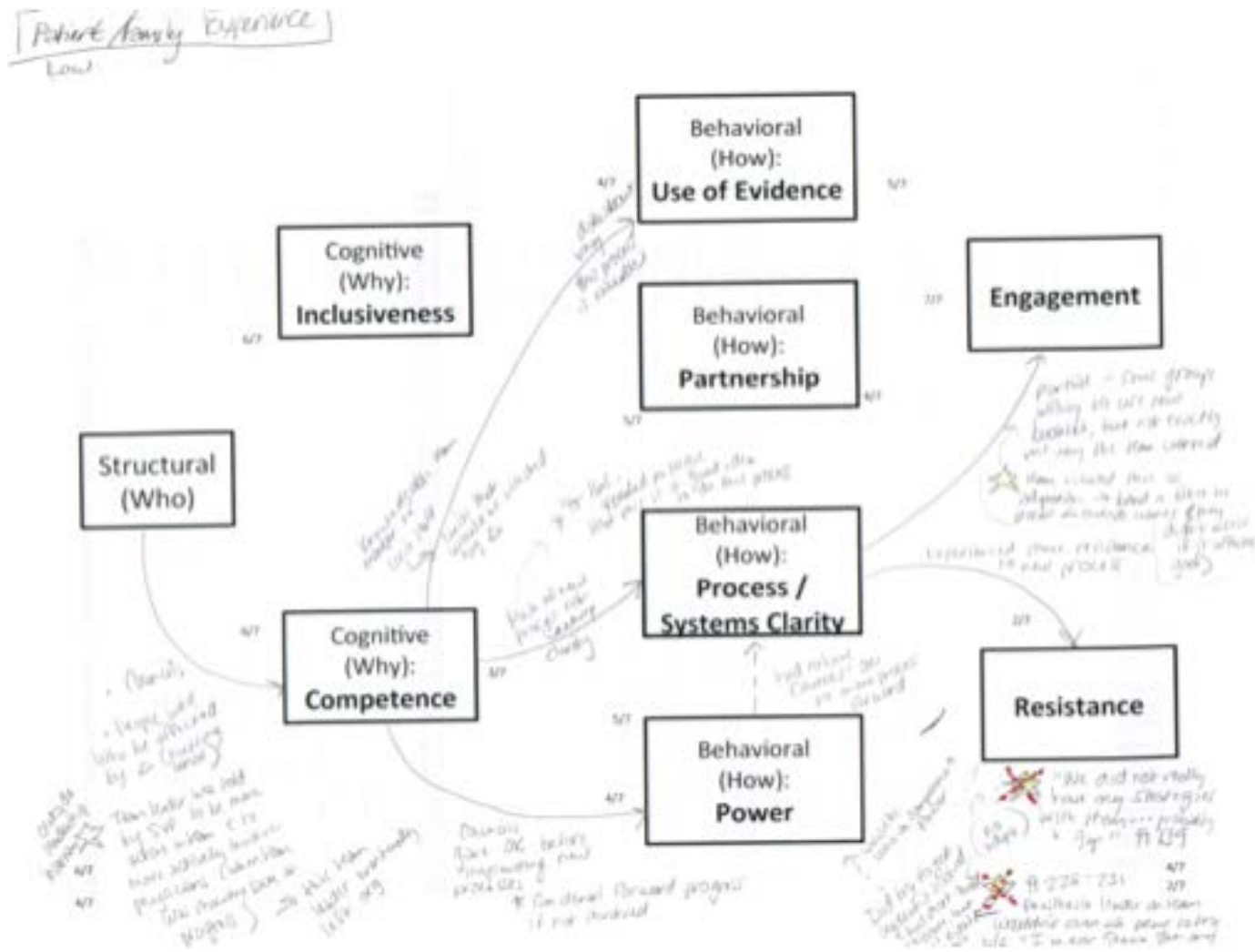


Figure G3: High Performing Teams Comparison - Aggregate Boundary Management Notes

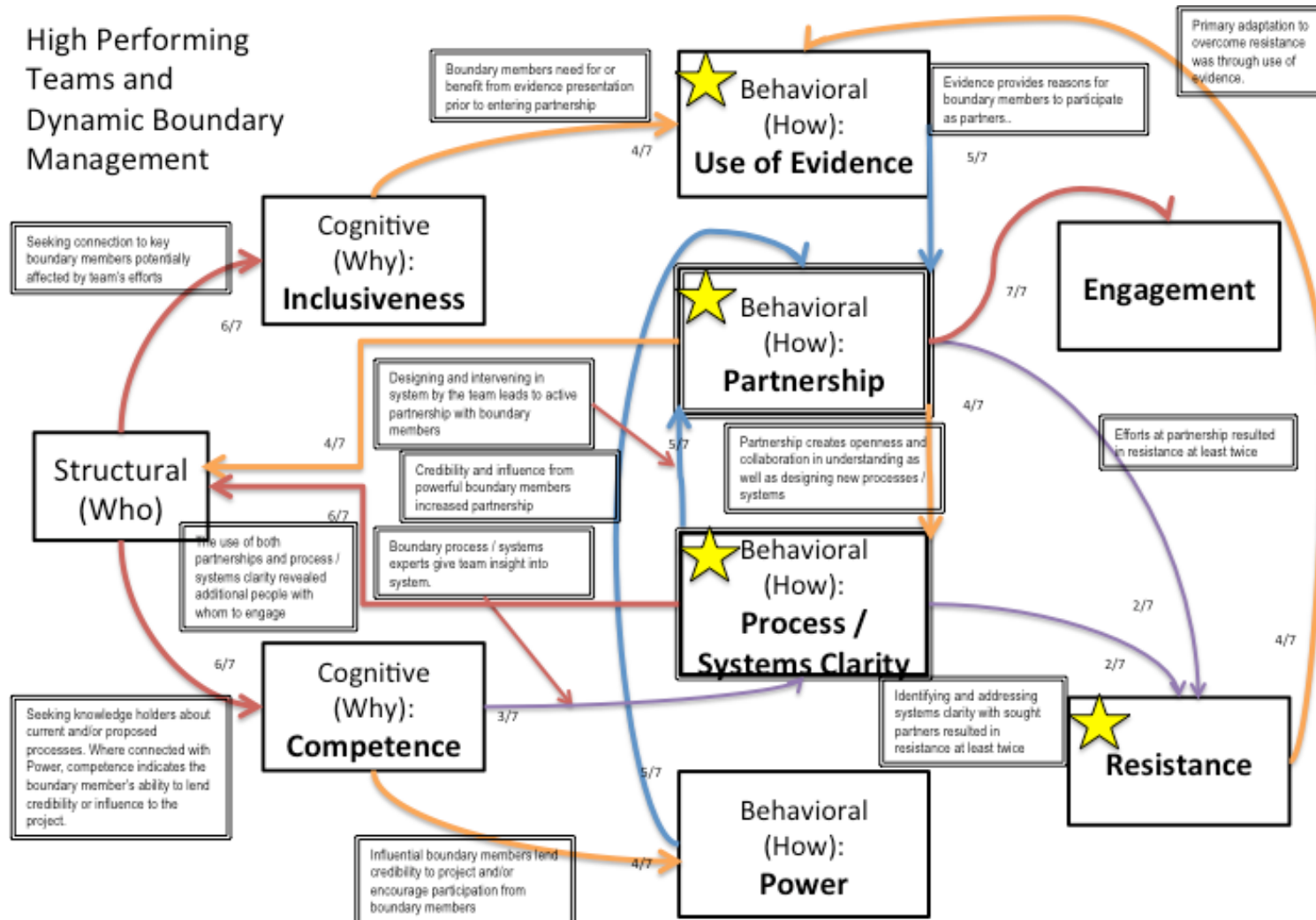
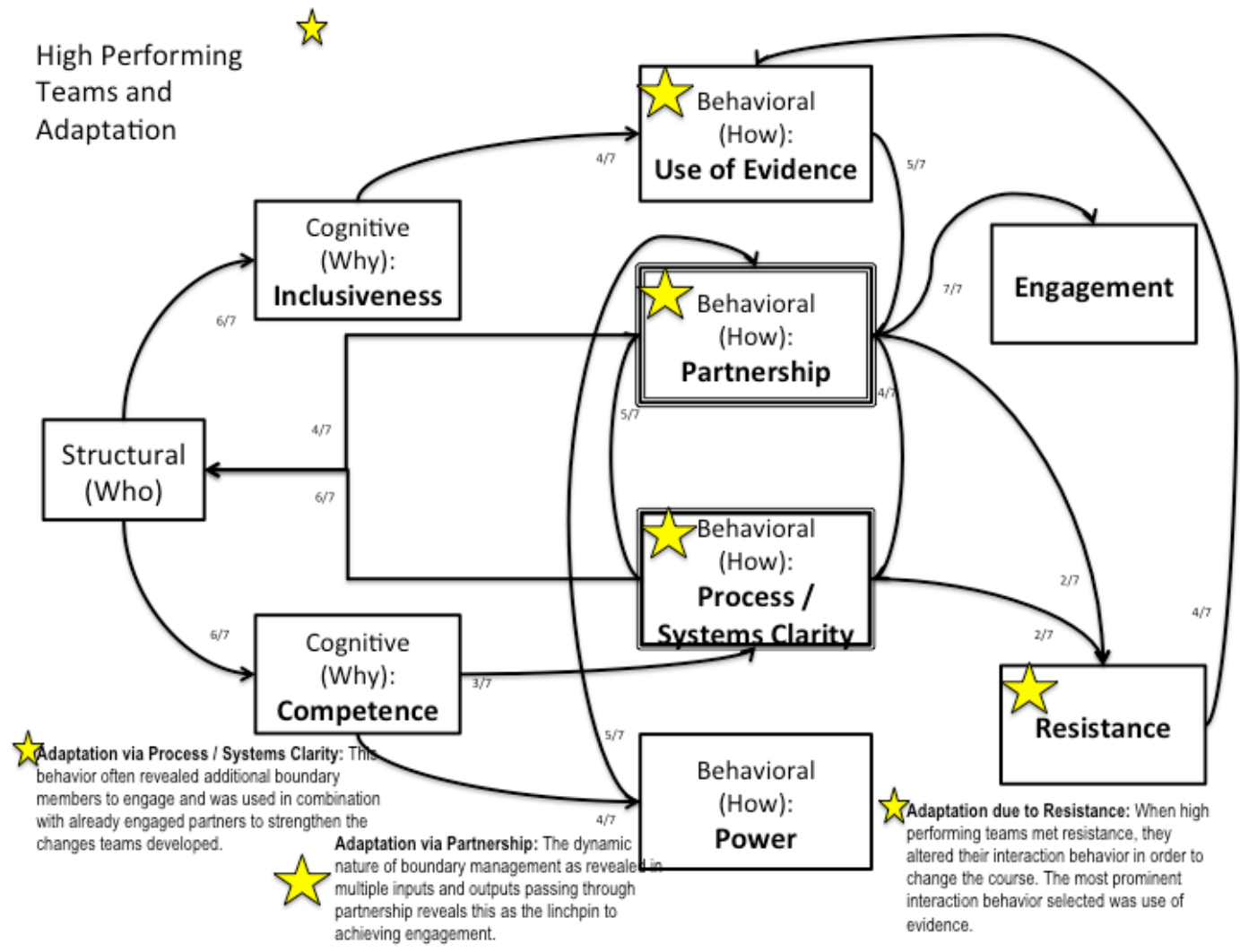


Figure G4: High Performing Teams Comparison – Aggregate Adaptation Notes



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